

THE TOPOLOGY IS THE SINK: TOPOLOGICAL HALLUCINATION DETECTORS REDUCE TO FIRST-ORDER ATTENTION STATISTICS

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ABSTRACT

The recent TOHA (ACL 2026) is part of a class of hallucination detectors based on persistent homology (PH) of attention graphs. In this paper, we show that, in decoder-only models, such topological features are equivalent to first-order attention statistics, usually dominated by the attention sink, either exactly or up to one computable number. For instance, for any symmetric dissimilarity and vertex s , we define the coning defect δ_s . Classical cone and stability arguments show that the whole Vietoris–Rips barcode is at most bottleneck distance δ_s away from a “star diagram” read off the distances to s . Any stable vectorization is thus a function of one column of the attention matrix, up to δ_s . This applies verbatim to TOHA, and proves that its prompt–response topological divergence is equal to one minus the response tokens’ mean largest prompt attention whenever a prompt-level coning defect vanishes, and is within that defect otherwise. With 20,580,160 per-head graphs from 16 settings (7 open models of 1.1B–7.62B parameters; TruthfulQA, HaluEval, and an on-policy TriviaQA benchmark we build), this identity is exact on 57–88% of them (including in two models with no first-token sink, where it is the largest prompt attention) with no violation. The sink is a property of the model, not of the theorem. For whole-graph PH, adding TOHA’s score to its first-order counterpart changes AUC by at most 0.007 (equivalent within ± 0.015 in 16 of 16 settings). Supervised probes on the two are equivalent in 16 of 16. On TruthfulQA and on-policy TriviaQA, none of 0D PH, sink-deflated PH, or the per-head coning defect add more than 0.009 AUC to cheap non-topological probes, and TOHA adds at most 0.013. 838,356 layer graphs exhibit no bounds violations. All models with a first-token sink have 36–98% that are exactly coned, with zero 1D persistence on all of them. Median per-layer rank correlation between length-normalized 0D persistence and sink attention is at least 0.98. We release the extractor, the evaluator, and per-layer features for a subset of settings.

1 INTRODUCTION

A growing class of hallucination detectors views a transformer’s attention matrix as a weighted graph, and then summarize it using persistent homology (PH), especially the barcodes of its connected components (H_0) and loops (H_1) along a Vietoris–Rips filtration (Kushnareva et al., 2021; Cherniavskii et al., 2022; Perez & Reinauer, 2022; Kostenok et al., 2023; Bazarova et al., 2026; Samaga et al., 2026). Their intuition is that perhaps global features in the flow of information could be detected by topology, which cannot be captured by statistics on tokens alone. In a parallel development, it has been shown that decoder-only language models devote a high proportion of their attention to the first token, known as the attention sink (Xiao et al., 2024; Sun et al., 2024; Cancedda, 2024; Gu et al., 2025; Barbero et al., 2025). The connection between these fields has only been made recently, which indicates that sinks convey most of the information needed by attention-based hallucination detectors (Binkowski et al., 2026), but no-one has investigated what role they play in the topology of attention graphs.

In this paper we show that, for topological detectors, the two literatures study the same object to a large and measurable degree. Every sublevel set of the Vietoris–Rips filtration is a cone with apex

s when one vertex s is closer to every other vertex than those vertices are to each other. All higher homology disappears and the minimum spanning tree (which computes H_0) will be the star around s . This is where real attention graphs live, or almost so, thanks to the presence of a sink. This means that in most layers, the “topology” of the causal attention graph is precisely a function of how much attention each token sends to the sink. In the rest of the layers, it depends on a single computable number, the coning defect. This is also true for the published detector which restricts topology to the prompt–response interface: TOHA’s divergence is a first-order attention statistic plus a computable correction, and on real heads the correction adds at most 0.007 AUC.

Contributions.

1. **A whole-barcode sink reduction (§3).** The ingredients are classical: a flag complex with a dominating vertex is a cone (Barmak & Minian, 2012; Boissonnat & Pritam, 2020), and persistence is stable (Cohen-Steiner et al., 2007; Chazal et al., 2009; 2014). We combine them into one-sided bounds in a single computable number. For any symmetric dissimilarity and vertex s , bars of dimension $k \geq 1$ have length at most δ_s , and 0D total persistence is sandwiched around the star weight (Theorem 1). This means that the entire barcode lies within bottleneck distance δ_s of the star diagram. All stable vectorizations are thus equivalent to a single column of distances (Corollary 1). For causal attention, δ_0 has a closed form (Corollary 2).
2. **The published topological detector is a first-order detector (§3, §5.4).** Proposition 1: The topological divergence of TOHA is one minus the average max-prompt attention of the response tokens on prompt-coned heads or is within a computable defect of it. The qualitative observation by Binkowski et al. (2026) is made exact. There are no violations of the identity on 20,580,160 real head graphs, and adding TOHA to its first-order counterpart changes AUC by 0.000 to +0.007.
3. **Verification of the whole-graph reduction (§5.1, Appendix D).** There are no bound violations in 838,356 graphs, and graphs are exactly coned in 36–98% of cases in sink-dominated models. In a sink dose-response study, we explain how 1D features become weak because of coning, while H_1 is empty on all coned graphs.
4. **A detection study using strong baselines (§5.3).** We compare 0D/1D PH, sink features, sink-deflated PH, the coning defect, and TOHA against length, lexical, log-probability, LLM-Check (Sriramanan et al., 2024), Lookback-Lens (Chuang et al., 2024), and hidden-state probes. We report results from TruthfulQA, HaluEval, and an on-policy TriviaQA benchmark using grouped cross-validation, early- and late-fusion incremental tests, and equivalence tests.

2 BACKGROUND AND RELATED WORK

Topology of attention. Kushnareva et al. (2021) and Cherniavskii et al. (2022) introduced barcodes of symmetrized BERT attention graphs as features for artificial-text detection and acceptability; Perez & Reinauer (2022) and Kostenok et al. (2023) extended them to classification and uncertainty estimation. TOHA (Bazarova et al., 2026) calculates an MST-based distance between prompt and response subgraphs for hallucination detection, HalluZig (Samaga et al., 2026) relies on zigzag persistence across layers, and CHARM (Frasca et al., 2026) learns a message-passing model over attention graphs. Gardinazzi et al. (2025) study zigzag persistence of hidden representations. All 0D summaries in this family are MST statistics (Fact 1; Gower & Ross, 1969). Our results concern what those statistics measure in causal models.

Attention sinks. Xiao et al. (2024) observed that streaming LMs keep the first tokens’ key-values because a large share of attention lands there; Sun et al. (2024) tied sinks to massive activations, Cancedda (2024) to low-rank “dark signals”, and Gu et al. (2025) characterized when they emerge during pre-training. Barbero et al. (2025) argued they prevent over-mixing; related “no-op” attention to special tokens was observed in BERT (Clark et al., 2019) and vision transformers (Darcet et al., 2024). Binkowski et al. (2026) use sinks with large-norm values directly as a hallucination signal (SinkProbe) and argue that earlier attention-based detectors rely on them implicitly. We make this precise for topological detectors, turning sink strength into a statement about the topology of attention graphs.

Hallucination detection. Beyond topology, detectors use sampling consistency (Manakul et al., 2023; Farquhar et al., 2024), internal probes (Azaria & Mitchell, 2023; Burns et al., 2023; Marks & Tegmark, 2024; Li et al., 2023b; Kossen et al., 2024; Orgad et al., 2025; Du et al., 2024), covariance spectra (Chen et al., 2024), attention-kernel log-determinants (Sriramanan et al., 2024), and context-vs-generated attention ratios (Chuang et al., 2024). Recent audits find that several benchmarks admit near-perfect detection from surface form alone (Hussain & Kantarcioglu, 2026; Janiak et al., 2025), echoing annotation artifacts in NLI (Gururangan et al., 2018; McCoy et al., 2019). We include lexical controls and an on-policy benchmark.

3 THEORY: ATTENTION SINKS CONE THE VIETORIS–RIPS FILTRATION

Let $D \in \mathbb{R}_{\geq 0}^{N \times N}$ be symmetric with zero diagonal. The Vietoris–Rips filtration is $K_\epsilon = \{\sigma \subseteq [N] : \max_{u,v \in \sigma} D_{uv} \leq \epsilon\}$; $\text{dgm}_k(D)$ is the barcode of $H_k(K_\epsilon; \mathbb{F})$ over a field, and $P_k(D)$ is its total persistence (sum of finite bar lengths). For attention, $A^{(l)}$ is the head-averaged, row-stochastic, lower-triangular attention matrix of layer l and, following the literature, $D = 1 - \max(A, A^\top)$ with zero diagonal.

Fact 1 (Gower & Ross, 1969). *All 0D bars are born at 0, and their finite death times are, as a multiset, the edge weights of a minimum spanning tree (MST) of D . In particular $P_0(D) = w(\text{MST}(D))$ is the total persistence (sum of finite bar lengths) of $\text{dgm}_0(D)$.*

Definition 1 (Coning defect, star weight). For $s \in [N]$, the *coning defect* is $\delta_s(D) = \max_{u \neq v, u, v \neq s} [\max(D_{su}, D_{sv}) - D_{uv}]_+$ (zero if $N < 3$), and the *star weight* is $S_s(D) = \sum_{u \neq s} D_{su}$. We call D *coned at s* if $\delta_s(D) = 0$, i.e. s is at least as close to both endpoints of every edge as they are to each other.

Theorem 1 (Sink-coning reduction). *For every symmetric D and every vertex s , with $\delta = \delta_s(D)$:*

- (a) *every bar of $\text{dgm}_k(D)$, $k \geq 1$, has length at most δ ; hence $P_k(D) \leq \delta \cdot |\text{dgm}_k(D)|$;*
- (b) *$S_s(D) - (N-1)\delta \leq P_0(D) \leq S_s(D)$, and $\max_u D_{su} - \delta \leq \max \text{dgm}_0^{\text{fin}}(D) \leq \max_u D_{su}$;*
- (c) *if D is coned at s , then $H_k(K_\epsilon) = 0$ for all $k \geq 1$ and all ϵ , the star at s is an MST, and $P_0(D) = S_s(D)$.*

Proof sketch (full proof in Appendix A). Let $D'_{uv} = \max(D_{uv}, D_{su}, D_{sv})$ for $u, v \neq s$ and $D'_{su} = D_{su}$. Then $D \leq D' \leq D + \delta$ entrywise and D' is coned at s . Since D' is coned, any simplex $\sigma \not\ni s$ with $|\sigma| \geq 2$ has the same filtration value as $\sigma \cup \{s\}$, and a vertex not yet joined to s has no edges; so each K'_ϵ is a cone with apex s plus isolated points, and $H_k(K'_\epsilon) = 0$ for $k \geq 1$. The inclusions $K_b \subseteq K'_{b+\delta} \subseteq K_{b+\delta}$ factor the persistence map $H_k(K_b) \rightarrow H_k(K_{b+\delta})$ through zero, so no bar outlives δ . For (b), the star is a spanning tree of D (upper bound) and an MST of D' ; MST weight and bottleneck are $(N-1)$ - and 1-Lipschitz in the sup-norm. The ingredients are standard: Lemma 1 is the dominating-vertex (cone) argument behind strong and edge collapses (Barmak & Minian, 2012; Boissonnat & Pritam, 2020), and the last step is a one-sided form of the interleaving argument behind persistence stability (Chazal et al., 2009). $\square$

The bound in (a) is attained: if $D_{su} = 1$ for all u and four further vertices form a 4-cycle with side $\frac{1}{2}$ and diagonals 1, then $\delta_s = \frac{1}{2}$ and H_1 has the single bar $[\frac{1}{2}, 1)$.

Theorem 1 controls summaries. The same comparison controls the whole barcode. Call $\Sigma_s(D)$ the *star diagram*: in degree 0 the bars $[0, D_{su})$, $u \neq s$, plus one infinite bar, and nothing in degree $k \geq 1$.

Corollary 1 (The whole barcode is the star, up to δ_s). *For every k , $d_B(\text{dgm}_k(D), \Sigma_s(D)_k) \leq \delta_s$, with equality of the diagrams when $\delta_s = 0$. More precisely, the j -th smallest finite 0D death of D lies in $[D_{s(j)} - \delta_s, D_{s(j)}]$, where $D_{s(j)}$ is the j -th smallest entry of $(D_{su})_{u \neq s}$, and β_0 of K_ϵ lies between $1 + \#\{u : D_{su} > \epsilon + \delta_s\}$ and $1 + \#\{u : D_{su} > \epsilon\}$. Hence every vectorization that is L -Lipschitz in d_B (e.g. persistence landscapes; Bubenik, 2015) is within $L\delta_s$ of a function of the column $(D_{su})_u$. Vectorizations that are stable in 1-Wasserstein distance (e.g. persistence images; Adams et al., 2017) are within $L((N-1)\delta_s + \frac{1}{2}\delta_s \sum_{k \geq 1} |\text{dgm}_k(D)|)$ of such a function. On coned graphs, every vectorization (such as persistence entropy or Betti curves) is precisely such a function.*

Corollary 2 (Causal attention). *Let A be lower-triangular and row-stochastic, $D = 1 - \max(A, A^\top)$, and $s = 0$ the first token. Then $D_{0u} = 1 - A_{u0}$, so $S_0 = (N - 1)(1 - \bar{m})$ with $\bar{m} = \frac{1}{N-1} \sum_{u \geq 1} A_{u0}$ the mean sink attention, and*

$$\delta_0 = \max_{1 \leq v < u} [A_{uv} - \min(A_{u0}, A_{v0})]_+.$$

Thus $P_0 \in [(N - 1)(1 - \bar{m} - \delta_0), (N - 1)(1 - \bar{m})]$ and all $H_{k \geq 1}$ bars have length at most δ_0 .

Remark 1 (Choice of distance). The graph is exactly coned iff, for all $1 \leq v < u$, $A_{uv} \leq A_{u0}$ and $A_{uv} \leq A_{v0}$: no token attends to an earlier non-sink token more than either of the two attends to the sink. Coning is an order condition, so it is invariant under any entrywise strictly increasing transform f of D with $f(0) = 0$ (e.g. $-\log$ of the symmetrized attention, or ranks without ties, in place of $1 - A$). The set of exactly coned graphs, and the vanishing of their higher homology, does not depend on how attention is turned into a distance. Only the magnitude of δ_s on non-coned graphs does. (A merely non-decreasing f , such as clipping, can create coning.)

Corollary 2 makes four testable predictions: **(C1)** in coned layers 0D total persistence equals $(N-1)(1 - \bar{m})$, so after dividing by $N - 1$ it is an affine function of sink attention; **(C2)** H_1 is empty whenever $\delta_0 = 0$; **(C3)** 1D persistence can only be non-zero where the sink is violated, with bars no longer than δ_0 ; **(C4)** removing the sink and renormalizing (deflation) exposes whatever non-sink topology exists.

Random weights and directed complexes. For i.i.d. weights, on the other hand, 1D total persistence is known to scale linearly with N (Hiraoka & Shirai, 2017; Hino & Kanazawa, 2019), and we give an elementary explicit bound (Proposition 2, Appendix A): $\mathbb{E}[P_1] \geq \frac{3}{8}N - \sqrt{3N} + O(1)$. We prove that the directed flag complex of the digraph is not a way out for causal attention: the digraph is a transitive tournament, so its weighted directed flag complex is isomorphic to $\text{VR}(D)$ as a filtered complex (Proposition 3, Appendix A.4).

The published topological detector reduces to a first-order statistic. TOHA (Bazarova et al., 2026) scores an attention head by the topological divergence between the response tokens R and the prompt tokens P . Its per-head score is $d(R, P) = \text{MTopDiv}(R, P)/|R|$, where MTopDiv is the total 0D persistence of the head’s graph $D = 1 - \max(A, A^\top)$ after all prompt–prompt distances are set to 0 (equivalently, the minimum spanning forest attaching R to P). TOHA averages d over at most ten heads chosen on labeled data. For $u \in R$ let $\pi_u = \max_{v \in P} A_{uv}$ be u ’s largest attention to any prompt token.

Proposition 1 (TOHA is max-prompt attention, up to a defect). *Let A be lower-triangular and row-stochastic, and let $P = \{0, \dots, p - 1\}$ and $R = \{p, \dots, N - 1\}$. With $\delta_P = \max_{w < u \in R} [A_{uw} - \min(\pi_u, \pi_w)]_+$,*

$$\frac{1}{|R|} \sum_{u \in R} (1 - \pi_u) - \delta_P \leq d(R, P) \leq \frac{1}{|R|} \sum_{u \in R} (1 - \pi_u),$$

with equality when $\delta_P = 0$. Moreover $1 - \pi_u \leq 1 - A_{u0}$, with equality iff the first token is among u ’s most-attended prompt tokens. On a head where this holds for every $u \in R$ and $\delta_P = 0$, $d(R, P) = 1 - \frac{1}{|R|} \sum_{u \in R} A_{u0}$, the response tokens’ mean sink attention.

Proof sketch. Zeroing the prompt block merges P at scale 0, so the remaining finite 0D bars are the MST weights of the quotient graph on $\{p^*\} \cup R$, where $D_{p^*u} = \min_{v \in P} D_{uv} = 1 - \pi_u$ (Fact 1). Apply Theorem 1(b) at the apex p^* ; as in Corollary 2, its coning defect is δ_P and its star weight is $\sum_u (1 - \pi_u)$. Divide by $|R|$. $\square$

Binkowski et al. (2026) note that sink nodes are “likely to appear in the MST” of MTopDiv . Proposition 1 makes this exact, and it indicates where the topology can enter: only through δ_P , which is non-zero only when a response token attends to an earlier response token more than one of the two attends to its best prompt token. We test the proposition on real heads, and TOHA against its first-order counterparts, in §5.4.

4 EXPERIMENTAL SETUP

Models and benchmarks. We use Qwen2.5-1.5B/3B/7B-Instruct, Phi-3-mini-4k-instruct, TinyLlama-1.1B-Chat, SmolLM-1.7B-Instruct, and Mistral-7B-Instruct-v0.2: 7 open models spanning 1.1B–7.62B parameters, with additional Mistral-7B analysis in Appendix H. Mistral runs with its hand-written [INST] string and its own chat template. The model and human prompts have one trailing token difference: the end-of-sequence marker for the model’s chat template. The differences between them are a consistency check since the two use the same template. That token falls inside the response, so the two agree to 0.023 AUC on all banks except answer-only PH, which is restricted to the response and differs by 0.071. **TruthfulQA** (Lin et al., 2022): all 817 questions, each paired with its best answer and first incorrect answer (1,634 rows) under each model’s chat format (Appendix B). **HaluEval QA** (Li et al., 2023a): the first 1,000 (Qwen2.5-3B) or 500 (Qwen2.5-1.5B) knowledge-grounded questions with right and hallucinated answers. They both follow the model reading dataset-written answers. **TriviaQA on-policy**: 2,000 random validation questions (Joshi et al., 2017); each model answers greedily (short-phrase instruction, ≤ 24 tokens); answer labeled hallucinated iff it includes a normalized gold alias. The model-provided answer is re-read to extract features. The table contains per-model accuracies (Table 6). All models run in bfloat16 with eager attention.

Features. Per layer (head-averaged A): 0D (max lifetime, total persistence); 1D (four summaries, divided by N); SINK (S_0 and $\max_u D_{0u}$, i.e. exactly the two quantities Corollary 2 says 0D reduces to); DEFLATED PH (0D+1D after deleting token 0 and renormalizing rows); ANSWER-ONLY PH (induced subgraph on answer tokens); row entropy and row max. Per head: 0D (MST total and max via a dense Prim’s algorithm), sink mass, entropy, Lookback ratio (Chuang et al., 2024), and the LLM-Check attention score (Sriramanan et al., 2024). TOHA (per head, separate forward pass, same rows): the exact score d of Bazarova et al. (2026) via a dense Prim’s algorithm on the prompt-contracted graph, the $\bar{\pi}$ -score, the response sink score, and δ_P (Proposition 1). Other: answer-token log-probability and predictive-entropy statistics; mean-pooled answer hidden states at the middle layer; TF-IDF word and character n -grams of the answer text (LEXICAL).

Protocol. All numbers use one evaluator: standardized features and L_2 logistic regression, with $C = 1$ for banks of at most 64 dimensions and otherwise $C \in \{0.01, 0.1, 1\}$ chosen by inner grouped 3-fold CV. The outer loop is 10-fold StratifiedGroupKFold grouped by question, repeated over 3 seeds (42–44), with out-of-fold probabilities averaged. We test incremental value in two ways. Early fusion (“X|Y”) concatenates the banks. Late fusion (“ $X \oplus Y$ ”) stacks the two banks’ out-of-fold logits in a second-level logistic regression under the same CV, so that a small bank cannot be drowned out by a wide one. Differences in pooled AUC use a 10,000-resample question-level cluster bootstrap. We call two banks equivalent if the 90% interval lies inside ± 0.015 AUC (two one-sided tests, TOST, at $\alpha = 0.05$). We report TOST power. Paired benchmarks also report within-pair accuracy. The theory checks use all (example, layer) graphs. Ripser (Bauer, 2021) represents its filtration values as 32-bit floats, so we check the bar-length and max-death bounds at tolerance 10^{-5} , and the P_0 sandwich, a sum of up to N such values, at 10^{-3} .

5 RESULTS

5.1 THE REDUCTION HOLDS ON REAL ATTENTION

Table 1 checks Theorem 1 on every layer-level graph of every example: 838,356 graphs in total. There are 0 violations of any bound. In the 13 settings whose models place 0.42 to 0.75 of attention on token 0, 36–98% of graphs are exactly coned ($\delta_0 = 0$). On each of these graphs, H_1 is empty as required by prediction C2. For comparison, δ_0 and 1D total persistence have Spearman correlation 0.07 to 0.89 across graphs (C3); this includes the 2 settings above 90% coned where both measures are nearly constant (99.99% of graphs have no 1D bars) and thus the correlation is not weak but instead degenerate. Elsewhere it is at least 0.32. Both P_0 and the star weight $S_0 = (N-1)(1 - \bar{m})$ depend on length, so their raw per-layer rank correlation (median ≥ 0.999) depends in part on length. Dividing both by $N - 1$ gives the median per-layer rank correlation between $P_0/(N - 1)$ and $1 - \bar{m}$ at least 0.98 for each sink setting (C1). Length-normalized 0D total persistence, up to rank, is thus one minus the mean attention paid to the first token in most layers. The single worst individual

Table 1: **Theorem 1 on real attention.** *cells*: (example, layer) graphs; *viol.*: violations of any bound in Theorem 1(a)–(b) with $s = 0$; *coned*: share with $\delta_0 = 0$ exactly; $H_1 = \emptyset$ | *coned*: share of coned graphs with empty H_1 (C2 predicts 100%). Last two columns: median over layers of the Spearman correlation across examples between P_0 and S_0 (ρ_{raw}), and between $P_0/(N-1)$ and $1 - \bar{m} = S_0/(N-1)$ (ρ_N), which removes the shared dependence on sequence length.

Benchmark	Model	$\bar{N}$	graphs	viol.	sink attn.	coned	$H_1 = \emptyset$ coned	ρ_{raw}	ρ_N
TruthfulQA	Qwen2.5-3B	44.2	58,824	0	0.50	62.2%	100.0%	1.000	1.000
TruthfulQA	Qwen2.5-1.5B	44.2	45,752	0	0.53	64.5%	100.0%	1.000	1.000
TruthfulQA	Phi-3-mini	29.0	52,288	0	0.69	88.4%	100.0%	1.000	1.000
TruthfulQA	TinyLlama-1.1B	42.3	35,948	0	0.53	36.2%	100.0%	1.000	0.982
TruthfulQA	SmolLM-1.7B	32.4	39,216	0	0.02	0.0%	–	0.967	0.412
TruthfulQA	Mistral-7B	33.0	52,288	0	0.75	98.2%	100.0%	1.000	1.000
TruthfulQA	Mistral-7B (chat)	34.0	52,288	0	0.75	98.2%	100.0%	1.000	1.000
TruthfulQA	Qwen2.5-7B	44.2	45,752	0	0.01	0.0%	–	0.976	0.513
HaluEval	Qwen2.5-3B	126.9	72,000	0	0.42	52.7%	100.0%	1.000	1.000
HaluEval	Qwen2.5-1.5B	128.1	28,000	0	0.45	62.5%	100.0%	1.000	1.000
TriviaQA (on-policy)	Qwen2.5-3B	68.6	72,000	0	0.48	63.0%	100.0%	1.000	1.000
TriviaQA (on-policy)	Qwen2.5-1.5B	68.8	56,000	0	0.52	50.6%	100.0%	1.000	1.000
TriviaQA (on-policy)	Phi-3-mini	46.1	64,000	0	0.69	87.1%	100.0%	1.000	1.000
TriviaQA (on-policy)	TinyLlama-1.1B	72.0	44,000	0	0.50	36.4%	100.0%	1.000	0.991
TriviaQA (on-policy)	Mistral-7B	58.7	64,000	0	0.64	65.8%	100.0%	1.000	1.000
TriviaQA (on-policy)	Qwen2.5-7B	68.3	56,000	0	0.01	0.0%	–	0.986	0.489

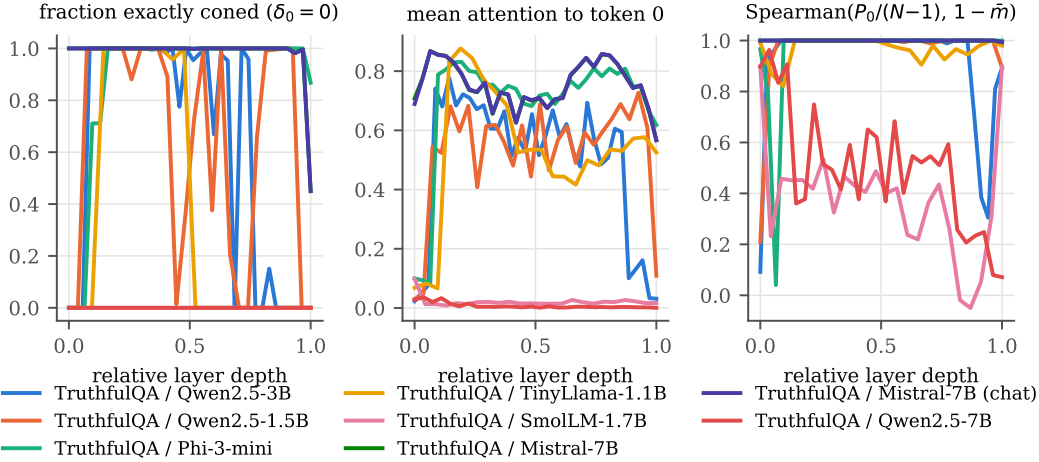


Figure 1: **Layerwise structure (TruthfulQA).** Coning (left) tracks sink attention (middle): layers with little sink attention, typically the first few and, in some models, the last few, are rarely coned, and the models without a first-token sink (Qwen2.5-7B and SmolLM-1.7B) are never coned at $s = 0$. Right: length-normalized rank correlation between P_0 and sink attention. It is near 1 in the layers of sink models, dips in some open layers, and is low throughout the two without a sink.

layer reaches only -0.14 to 0.99 per setting. Figure 1 locates coning: it happens in the middle of the network, where attention to the sink is high. The first few layers are never sink-dominated, and in some models (TinyLlama, the Qwen models) neither is a block of late layers; these layers are rarely coned. There the length-normalized correlation can fall, as Theorem 1(b) allows once $(N-1)\delta_0$ is comparable to the spread of S_0 across examples.

Models without a first-token sink as a contrast. We have 2 models with essentially no attention on token 0, Qwen2.5-7B and SmolLM-1.7B, with 0.01 to 0.02 attention there. None of their graphs are coned at $s = 0$, so Theorem 1 at the first token gives them no useful guarantee. The two are not alike: allowing any apex, 0% of SmolLM-1.7B’s graphs are coned, so no vertex cones any of them, against 10% of Qwen2.5-7B’s, whose attention is strongly concentrated, just not on token 0 (§5.4). One model has no cone structure to find; for the other, $s = 0$ is the wrong apex. Their H_1 is non-empty on 55% to 87% of graphs, and the raw $\rho(P_0, S_0)$ is high (0.967 to 0.986), but nearly all of that comes from shared lengths: the length-normalized correlation is 0.41 to 0.51. Both differ from the sink-dominated models in weights, data and template, so this is a contrast, not a controlled

Table 2: **Pooled out-of-fold ROC-AUC** (3-seed repeated grouped 10-fold CV; seed SD ≤ 0.008). TQA: TruthfulQA; HE: HaluEval; TrQA: on-policy TriviaQA. Lex.: answer-text TF-IDF; LogP: log-probability statistics; LLM-C.: LLM-Check; Sink: $(S_0, \max_u D_{0u})$ per layer; 0D/1D: persistence per layer; subscript h : per head (Sink $_h$: sink mass; 0D $_h$: MST total and max); Lookb.: Lookback ratios; Hidden: mid-layer probe; Non-topo.: LogP+LLM-C.+Lookb.+Hidden+row statistics. Best per row in bold. Full table in Appendix J.

Setting	Lex.	LogP	LLM-C.	Sink	0D	1D	Sink $_h$	0D $_h$	Lookb.	Hidden	Non-topo.
TQA Qw3B	.820	.630	.693	.737	.736	.660	.851	.853	.897	.913	.926
TQA Qw1.5B	.820	.638	.666	.723	.726	.586	.830	.838	.851	.916	.917
TQA Phi-3	.820	.578	.653	.651	.646	.526	.871	.868	.892	.933	.936
TQA TinyLL	.820	.637	.614	.643	.654	.592	.808	.806	.850	.883	.885
TQA SmolLM	.820	.588	.604	.611	.583	.543	.720	.800	.863	.894	.899
TQA Mis7B	.820	.641	.652	.685	.684	.501	.903	.893	.933	.956	.959
TQA Mis7B (chat)	.820	.651	.653	.688	.688	.501	.907	.900	.934	.958	.960
TQA Qw7B	.820	.674	.667	.749	.679	.610	.821	.850	.922	.933	.950
HE Qw3B	.981	.993	.745	.829	.811	.710	.958	.971	.998	1.00	1.00
HE Qw1.5B	.979	.995	.791	.820	.789	.747	.922	.949	.999	1.00	1.00
TrQA Qw3B	.596	.839	.688	.717	.732	.601	.820	.807	.861	.841	.870
TrQA Qw1.5B	.609	.823	.676	.707	.716	.574	.767	.768	.824	.855	.873
TrQA Phi-3	.635	.877	.676	.707	.710	.546	.826	.807	.840	.867	.883
TrQA TinyLL	.740	.724	.688	.720	.720	.659	.814	.775	.823	.827	.842
TrQA Mis7B	.677	.728	.686	.696	.699	.611	.812	.775	.829	.833	.841
TrQA Qw7B	.594	.847	.725	.714	.668	.582	.801	.798	.879	.854	.881

intervention; the within-model intervention is deflation (C4, below). The reduction is a property of sink-dominated attention, not of the pipeline.

5.2 WHAT TOPOLOGY ADDS BEYOND THE SINK

If 0D persistence reduced exactly to the star, a classifier on the two sink features would match one on the two 0D features in every layer. Table 14 (“Sink vs. 0D”) shows the two banks differ by -0.015 to $+0.031$ AUC across the 13 sink settings evaluated, and are statistically equivalent within ± 0.015 in 8 of them (the TOST power of this comparison is modest, so non-equivalence is not evidence of a difference). Adding 0D to the sink bank gains 0.000 to $+0.028$.

The residual lives exactly where the theory says it can. We split layers without using labels: coned layers ($\geq 90\%$ of graphs coned) and open layers ($< 50\%$). Adding the coned layers’ 0D features to SINK changes AUC by -0.001 to $+0.004$, whereas the open layers account for $+0.003$ to $+0.027$ (Appendix J). Almost everything 0D adds beyond sink attention comes from the layers where the sink does not dominate.

Per head, the reduction is tighter. Per-head sink mass and per-head 0D are far stronger than their head-averaged versions (Table 2), but adding per-head MST total ($P_{0,h}$) to per-head sink mass changes AUC by only -0.001 to $+0.011$, which is equivalent within ± 0.015 in 13 of the 13 sink settings with per-head features.

The coning defect is a signal, but not a new one. Theorem 1 identifies δ_0 as the unique locus where topology may diverge from the sink, so we evaluate it as a signal. Head-averaged, it is weak (AUC 0.50 to 0.67). Per head, it is strong (0.69 to 0.96), with a small improvement when stacked on top of per-head sink mass ($+0.004$ to $+0.029$ AUC; 95% interval above 0 in 16 of 16 settings). Stacking on top of the non-topological probes yields smaller gains (0.000 to $+0.009$; above 0 in 5, equivalent within ± 0.015 in 16 of 16).

Deleting the sink exposes some, redundant, structure. Deflated PH (C4) gives a modest gain of $+0.029$ to $+0.067$ over SINK on TruthfulQA (95% interval excludes 0 in 6 of 6 sink settings) but less on the on-policy benchmark ($+0.002$ to $+0.043$; 2 of 5). The same holds for answer-only PH (Appendix J). Attention graphs do contain some non-sink geometry, particularly on teacher-forced text. But this geometry is captured by non-topological probes (§5.3).

Table 3: **TOHA against its first-order counterparts** (Proposition 1). $\delta_P=0$: share of (example, head) graphs that are exactly prompt-coned, on which d equals the $\bar{\pi}$ -score. $\rho(d, \bar{\pi})$: median over heads of the rank correlation across examples. AUC columns: TOHA’s head-selection algorithm run on d (TOHA), on the $\bar{\pi}$ -score (TOHA $[\bar{\pi}]$), and on the response sink score (TOHA $[\bar{a}_0]$); then supervised probes on all heads of each score; then the non-topological bank. Δ columns and marks as in Table 14; $\oplus$ is late fusion.

Setting	$\delta_P=0$	$\rho(d, \bar{\pi})$	TOHA	TOHA $[\bar{\pi}]$	TOHA $[\bar{a}_0]$	TOHA $_{\text{sup}}$	$\bar{\pi}_{\text{sup}}$	$\bar{a}_0_{\text{sup}}$	Non-topo.	$\bar{\pi}$ -TOHA	$\bar{a}_0$ -TOHA	TOHA $\oplus\bar{\pi}$	TOHA $\oplus\text{NT}$
TQA Qw3B	57%	.976	.668	.670	.649	.892	.892	.892	.926	+0.003 $\equiv$	-0.018*	+0.003 $\equiv$	+0.003 $\equiv$
TQA Qw1.5B	58%	.978	.646	.649	.628	.858	.857	.857	.917	+0.003 $\equiv$	-0.018*	+0.005 $\equiv$	+0.002 $\equiv$
TQA Phi-3	74%	.996	.751	.742	.715	.898	.901	.905	.936	-0.009*	-0.036*	0.000 $\equiv$	0.000 $\equiv$
TQA TinyLL	74%	.995	.666	.655	.618	.845	.843	.848	.885	-0.012*	-0.049*	+0.002 $\equiv$	+0.001 $\equiv$
TQA SmolLM	58%	.954	.701	.695	.595	.862	.860	.841	.899	-0.007 $\equiv$	-0.107*	+0.005 $\equiv$	+0.002 $\equiv$
TQA Mis7B	81%	.998	.751	.764	.766	.932	.934	.936	.959	+0.013*	+0.015*	0.000 $\equiv$	+0.001 $\equiv$
TQA Mis7B (chat)	80%	.998	.753	.762	.761	.934	.936	.937	.960	+0.009 $\equiv$	+0.007 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$
TQA Qw7B	57%	.980	.705	.709	.544	.925	.927	.906	.950	+0.004 $\equiv$	-0.161*	+0.001 $\equiv$	+0.004 $\equiv$
HE Qw3B	68%	.987	.967	.982	.961	.999	.999	1.00	1.00	+0.015*	-0.007 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
HE Qw1.5B	69%	.988	.970	.988	.972	.999	.999	.999	1.00	+0.018*	+0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TrQA Qw3B	76%	.991	.728	.736	.742	.862	.863	.870	.870	+0.009	+0.014*	+0.002 $\equiv$	+0.012*
TrQA Qw1.5B	77%	.992	.712	.714	.708	.828	.827	.836	.873	+0.002 $\equiv$	-0.005	+0.003 $\equiv$	+0.004 $\equiv$
TrQA Phi-3	88%	.998	.724	.731	.733	.847	.851	.850	.883	+0.007 $\equiv$	+0.008	0.000 $\equiv$	+0.001 $\equiv$
TrQA TinyLL	66%	.994	.714	.725	.693	.833	.826	.832	.842	+0.011*	-0.021*	+0.007 $\equiv$	+0.010*
TrQA Mis7B	73%	.992	.711	.704	.721	.821	.817	.822	.841	-0.007	+0.010	+0.005 $\equiv$	+0.009 $\equiv$
TrQA Qw7B	76%	.994	.754	.790	.617	.876	.880	.842	.881	+0.036*	-0.137*	+0.001 $\equiv$	+0.012*

One-dimensional persistence. In most real graphs there are no 1D bars at all (Table 1). Contrast that with i.i.d. weights, which exhibit linear growth in 1D persistence. The change to add 1D to 0D causes an AUC change of -0.008 to +0.033 and is equivalent within ± 0.015 in 9 of 13 sink settings. See Figure 4 for examples: under a strong sink, a loop can be seen by H_1 only if it breaks the cone.

5.3 HALLUCINATION DETECTION AGAINST STRONG BASELINES

Table 2 places attention-graph topology among standard detectors. Three patterns hold across benchmarks.

Cheap non-topological probes dominate topology. On TruthfulQA and the on-policy benchmark, the mean-pooled hidden-state probe (0.83 to 0.96) and per-head Lookback ratios (0.82 to 0.93) outperform all topological banks, and combined with log-probability, LLM-Check, and row statistics score 0.84 to 0.96. With this combination, adding 0D PH changes AUC from 0.000 to +0.001 (equivalent within ± 0.015 in 16 of the 16 settings with per-head features), and adding sink-deflated PH changes AUC from -0.001 to +0.002 (equivalent in 16 of 16). To check whether early fusion hides a small bank inside a $\sim 2,000$ -dimensional one, we repeat the test with late fusion. Stacking the 0D score onto the non-topological score changes AUC by 0.000 to +0.003 (equivalent in 16 of 16 settings; the 95% interval lies above 0 in 0 of them), and stacking the deflated-PH score changes it by 0.000 to +0.005 (equivalent in 16 of 16).

Which cheap probe wins depends on whether the text is teacher-forced. Log-probability is weak (0.58 to 0.67) on TruthfulQA, which reflects the benchmark’s focus on imitative falsehoods. 0D PH (0.58 to 0.74) outscores 7 of 8 models, significantly for 5. On-policy, log-probability is strong (0.72 to 0.88) and significantly beats 0D (0.67 to 0.73) for 4 of 6 models. Single-teacher-forced benchmarks cannot be used to draw conclusions about topology vs. confidence.

Resolution per-head trumps topology. Head-averaged features are weaker than per-head features of any type, and per-head entropy lies within -0.016 to +0.044 AUC of per-head 0D (Appendix J).

5.4 THE PUBLISHED TOPOLOGICAL DETECTOR IS FIRST-ORDER

We test Proposition 1 on every head of every example: 20,580,160 head graphs from 16 settings (Table 3). We then run TOHA’s head-selection algorithm (Bazarova et al., 2026, Alg. 1) inside our cross-validation, selecting heads on the whole training fold (larger than TOHA’s probe set), once on TOHA’s score d and, unchanged, on the two first-order scores the proposition names: the $\bar{\pi}$ -score $\frac{1}{|R|} \sum_u (1 - \pi_u)$ and the response sink score $1 - \bar{a}_0 = \frac{1}{|R|} \sum_u (1 - A_{u0})$.

The identity holds on real heads. The sandwich has 0 violations. The score d is our own implementation, so we also verify it against the authors’ released code: on 21,696 head graphs from 2 settings, our MTOP-DIV routine (Bazarova et al., 2026) using ripser instead of our dense Prim’s algorithm,

agrees to 1.2×10^{-8} . There are 57–88% exact prompt cones for each setting. In these head graphs, d equals the $\bar{\pi}$ -score up to float rounding (largest gap $6 \cdot 10^{-8}$). We see that the median per-head rank correlation across examples between d and the $\bar{\pi}$ -score is 0.954 to 0.998. It is 0.78 to 0.99 in the sink models, but -0.16 to -0.09 in Qwen2.5-7B and SmolLM-1.7B, which have no first-token sink: there d tracks the largest prompt attention just as tightly while being, if anything, mildly anti-correlated with attention to token 0. The reduction is to max-prompt attention; that this coincides with the sink is a property of the model, not of the theorem. Shifting the sink’s attention logit shifts these quantities (Appendix G).

Topology has little effect on the first-order score. Stacking d onto the $\bar{\pi}$ -score changes AUC by 0.000 to +0.007. Equivalent to zero within ± 0.015 in 16 of 16; the 95% interval is above 0 in 5. Using supervised probes on all heads (TOHA’s own setting for feature comparison), the $\bar{\pi}$ -score is -0.007 to +0.004 different from d (equivalent in 16 of 16). The response sink score is -0.034 to +0.008 different from d (equivalent in 12). Exceptions are the models without a first-token sink (Qwen2.5-7B and SmolLM-1.7B), where the sink score is worse, and one on-policy setting which crosses the margin on the positive side. TOHA’s discrete head selection is noisier: running on the $\bar{\pi}$ -score changes AUC by -0.012 to +0.036 (significantly higher in 6 settings, lower in 2, non-inferior at margin 0.015 in 13 of 16). Running on the sink score, it loses up to 0.049 AUC in the sink models (0.107 to 0.161 in Qwen2.5-7B and SmolLM-1.7B, which have no first-token sink): it selects heads which are prompt-coned on 60–95% of examples, while the sink is only the top prompt token on 19–79% of responses. Stacking the 0D score on top of the non-topological score shifts the AUC by 0.000 to +0.012; by the above, whatever it adds is carried by the $\bar{\pi}$ -score.

Why “attention to $\langle s \rangle$ ” looked weak. Bazarova et al. (2026, Table 9) report that attention to the first token is a much weaker feature than their divergence. Proposition 1 identifies the quantity that d actually tracks: per head, the attention of the response tokens to their most-attended prompt token. In the sink models, that token is the sink for every response token in 40–83% of head graphs.

Benchmark artifacts. A TF-IDF model on the answer text alone attains 0.82 AUC on TruthfulQA and 0.98 on HaluEval, but only 0.59 to 0.74 on-policy: teacher-forced benchmarks reward recognizing how the dataset’s answers were written (Hussain & Kantarcioglu, 2026; Janiak et al., 2025). We draw no conclusions from HaluEval, where lexical separability is near perfect.

6 DISCUSSION

What attention-graph topology measures. In decoder-only models, Vietoris–Rips persistence of an attention graph, all stable summaries of it, and TOHA’s prompt–response divergence (with the prompt as apex) depend on one scalar per graph: the coning defect of the first token. Where it is zero, as in most middle layers of every sink model, 0D total persistence equals $(N-1)(1-\bar{m})$, a function of length and sink attention alone, and higher homology is empty. Where it is positive, Theorem 1 gives an upper bound on the deviation of topology from the sink; in our data that is mostly the first layers; practical guidelines follow in Appendix E.

Limitations. Our scales go 1.1B–7.62B; sink strength varies with scale and training (Gu et al., 2025), so the proportion of coned layers is model-specific, though Theorem 1 is not. We cover Vietoris–Rips persistence of symmetrized attention (head-averaged or per head), its causal directed-flag equivalent, its stable vectorizations, and TOHA. HalluZig’s zigzag uses graph (1-skeleton) homology, where a cone graph still has cycles, so Theorem 1 does not bound its H_1 (Samaga et al., 2026); learned graph models (Frasca et al., 2026) are also outside it. TOHA is designed for RAG; only HaluEval supplies a context passage here. Our on-policy labels use alias string matching, which mislabels paraphrases and hedged answers; an LLM-judge audit (200 answers, 2 settings) finds 5.5% to 14.0% disagreement and moves every bank by at most 0.063 AUC without changing the ordering, though the judge is itself a model (Janiak et al., 2025). TinyLlama’s on-policy answers lack the BOS token that extraction adds. Intervals resample questions under a fixed, seed-averaged predictor, ignoring refit variance, so results are contingent on the trained models; prompt formats also differ across models (Appendix B).

REPRODUCIBILITY STATEMENT

Run order is given in `sinktda/run_fix.sh` and `sinktda/run_toha.sh`. All code is released at <https://anonymous.4open.science/r/ICLR-BE07/sinktda/extract.py> (all features), `sinktda/evaluate.py` (every AUC, interval, and theory check), `sinktda/synthetic.py` (Figure 4), and `sinktda/report.py` (every table and figure). For each layer we give per-layer features for 4 settings (TQA Mis7B, TQA Mis7B (chat), TQA Qw7B and TrQA Qw7B) and for the on-policy generations for TrQA Mis7B and TrQA Qw7B. All per-head and hidden-state dumps exceed the 8 MB per-file limit of the anonymous host, and the dumps for the other settings have been lost with the machine producing them. These can be recreated using `sinktda/extract.py` using the model and setting identifiers in Appendix B. Seeds, prompts, dtype and model identifiers are listed in Appendix B; proofs are in Appendix A.

LLM USAGE STATEMENT

We use LLMs to assist with implementing the experiment code, checking the statistics that we report against our result files, and writing and editing this paper. All claims, numbers, proofs, and citations have been checked by the authors. We take responsibility for the entirety of this paper.

REFERENCES

- Henry Adams, Tegan Emerson, Michael Kirby, Rachel Neville, Chris Peterson, Patrick Shipman, Sofya Chepushtanova, Eric Hanson, Francis Motta, and Lori Ziegelmeier. Persistence images: A stable vector representation of persistent homology. *Journal of Machine Learning Research*, 18(8):1–35, 2017.
- Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it’s lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 967–976, 2023.
- Federico Barbero, Álvaro Arroyo, Xiangming Gu, Christos Perivolaropoulos, Michael Bronstein, Petar Veličković, and Razvan Pascanu. Why do LLMs attend to the first token? In *Conference on Language Modeling (COLM)*, 2025. arXiv:2504.02732.
- Jonathan Ariel Barmak and Elias Gabriel Minian. Strong homotopy types, nerves and collapses. *Discrete & Computational Geometry*, 47(2):301–328, 2012.
- Ulrich Bauer. Ripser: Efficient computation of Vietoris–Rips persistence barcodes. *Journal of Applied and Computational Topology*, 5(3):391–423, 2021.
- Alexandra Bazarova, Andrei Volodichev, Aleksandr Yugay, Andrey Shulga, Alina Ermilova, Konstantin Polev, Julia Belikova, Rauf Parchiev, Dmitry Simakov, Maxim Savchenko, Andrey Savchenko, Serguei Barannikov, and Alexey Zaytsev. Hallucination detection in LLMs with topological divergence on attention graphs. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL Volume 1: Long Papers)*, pp. 15449–15470, 2026. arXiv:2504.10063.
- Jakub Binkowski, Kamil Adamczewski, and Tomasz Kajdanowicz. Attention sinks as internal signals for hallucination detection in large language models. *arXiv preprint arXiv:2604.10697*, 2026.
- Jean-Daniel Boissonnat and Siddharth Pritam. Edge collapse and persistence of flag complexes. In *36th International Symposium on Computational Geometry (SoCG)*, volume 164 of *LIPIcs*, pp. 19:1–19:15, 2020.
- Peter Bubenik. Statistical topological data analysis using persistence landscapes. *Journal of Machine Learning Research*, 16:77–102, 2015.
- Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*, 2023.

- Nicola Cancedda. Spectral filters, dark signals, and attention sinks. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- Frédéric Chazal, David Cohen-Steiner, Marc Glisse, Leonidas J. Guibas, and Steve Y. Oudot. Proximity of persistence modules and their diagrams. In *Proceedings of the 25th Annual Symposium on Computational Geometry (SoCG)*, pp. 237–246, 2009.
- Frédéric Chazal, Vin de Silva, and Steve Oudot. Persistence stability for geometric complexes. *Geometriae Dedicata*, 173(1):193–214, 2014.
- Chao Chen, Kai Liu, Ze Chen, Yi Gu, Yue Wu, Mingyuan Tao, Zhihang Fu, and Jieping Ye. INSIDE: LLMs’ internal states retain the power of hallucination detection. In *International Conference on Learning Representations (ICLR)*, 2024.
- Daniil Cherniavskii, Eduard Tulchinskii, Vladislav Mikhailov, Irina Proskurina, Laida Kushnareva, Ekaterina Artemova, Serguei Barannikov, Irina Piontkovskaya, Dmitri Piontkovski, and Evgeny Burnaev. Acceptability judgements via examining the topology of attention maps. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022.
- Yung-Sung Chuang, Linlu Qiu, Cheng-Yu Hsieh, Ranjay Krishna, Yoon Kim, and James R. Glass. Lookback lens: Detecting and mitigating contextual hallucinations in large language models using only attention maps. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 1419–1436, 2024.
- Kevin Clark, Urvashi Khandelwal, Omer Levy, and Christopher D. Manning. What does BERT look at? an analysis of BERT’s attention. In *Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pp. 276–286, 2019.
- David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007.
- Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. In *International Conference on Learning Representations (ICLR)*, 2024.
- Xuefeng Du, Chaowei Xiao, and Yixuan Li. HaloScope: Harnessing unlabeled LLM generations for hallucination detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Herbert Edelsbrunner and John L. Harer. *Computational Topology: An Introduction*. American Mathematical Society, 2010.
- Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.
- Fabrizio Frasca, Guy Bar-Shalom, Yftah Ziser, and Haggai Maron. Neural message-passing on attention graphs for hallucination detection. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2509.24770.
- Alan M. Frieze. On the value of a random minimum spanning tree problem. *Discrete Applied Mathematics*, 10(1):47–56, 1985.
- Yuri Gardinazzi, Karthik Viswanathan, Giada Panerai, Alessio Ansuini, Alberto Cazzaniga, and Matteo Biagetti. Persistent topological features in large language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 18811–18830, 2025. arXiv:2410.11042.
- John C. Gower and Gavin J. S. Ross. Minimum spanning trees and single linkage cluster analysis. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 18(1):54–64, 1969.
- Xiangming Gu, Tianyu Pang, Chao Du, Qian Liu, Fengzhuo Zhang, Cunxiao Du, Ye Wang, and Min Lin. When attention sink emerges in language models: An empirical view. In *International Conference on Learning Representations (ICLR)*, 2025.
- Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. Annotation artifacts in natural language inference data. In *Proceedings of NAACL-HLT*, pp. 107–112, 2018.

- Masanori Hino and Shu Kanazawa. Asymptotic behavior of lifetime sums for random simplicial complex processes. *Journal of the Mathematical Society of Japan*, 71(3):765–804, 2019.
- Yasuaki Hiraoka and Tomoyuki Shirai. Minimum spanning acycle and lifetime of persistent homology in the Linial–Meshulam process. *Random Structures & Algorithms*, 51(2):315–340, 2017.
- Khizar Hussain and Murat Kantarcioglu. PARALLAX: Separating genuine hallucination detection from benchmark construction artifacts. *arXiv preprint arXiv:2605.17028*, 2026.
- Denis Janiak, Jakub Binkowski, Albert Sawczyn, Bogdan Gabrys, Ravid Shwartz-Ziv, and Tomasz Kajdanowicz. The illusion of progress: Re-evaluating hallucination detection in LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025.
- Mandar Joshi, Eunsol Choi, Daniel Weld, and Luke Zettlemoyer. TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1601–1611, 2017.
- Jannik Kossen, Jiatong Han, Muhammed Razzak, Lisa Schut, Shreshth Malik, and Yarin Gal. Semantic entropy probes: Robust and cheap hallucination detection in LLMs. *arXiv preprint arXiv:2406.15927*, 2024.
- Elizaveta Kostenok, Daniil Cherniavskii, and Alexey Zaytsev. Uncertainty estimation of transformers’ predictions via topological analysis of the attention matrices. *arXiv preprint arXiv:2308.11295*, 2023.
- Laida Kushnareva, Daniil Cherniavskii, Vladislav Mikhailov, Ekaterina Artemova, Serguei Baranikov, Alexander Bernstein, Irina Piontkovskaya, Dmitri Piontkovski, and Evgeny Burnaev. Artificial text detection via examining the topology of attention maps. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021.
- Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. HaluEval: A large-scale hallucination evaluation benchmark for large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023a.
- Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023b.
- Stephanie Lin, Jacob Hilton, and Owain Evans. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL Volume 1: Long Papers)*, pp. 3214–3252, 2022.
- Daniel Lütgehetmann, Dejan Govc, Jason P. Smith, and Ran Levi. Computing persistent homology of directed flag complexes. *Algorithms*, 13(1):19, 2020.
- Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 9004–9017, 2023.
- Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *Conference on Language Modeling (COLM)*, 2024.
- R. Thomas McCoy, Ellie Pavlick, and Tal Linzen. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 3428–3448, 2019.
- Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. LLMs know more than they show: On the intrinsic representation of LLM hallucinations. In *International Conference on Learning Representations (ICLR)*, 2025.
- Ilan Perez and Raphael Reinauer. The topological BERT: Transforming attention into topology for natural language processing. *arXiv preprint arXiv:2206.15195*, 2022.

- Shreyas N. Samaga, Gilberto Gonzalez Arroyo, and Tamal K. Dey. HalluZig: Hallucination detection using zigzag persistence. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2026. arXiv:2601.01552.
- Gaurang Sriramanan, Siddhant Bharti, Vinu Sankar Sadasivan, Shoumik Saha, Priyatham Kattakinda, and Soheil Feizi. Llm-check: Investigating detection of hallucinations in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.
- Mingjie Sun, Xinlei Chen, J. Zico Kolter, and Zhuang Liu. Massive activations in large language models. In *Conference on Language Modeling (COLM)*, 2024.
- Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*, 2024.

A PROOFS

Throughout, D is a symmetric $N \times N$ matrix with non-negative entries and zero diagonal, K_ϵ is the Vietoris–Rips complex at scale ϵ , $\text{filt}_D(\sigma) = \max_{u,v \in \sigma} D_{uv}$ ($\text{filt}_D(\{u\}) = 0$), and homology is taken over a field. Bars are half-open intervals $[b, d)$.

A.1 FACT 1

All vertices enter at 0. Processing edges in non-decreasing weight order, an edge either joins two components, in which case (elder rule) one 0D class dies at its weight and Kruskal’s algorithm adds the edge to the spanning forest, or it closes a cycle, in which case no 0D class dies and Kruskal discards it. Hence the $N - 1$ finite death times are the MST edge weights (Gower & Ross, 1969; Edelsbrunner & Harer, 2010).

A.2 THEOREM 1

Lemma 1 (Coned filtrations). *If D is coned at s , then for every $\epsilon \geq 0$, $K_\epsilon = C_\epsilon \sqcup I_\epsilon$, where C_ϵ is a simplicial cone with apex s and I_ϵ is a set of isolated vertices. Consequently $\tilde{H}_k(K_\epsilon) = 0$ for every $k \geq 1$.*

Proof. Let $\sigma \in K_\epsilon$ with $s \notin \sigma$ and $|\sigma| \geq 2$. For each $u \in \sigma$ choose $v \in \sigma \setminus \{u\}$; coning gives $D_{su} \leq D_{uv} \leq \text{filt}_D(\sigma) \leq \epsilon$. Therefore $\text{filt}_D(\sigma \cup \{s\}) = \text{filt}_D(\sigma)$ and $\sigma \cup \{s\} \in K_\epsilon$. Now let $u \neq s$ be a vertex with $D_{su} > \epsilon$. If u had an edge $\{u, v\} \in K_\epsilon$ then $D_{su} \leq D_{uv} \leq \epsilon$, a contradiction; so u is isolated. Let I_ϵ be these vertices and C_ϵ the subcomplex on the remaining vertices. Every vertex of C_ϵ is adjacent to s , and by the first part every simplex of C_ϵ extends by s , so C_ϵ is the cone over its link of s and is contractible. H_k of a disjoint union is the direct sum, and isolated points have no homology in degree ≥ 1 . $\square$

Proof of Theorem 1. Fix s and $\delta = \delta_s(D)$. Define $D'_{uv} = \max(D_{uv}, D_{su}, D_{sv})$ for $u, v \neq s$, $u \neq v$; $D'_{su} = D'_{us} = D_{su}$; $D'_{uu} = 0$.

Step 1: $D \leq D' \leq D + \delta$ and D' is coned at s . The first inequality is immediate. For $u, v \neq s$, $D'_{uv} - D_{uv} = [\max(D_{su}, D_{sv}) - D_{uv}]_+ \leq \delta$. Coning: $\max(D'_{su}, D'_{sv}) = \max(D_{su}, D_{sv}) \leq D'_{uv}$.

Step 2: interleaving. Since filtration values are maxima of entries, $\text{filt}_D \leq \text{filt}_{D'} \leq \text{filt}_D + \delta$. Hence for every b , $K_b \subseteq K'_{b+\delta} \subseteq K_{b+\delta}$, where K' is the filtration of D' .

Step 3: (a). The persistence map $H_k(K_b) \rightarrow H_k(K_{b+\delta})$ is induced by the composite inclusion in Step 2 and therefore factors through $H_k(K'_{b+\delta})$, which is zero for $k \geq 1$ by Lemma 1. The rank of $H_k(K_b) \rightarrow H_k(K_{b'})$ equals the number of bars $[b_i, d_i)$ with $b_i \leq b$ and $d_i > b'$. A bar with $d_i - b_i > \delta$ would contribute to this rank at $b = b_i$, $b' = b_i + \delta$, a contradiction. So every bar has length at most δ , and summing gives the bound on P_k .

Step 4: (b). By Fact 1, $P_0(D) = w_D(T_D)$ for an MST T_D of D . The star $T_s = \{\{s, u\}\}_{u \neq s}$ is a spanning tree, so $P_0(D) \leq w_D(T_s) = S_s(D)$. For the lower bound, T_s is an MST of D' , even with ties. Root any spanning tree T at s and let e_u be the edge from $u \neq s$ to its parent $p(u)$. If $p(u) = s$ then $D'_{e_u} = D'_{su}$; otherwise $D'_{e_u} = D'_{up(u)} \geq \max(D'_{su}, D'_{sp(u)}) \geq D'_{su}$ because D' is coned. The map $u \mapsto e_u$ is a bijection onto the edges of T , so $w_{D'}(T) \geq \sum_{u \neq s} D'_{su} = w_{D'}(T_s)$ and $\max_{e \in T} D'_e \geq \max_u D'_{su}$. Hence $w_{D'}(T_{D'}) = w_{D'}(T_s) = S_s(D)$, and $P_0(D) = w_D(T_D) \geq w_{D'}(T_D) - (N-1)\delta \geq w_{D'}(T_{D'}) - (N-1)\delta = S_s(D) - (N-1)\delta$. The maximum finite 0D death is the bottleneck value $\min_T \max_{e \in T} w(e)$, which is attained by any MST; it is at most $\max_u D_{su}$ (the star is a spanning tree) and is 1-Lipschitz in the sup-norm, which gives the lower bound because the bottleneck of D' is $\max_u D_{su}$ (by the rooted-tree argument above).

Relation to known results. Lemma 1 is the classical observation that a flag complex in which one vertex is adjacent to every non-isolated vertex, and to every simplex, is a cone; this is the dominated-vertex mechanism behind strong collapses (Barmak & Minian, 2012) and edge collapses of flag filtrations (Boissonnat & Pritam, 2020). Step 3 is a one-sided instance of the interleaving argument

behind the stability of persistence (Cohen-Steiner et al., 2007; Chazal et al., 2009; 2014). The two-sided stability theorem would give bars of length at most 2δ , whereas the one-sided comparison $D \leq D'$ gives δ , which is attained (Section 3).

Step 5: (c). If $\delta = 0$ then $D' = D$, and the claims follow from Lemma 1 and Step 4. $\square$

Proof of Corollary 1. Let D' be as in the proof of Theorem 1. Since D' is coned at s , Lemma 1 and Step 4 give $\text{dgm}_k(D') = \Sigma_s(D)_k$ for all k : no bars in degree $k \geq 1$, and 0D deaths equal to the star weights $D'_{su} = D_{su}$. For $1 \leq j \leq N - 1$, the j -th smallest finite 0D death of a matrix E is $w_j(E) = \min\{\epsilon : \beta_0(K_\epsilon^E) \leq N - j\}$. If $E \leq E'$ entrywise then $K_\epsilon^{E'} \subseteq K_\epsilon^E$ on the same vertex set, so $\beta_0(K_\epsilon^E) \leq \beta_0(K_\epsilon^{E'})$ and $w_j(E) \leq w_j(E')$; and $w_j(E + c) = w_j(E) + c$ on off-diagonal entries. From $D \leq D' \leq D + \delta_s$ we get $w_j(D) \leq D_{s(j)} \leq w_j(D) + \delta_s$. The same inclusions give the Betti-0 sandwich, because $\beta_0(K'_\epsilon) = 1 + \#\{u : D_{su} > \epsilon\}$. Matching the j -th deaths shows that $d_B(\text{dgm}_0(D), \Sigma_s(D)_0) \leq \delta_s$ and $W_1 \leq (N - 1)\delta_s$. In degree $k \geq 1$, every bar has length at most δ_s (Theorem 1(a)), so matching it to the diagonal costs at most $\delta_s/2$ in ℓ^∞ . The Lipschitz statements follow from the stability of landscapes in d_B (Bubenik, 2015) and of persistence images in W_1 (Adams et al., 2017). If $\delta_s = 0$ then $D = D'$. $\square$

Proof of Proposition 1. Let $\tilde{D}$ equal D except that $\tilde{D}_{vv'} = 0$ for $v, v' \in P$. At scale 0, Kruskal's algorithm joins all of P into one component with $|P| - 1$ zero-length bars. Every later merge is a merge in the quotient graph Q on $\{p^*\} \cup R$ with $Q_{p^*u} = \min_{v \in P} \tilde{D}_{vu}$ and $Q_{uw} = D_{uw}$. So $\text{MTopDiv}(R, P) = P_0(Q)$. For $u \in R$ and $v \in P$ we have $v < u$ and $D_{uv} = 1 - A_{uv}$, so $Q_{p^*u} = 1 - \pi_u$. For $w < u$ in R , $Q_{uw} = 1 - A_{uw}$, and $\max(Q_{p^*u}, Q_{p^*w}) - Q_{uw} = A_{uw} - \min(\pi_u, \pi_w)$, so $\delta_{p^*}(Q) = \delta_P$. Theorem 1(b) with $N - 1 = |R|$ gives the sandwich, with equality when $\delta_P = 0$ by part (c). Finally $\pi_u \geq A_{u0}$ because $0 \in P$. $\square$

We checked both statements numerically. The fast MTop-Div (a dense Prim's algorithm on Q) agrees with ripser run on $\tilde{D}$ to $3 \cdot 10^{-8}$ on 400 random causal matrices. On these, and on 300 further random matrices checked for the elementwise death bound and the bar bound in degrees 1 and 2, there are 0 violations (sinktda/check_theory.py).

Proof of Corollary 2. For $u > v$, $A_{vu} = 0$, so $D_{uv} = 1 - A_{uv}$; in particular $D_{0u} = 1 - A_{u0}$. For $1 \leq v < u$, $\max(D_{0u}, D_{0v}) - D_{uv} = A_{uv} - \min(A_{u0}, A_{v0})$. Substitute into Theorem 1. $\square$

A.3 LINEAR GROWTH OF 1D PERSISTENCE UNDER I.I.D. WEIGHTS

The usual reason for length normalizing 1D features is that there is growth in cycles with N . This is understood if weights are i.i.d.: 0D total persistence tends to $\zeta(3)$ (Frieze, 1985) and 1D total persistence of the random clique-complex process is $\Theta(N)$ (Hiraoka & Shirai, 2017; Hino & Kanazawa, 2019). Here is a simple bound that makes explicit the constant.

Proposition 2 (Linear growth under i.i.d. weights). *If $\{D_{uv}\}_{u < v}$ are i.i.d. Uniform $[0, 1]$, then $\mathbb{E}[P_1(N)] \geq \int_0^1 \left[\binom{N}{2}p - N - \binom{N}{3}p^3 \right]_+ dp = \frac{3}{8}N - \sqrt{3N} + O(1)$. (There is also an $O(N)$ upper bound Theorem 1.5 of Hino & Kanazawa, 2019.)*

Proof. With i.i.d. uniform weights, the edges with $D_{uv} \leq p$ form an Erdős-Rényi graph $G(N, p)$ and K_p is its clique (flag) complex. Every 1D bar is contained in $[0, 1]$, because K_1 is the full simplex. Writing $\beta_1(p)$ for the first Betti number of K_p , the number of bars alive at p is $\beta_1(p)$, so $P_1 = \int_0^1 \beta_1(p) dp$. For any finite complex with f_i faces of dimension i , the strong Morse inequality at degree 2 reads $\beta_2 - \beta_1 + \beta_0 \leq f_2 - f_1 + f_0$ (Edelsbrunner & Harer, 2010), hence $\beta_1 \geq f_1 - f_0 - f_2 + \beta_0 + \beta_2 \geq f_1 - f_0 - f_2$. Since also $\beta_1 \geq 0$, Jensen's inequality gives $\mathbb{E}\beta_1(p) \geq [\mathbb{E}f_1 - f_0 - \mathbb{E}f_2]_+ = \left[\binom{N}{2}p - N - \binom{N}{3}p^3 \right]_+$. Integrating over p gives the first inequality. With $p = x/\sqrt{N}$ and $dp = dx/\sqrt{N}$, the integrand divided by $\sqrt{N}$ is $N(x/2 - x^3/6) - \sqrt{N} + O(1)$ uniformly for $x \in [0, \sqrt{3} + 1]$. It is positive on an interval whose endpoints are $O(N^{-1/2})$ from 0 and $\sqrt{3}$, and it is negative for larger x . Hence $\int_0^1 [\cdot]_+ dp = N \int_0^{\sqrt{3}} \left(\frac{x}{2} - \frac{x^3}{6} \right) dx - \sqrt{3N} + O(1) = \frac{3}{8}N - \sqrt{3N} + O(1)$. Numerically, $(\frac{3}{8}N - \text{bound})/\sqrt{N} = 1.62, 1.67, 1.70, 1.72$ at $N = 2^8, 2^{10}, 2^{12}, 2^{14}$, approaching

$\sqrt{3} \approx 1.73$. The order $\Theta(N)$ itself is known (Hiraoka & Shirai, 2017; Hino & Kanazawa, 2019); the proposition only makes the constant explicit. $\square$

Numerically, the integral equals 0.791, 3.92, 11.9, 30.2, 70.1 at $N = 16, 32, 64, 128, 256$, against simulated means 1.91, 5.83, 15.5, 36.2, 79.3 under i.i.d. Uniform $[0, 1]$ weights (Figure 4, left). Over $N \in [16, 256]$ both grow faster than linearly (log-log slopes 1.55 and 1.31), a pre-asymptotic effect of the $-\sqrt{3N}$ term; both are $\Theta(N)$ as $N \rightarrow \infty$. (Figure 4, left). Both functions on $[16, 256]$ increase more than linearly (with log-log slopes 1.55 and 1.31), due to the $-\sqrt{3N}$ term in pre-asymptotics. Both are $\Theta(N)$ as $N \rightarrow \infty$.

A.4 DIRECTED FLAG COMPLEXES ON CAUSAL ATTENTION

Proposition 3 (Directed flag collapse). *Let A be lower-triangular with non-negative entries (self-loops are ignored). Let $\text{DFI}(A)$ be the filtered directed flag complex (Lütgehetmann et al., 2020) of the digraph with an edge $i \rightarrow j$ for every $j < i$ (a transitive tournament) and weight $1 - A_{ij}$, and let $\text{VR}(D)$ be as above with $D = 1 - \max(A, A^\top)$. These filtered complexes are isomorphic, so $\text{dgm}_k(\text{DFI}(A)) = \text{dgm}_k(\text{VR}(D))$ for all k . This remains true for any weighting $w(i \rightarrow j) = \phi(A_{ij})$ with $D_{ij} = \phi(\max(A_{ij}, A_{ji}))$ and ϕ non-increasing.*

Proof. A directed k -simplex is an ordered tuple $(v_0, \dots, v_k)$ with an edge $v_a \rightarrow v_b$ for all $a < b$. Edges exist only from later to earlier tokens, so a vertex set S supports exactly one directed simplex, its elements in decreasing order; every other ordering uses a missing forward edge. (A missing edge is not a zero-weight edge; encoding it as weight 0 would admit all orderings, a common implementation error.) This bijection preserves faces. Its filtration value is $\max_{a < b} (1 - A_{v_a v_b})$, and for $u > v$, $D_{uv} = 1 - A_{uv}$, so the undirected value $\max_{u, v \in S} D_{uv}$ is the same number. $\square$

We confirm the bijection and the filtration identity explicitly on 420 random causal matrices ($N \leq 8$, four families including tie-dense and underflowing matrices; 30,060 vertex sets; zero violations) and check that it does not hold for bidirectional attention, meaning that a set of size n has $n!$ orderings. The collapse comes from causality: for decoders, adding the directed flag complex brings nothing to the symmetrized one. This statement does not cover directional features that are not coming from flag complexes, like Kushnareva et al. (2021) for bidirectional encoders.

B IMPLEMENTATION AND PROTOCOL DETAILS

Extraction. Every setting runs in bfloat16 with eager attention and no quantization. The ones using the two 7B models (Mistral-7B on both benchmarks, its chat-format variant, and Qwen2.5-7B on both benchmarks) have been extracted on an NVIDIA RTX 5080 (CUDA), while others were extracted on Apple-silicon (MPS), where float16 produces non-finite values and degenerate generations for Qwen models and the extractor refuses non-finite logits. The paragraph on float32 below measures how much the platform itself moves a reported AUC. Model identifiers are in Table 7 (Hugging Face main revisions as of September 2026). For each example we run one forward pass with attentions and hidden states, average heads, and form D in float64. Ripser (Bauer, 2021) computes 0D/1D barcodes; since SciPy’s sparse MST can omit near-zero-weight edges ($A_{u0} \approx 1$), we compute 0D MST summaries using a batched dense Prim’s algorithm. δ_0 applies the closed form of Corollary 2; we brute force compute $\min_s \delta_s$ over all apexes. DEFLATED deletes token 0 and renormalizes each row over the remaining causal support. ANSWER-ONLY limits D to the answer tokens and does not renormalize. The Lookback ratio of head h is the mean over answer tokens t of $\bar{a}_{\text{ctx}} / (\bar{a}_{\text{ctx}} + \bar{a}_{\text{new}})$, where $\bar{a}$ is the mean attention from t to prompt tokens and to answer tokens up to t . The LLM-Check score of a head is the mean log diagonal attention (the log-determinant of the triangular attention kernel divided by N), summed over heads per layer. Log-probability features are the mean, minimum, and sum of answer-token log-probabilities, mean and maximum predictive entropy, and mean maximum softmax probability. Hidden states are mean-pooled over answer tokens at layer $\lfloor L/2 \rfloor$.

Prompts. For TruthfulQA we use each model’s own format: the Qwen chat template with an explicit system turn “You are a helpful assistant.”, `<|user|>...<|end|>` for Phi-3, `<|user|>...</s>` for TinyLlama, ChatML without a system turn for SmolLM, and `[INST]...[/INST]` for Mistral. HaluEval uses Knowledge: `...\\nQuestion: ...\\nAnswer:` without a chat template. Boundary is then calculated by tokenizing the prefix (not including the final space), which the full string concatenates to the first answer token. When the chat template itself outputs the BOS token (e.g., Mistral’s own template, Llama-3.1), it does not output a second one. Both the prefix and the full string are then tokenized in the same way, meaning the token ids of the prefix are a prefix of the full string’s. This is verified for all models and templates without a GPU in `sinktda/check_tokenization.py`. The on-policy prompt is the model’s chat template around “Answer the following question with a short phrase only, no explanation. Question: ...”. The token at position 0 is the first token of the format (BOS for TinyLlama and Mistral, `<|user|>` for Phi-3, whose tokenizer adds no BOS, `<|im_start|>` for Qwen and SmolLM, and “Knowledge” for Qwen on HaluEval). The theorem holds for any value, not necessarily a special token. We generate on-policy answers using the chat-formatted prompt with `add_special_tokens=False` and extract features with the tokenizer’s default special tokens. They only differ for TinyLlama, where the input to feature extraction begins with a BOS token. Mistral runs with its hand-written `[INST]` string and its own chat template. After fixing the tokenization issue above, the two prefixes are the same. The differences lie in the settings (the end-of-sequence token for the template). Banks that do not rely on the answer window agree to 0.011 AUC. The answer-restricted banks disagree more (answer-only PH by +0.071, deflated PH by +0.023), since that token becomes part of the answer subgraph and the deflated graph.

Sensitivity to inference precision. An earlier run of five settings in float16 lets us measure how much each bank depends on numerical precision (Table 4). We find that switching to bfloat16 alters the AUC of head-averaged topological banks by up to 0.058 (TQA Qw3B; at most 0.005 elsewhere), of sink features by up to 0.003, but of non-topological banks by at most 0.001. One more reason to report the precision along with any attention-topology result is that short 1D bars tend to be close to the scale of δ_0 and can be affected by small changes in attention.

Table 4: **AUC change from float16 to bfloat16** (same prompts and protocol; HaluEval Qwen2.5-3B also changed its prompt boundary, see above).

Setting	Sink	0D	1D	Defl.	0D _h	LogP	Lookb.	Hidden	Non-topo.
TQA Qw3B	−0.003	−0.018	+0.058	−0.001	−0.001	−0.001	0.000	0.000	−0.001
TQA Phi-3	−0.001	−0.001	−0.003	0.000	−0.001	0.000	0.000	0.000	0.000
TQA TinyLL	0.000	0.000	−0.001	−0.001	0.000	0.000	0.000	0.000	0.000
TQA SmolLM	0.000	−0.001	−0.001	+0.001	+0.002	0.000	0.000	0.000	0.000
HE Qw3B	+0.001	−0.005	+0.001	+0.002	−	−0.001	−	0.000	−

float32 against bfloat16. In the fp16/bf16 comparison above, we vary precision while keeping hardware fixed. Next, to isolate effects of precision versus platform, we repeated our extraction of 2 settings twice, once in bfloat16 and once in float32, on a single CUDA machine, and compared with the bfloat16 numbers the paper reports (which, for these two settings, were extracted on Apple-silicon MPS). The platform gap (bf16 CUDA against bf16 MPS, and so also torch and scikit-learn versions) is at most 0.005 AUC. The precision gap (float32 against bfloat16 on the same machine) is at most 0.007 (0.007 on the topological banks). Both are well within the ± 0.015 equivalence range, which reveals that the reported numbers transfer to other hardware and float32 does not change a conclusion (Table 5).

Table 5: **Platform and precision sensitivity (Δ AUC).** “bf16 CUDA – bf16 MPS” isolates hardware and library versions at fixed precision; “fp32 – bf16 (CUDA)” isolates precision at fixed hardware.

Setting	Contrast	Sink	0D	1D	Defl.	0D _h	LogP	Lookb.	Hidden	Non-topo.
TQA Qw1.5B	bf16 CUDA – bf16 MPS	+0.001	+0.005	+0.003	+0.003	−0.002	0.000	+0.001	0.000	0.000
TQA Qw1.5B	fp32 – bf16 (CUDA)	+0.002	+0.004	+0.001	−0.007	+0.005	+0.003	+0.001	−0.001	0.000
TQA TinyLL	bf16 CUDA – bf16 MPS	0.000	−0.001	+0.001	−0.001	+0.001	0.000	0.000	0.000	0.000
TQA TinyLL	fp32 – bf16 (CUDA)	0.000	+0.001	+0.002	+0.003	−0.001	0.000	0.000	0.000	0.000

On-policy benchmark. Table 6 presents the model generations produced and how they were evaluated.

Table 6: **TriviaQA on-policy benchmark.** 2,000 validation questions sampled with seed 0; greedy decoding, first line of the answer, at most 24 new tokens. An answer is correct if any normalized gold alias (lower-cased, punctuation and articles removed) is contained as a whole-word span in the normalized answer. Empty answers are dropped.

Model	questions	accuracy	rows used	mean answer tokens
Qwen2.5-3B	2,000	42.4%	2,000	5.5
Qwen2.5-1.5B	2,000	31.4%	2,000	5.8
Phi-3-mini	2,000	56.2%	2,000	6.0
TinyLlama-1.1B	2,000	26.4%	2,000	18.9
Mistral-7B	2,000	66.5%	2,000	14.0
Qwen2.5-7B	2,000	51.4%	2,000	5.3

Statistics. Bootstrap resamples are drawn over questions (both rows of a pair move together). TOST power at margin m is under a true difference of 0 and a normal sampling distribution with the bootstrap SE: $2\Phi(m/SE - z_{0.95}) - 1$. Average repeated-CV predictions (seeds 42, 43, 44) are used for bootstrapping (seed 0), i.e., the intervals report sampling of questions with a fixed, split-averaged predictor without refitting variance. Seed-to-seed SD of the pooled AUC is given separately (Appendix J). It is at most 0.008, with 24 out of the bank-setting cells exceeding 0.004. The C rule is a function of bank width, so $C = 1$ fits a shallow model to a given bank and tuned C fits a deep model to the same bank. Late fusion trains the second-level model on first-level out-of-fold scores as in regular stacking. Training rows’ first-level scores are obtained by training models which have seen the corresponding test questions, which is a weak, standard optimism that both arms of the comparison share.

C ADDITIONAL ANALYSES

Models. Table 7 lists the audited architectures. All models use grouped or multi-head attention; we average attention over query heads.

Table 7: **Audited models.**

Model	params	layers	heads	KV heads	hidden
Qwen/Qwen2.5-3B-Instruct	3.09B	36	16	2	2048
Qwen/Qwen2.5-1.5B-Instruct	1.54B	28	12	2	1536
microsoft/Phi-3-mini-4k-instruct	3.82B	32	32	32	3072
TinyLlama/TinyLlama-1.1B-Chat-v1.0	1.10B	22	32	4	2048
HuggingFaceTB/SmolLM-1.7B-Instruct	1.71B	24	32	32	2048
mistralai/Mistral-7B-Instruct-v0.2	7.24B	32	32	8	4096
Qwen/Qwen2.5-7B-Instruct	7.62B	28	28	4	3584

Length scaling of 1D persistence on real attention. Table 8 regresses log total 1D persistence (summed over layers) on log N with a class fixed effect, on rows with $P_1 > 0$, so that the exponent is not driven by a length difference between correct and hallucinated answers. Exponents range from 0.03 to 1.39, against 1.31 for i.i.d. uniform weights and 1.27 for random causal attention without a sink over the same range of N . 10 of 11 settings grow significantly more slowly than the i.i.d. reference (not HE Qw3B). The exponent exceeds 1 in HE Qw3B, HE Qw1.5B, so growth on real attention is not uniformly sub-linear. Summed over layers, almost every example has some 1D bar, because every model has some layers that are not coned; within an example, the coned layers contribute nothing to P_1 whatever the length (Table 1).

Table 8: **Length scaling of 1D persistence.** OLS of $\log P_1$ on $\log N$ and the class label; rows with $P_1 = 0$ are excluded and counted in the first column.

Setting	rows with $P_1 > 0$	$\hat{\alpha}$ (class FE)	95% CI	R^2	Spearman(P_1, N)
TQA Qw3B	100.0%	0.25	[0.23, 0.27]	0.27	0.57
TQA Qw1.5B	100.0%	0.57	[0.52, 0.61]	0.29	0.51
TQA Phi-3	99.9%	0.93	[0.82, 1.04]	0.15	0.39
TQA TinyLL	100.0%	0.32	[0.28, 0.36]	0.14	0.28
TQA SmolLM	100.0%	0.61	[0.54, 0.69]	0.13	0.36
HE Qw3B	100.0%	1.39	[1.31, 1.47]	0.36	0.60
HE Qw1.5B	100.0%	1.10	[1.07, 1.13]	0.84	0.91
TrQA Qw3B	100.0%	0.72	[0.68, 0.75]	0.46	0.68
TrQA Qw1.5B	100.0%	0.75	[0.72, 0.78]	0.59	0.68
TrQA Phi-3	100.0%	0.70	[0.65, 0.75]	0.29	0.49
TrQA TinyLL	100.0%	0.03	[0.02, 0.03]	0.03	0.15
i.i.d. uniform weights (simulation)	100%	1.31	—	—	—
random causal attention, no sink (simulation)	100%	1.27	—	—	—

Sequence lengths. Table 9 gives class-conditional lengths. TruthfulQA is length-balanced by construction; HaluEval’s hallucinated answers are systematically longer; on-policy, incorrect answers differ in length from correct ones because they are the model’s own.

Table 9: **Class-conditional sequence length** (mean $\pm$ SD, full sequence) and mean answer length.

Setting	N (correct)	N (hallucinated)	answer tokens (c./h.)	point-biserial r	Welch p
TQA Qw3B	44.4 $\pm$ 7.9	44.0 $\pm$ 8.0	12.6 / 12.1	-0.030	0.22
TQA Qw1.5B	44.4 $\pm$ 7.9	44.0 $\pm$ 8.0	12.6 / 12.1	-0.030	0.22
TQA Phi-3	29.2 $\pm$ 8.9	28.7 $\pm$ 8.8	11.8 / 11.4	-0.024	0.32
TQA TinyLL	42.5 $\pm$ 9.0	42.1 $\pm$ 8.9	12.1 / 11.7	-0.024	0.32
TQA SmolLM	32.6 $\pm$ 8.2	32.1 $\pm$ 8.2	10.6 / 10.2	-0.026	0.29
HE Qw3B	121.2 $\pm$ 39.7	132.7 $\pm$ 41.2	3.8 / 15.4	+0.141	$< 10^{-9}$
HE Qw1.5B	122.2 $\pm$ 39.5	133.9 $\pm$ 41.3	3.6 / 15.3	+0.144	$< 10^{-5}$
TrQA Qw3B	68.0 $\pm$ 10.4	69.0 $\pm$ 9.7	5.3 / 5.7	+0.052	0.022
TrQA Qw1.5B	67.5 $\pm$ 10.3	69.4 $\pm$ 10.0	5.4 / 5.9	+0.086	0.00013
TrQA Phi-3	45.6 $\pm$ 11.3	46.9 $\pm$ 11.2	5.6 / 6.5	+0.057	0.01
TrQA TinyLL	67.3 $\pm$ 12.5	73.7 $\pm$ 14.9	17.0 / 19.6	+0.196	$< 10^{-20}$

Margin sensitivity. Table 10 reports TOST p -values at four equivalence margins for the four comparisons of Figure 5; bold marks $p < 0.05$ (equivalence).

Table 10: **TOST p -values by margin m .** NT: all non-topological features.

Setting m	Sink vs. 0D				0D Sink				1D 0D				0D NT			
	.010	.015	.020	.025	.010	.015	.020	.025	.010	.015	.020	.025	.010	.015	.020	.025
TQA Qw3B	.029	.001	<.001	<.001	.040	.001	<.001	<.001	1.000	.995	.972	.882	<.001	<.001	<.001	<.001
TQA Qw1.5B	.109	.016	<.001	<.001	.692	.276	.040	.002	.311	.034	<.001	<.001	<.001	<.001	<.001	<.001
TQA Phi-3	.138	.013	<.001	<.001	.112	.003	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001	<.001
TQA TinyLL	.520	.234	.064	.012	.962	.809	.483	.169	.210	.021	<.001	<.001	<.001	<.001	<.001	<.001
TQA SmolLM	.939	.865	.748	.602	1.000	.996	.960	.805	.179	.036	.003	<.001	<.001	<.001	<.001	<.001
HE Qw3B	.910	.676	.351	.110	1.000	1.000	.982	.804	.906	.551	.148	.012	<.001	<.001	<.001	<.001
HE Qw1.5B	.993	.970	.896	.755	.679	.263	.041	.002	.980	.918	.786	.569	<.001	<.001	<.001	<.001
TrQA Qw3B	.792	.522	.239	.069	.982	.866	.561	.219	.106	.015	.002	<.001	<.001	<.001	<.001	<.001
TrQA Qw1.5B	.431	.176	.050	.010	.528	.211	.044	.005	.079	.011	<.001	<.001	<.001	<.001	<.001	<.001
TrQA Phi-3	.047	.002	<.001	<.001	.114	.005	<.001	<.001	.086	.003	<.001	<.001	<.001	<.001	<.001	<.001
TrQA TinyLL	.091	.021	.003	<.001	.071	.005	<.001	<.001	.083	.013	.002	<.001	<.001	<.001	<.001	<.001

Per-layer signal. Figure 2 shows single-feature AUCs by depth. P_0 and the star weight trace nearly identical curves in every model with a sink, while P_1 carries signal only in the layers that are not coned. These AUCs are orientation-free ($\max(\text{AUC}, 1 - \text{AUC})$) and therefore slightly optimistic; they are descriptive, not a detector.

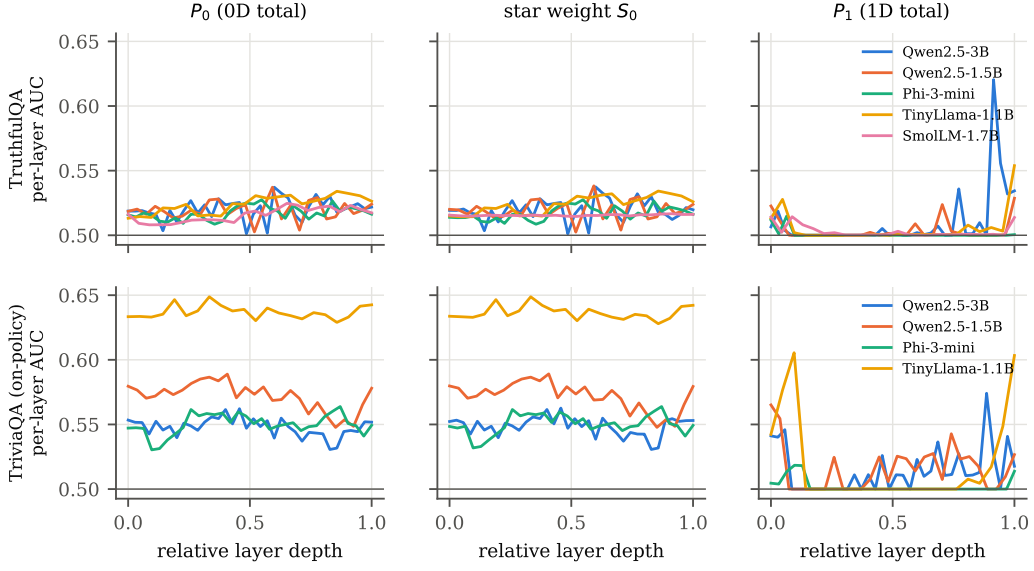


Figure 2: Single-feature AUC by layer.

Worked example. Figure 3 shows TruthfulQA question 67 with its incorrect answer (“Yes, vampires are real”) under TinyLlama-1.1B ($N=29$). In layer 5 the model sends 0.85 of its attention to token 0 on average, $\delta_0 = 0.000$, and accordingly $P_0 = 4.225$ equals the star weight $S_0 = 4.225$ and there are 0 H_1 bars. In layer 21 sink attention drops to 0.53 and $\delta_0 = 0.281$; now $P_0 = 12.287$ falls below $S_0 = 13.043$ (within the bound $(N-1)\delta_0 = 7.87$), and 10 H_1 bars appear, the longest of length $0.172 \leq \delta_0$. The theorem bounds loops but does not force them: in layers 2 and 11–18 of the same example δ_0 lies between 0.11 and 0.52, yet H_1 is empty.

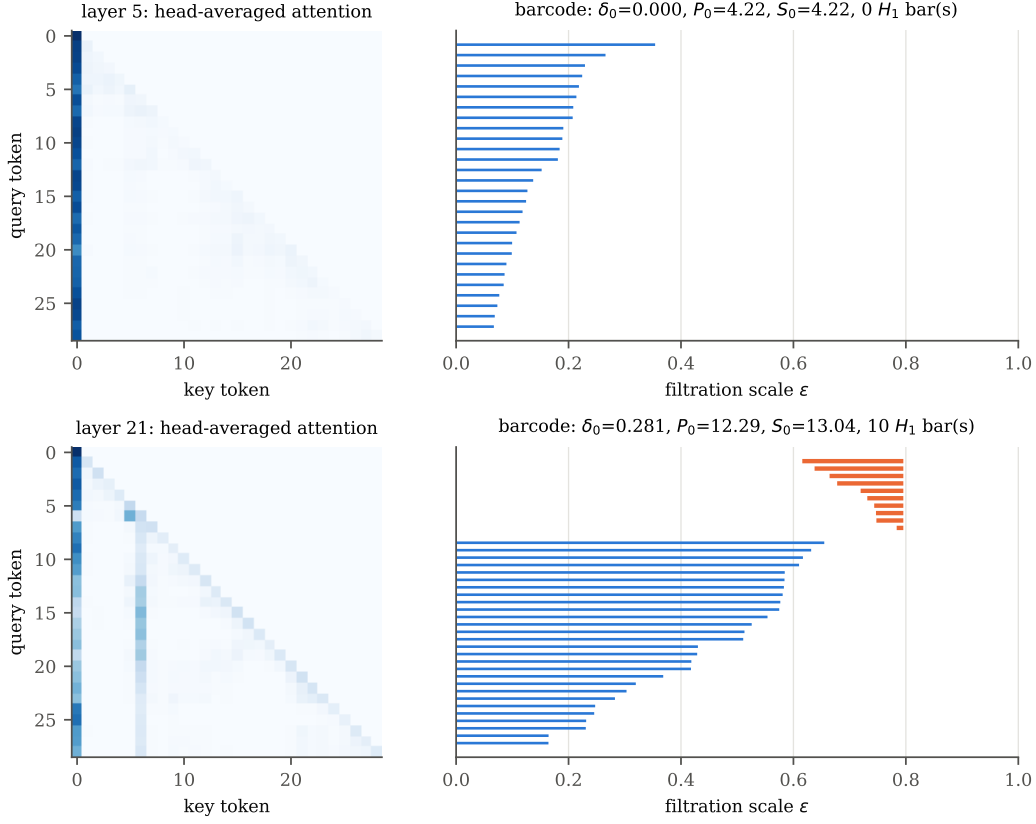


Figure 3: **A coned and a non-coned layer of the same example** (TinyLlama-1.1B, TruthfulQA). Left: head-averaged attention; the bright first column is the sink. Right: 0D bars (blue) and 1D bars (orange) of the Vietoris-Rips filtration of $D = 1 - \max(A, A^T)$.

D ONE-DIMENSIONAL PERSISTENCE

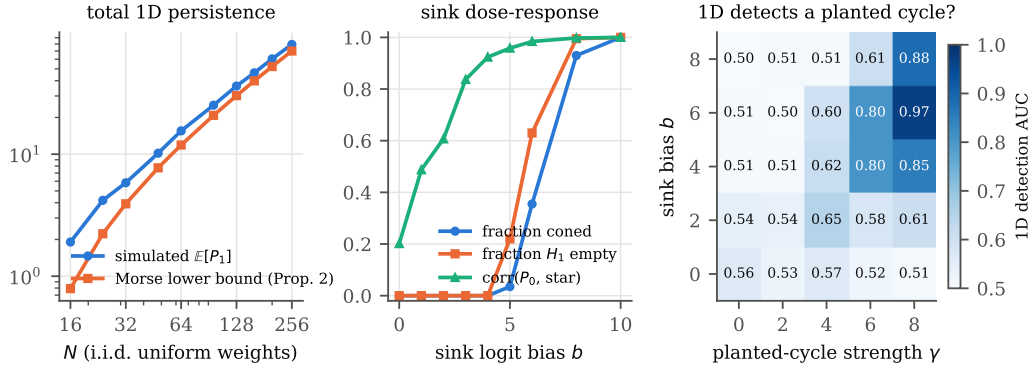


Figure 4: **Synthetic checks.** Left: Lower bound of Proposition 2 compared to simulated mean total 1D persistence under i.i.d. Uniform[0, 1] weights (20 draws per $N \leq 128$, 8 above). Middle: random causal attention ($N = 40$, 200 draws per point) with a logit bias b on token 0: graphs become exactly coned, H_1 empties, and P_0 becomes a function of the star weight, growing as the sink becomes stronger. Right: AUC of 1D total persistence for detecting a planted 6-cycle of logit strength γ (150 draws per class). Without a sink ($b = 0$) the planted cycle is lost among background cycles; under a strong sink ($b = 8$) it is invisible unless it is strong enough to break the cone.

Under i.i.d. weights, 1D persistence is thus growing faster than linearly in the lengths relevant for us and linearly asymptotically (Proposition 2; Figure 4, left). For real attention, almost no 1D bars exist (Table 1) and they appear only on graphs with a non-zero coning defect. Figure 4 (middle) simulates this change in a controlled manner: as we increase the logit bias of the sink, random causal attention goes from looking i.i.d.-like to doing exact coning, while H_1 empties out in lockstep.

This explains a common finding in practice, that 1D features add little to 0D. At full sample size, adding length-normalized 1D to 0D changes AUC by -0.008 to $+0.033$, which is equivalent within ± 0.015 in 9 of 13 sink settings; the exceptions are HE Qw1.5B, HE Qw3B, TQA Qw3B and TrQA Mis7B; of these, adding 1D has a significant positive effect on AUC in HE Qw1.5B, HE Qw3B and TQA Qw3B; the other exception has a negative point estimate but an interval that is too wide to be certain about equality. Note that the improvement seen in TruthfulQA Qwen2.5-3B is also the most sensitive result in terms of inference precision (Appendix B). We want to emphasize that this is not because loops are generally uninformative. For instance, in a strongly sinked model, if a loop exists, there must be some token paying more attention to another token than one of those two tokens is paying to the sink, and the duration of such a loop will be limited by how much extra attention one of them pays to the other. See Figure 4 (right). H_1 can readily find a planted 6-cycle under a moderate sink ($b = 6, \gamma = 8$: AUC 0.97). But a strong sink ($b = 8, \gamma \leq 4$: AUC ≤ 0.51) will mask it unless it is strong enough to shatter the cone. Without any sink ($b = 0$) H_1 will also fail (AUC ≤ 0.57), since the random attention will already have plenty of background cycles. 1D persistence is useful only in between.

E RECOMMENDATIONS FOR REPORTING ATTENTION-GRAPH TOPOLOGY

(i) Compare a topological detector against the first-order statistic named by the reduction, and report sink attention and δ_0 (or the relevant defect, such as δ_P for TOHA). (ii) Compare topological detectors against per-head non-topological statistics (sink mass, entropy, Lookback ratios) at the same resolution, as well as against a hidden-state probe. (iii) Evaluate on the model’s own generations rather than using teacher-forced benchmarks. These mix detection and recognition of dataset authorship. (iv) If we want to go beyond the sink, either remove the sink first (deflation) or consider only answer tokens. These reveal real non-sink geometry, but in our settings it was not new information.

F VALIDATION OF THE TOHA IMPLEMENTATION AND THE ON-POLICY LABELS

Against the authors’ released code. Our per-head score d is our own implementation, so we check it against the routine the TOHA authors release (Bazarova et al., 2026; `src/methods/mtopdiv/utis.py`), which builds the same distance matrix and calls ripser rather than our dense Prim’s algorithm. Over 21,696 head graphs from 2 settings the two agree to 1.2×10^{-8} (Table 11). So the reduction in Proposition 1 is not an artefact of how we compute d .

Table 11: **Our d against the authors’ MTop-Div code.** Δ is the per-head difference; both are divided by $|R|$.

Setting	head graphs	max $ \Delta $	mean $ \Delta $	Pearson r
TQA Mis7B	12,288	0	0	1.00000000
TQA Qw7B	9,408	1.2×10^{-8}	1.1×10^{-11}	1.00000000

LLM-judge audit of the on-policy labels. The on-policy label is an alias string match, which mislabels paraphrases and hedged answers. We sampled 200 answers per setting (seed 0) in 2 settings and had Qwen2.5-7B-Instruct grade them from the question, the gold aliases and the answer. Disagreement is 5.5% to 14.0% (Table 12), and is mostly the string match being too lax. Re-scoring the same out-of-fold predictions on those rows under the judge’s labels shifts every bank by at most 0.063 AUC and leaves the dominance orderings the paper relies on intact (NONTPO $\geq$ 0D, NONTPO $\geq$ DEFL, HIDDEN $\geq$ 0D, NONTPO $\geq$ PH-0D), so the conclusions do not rest on the labeling. The standard error of an AUC is near 0.03 with $n = 200$ audited rows, so near-ties

between other banks can and do reorder. Only the four orderings above are claimed. The judge is itself a language model, not a human annotator. One of the audited settings is graded by the same model that produced its answers, so 14.0% should be read as a floor rather than an estimate.

Table 12: **LLM-judge audit of the on-policy labels.** “match 0/judge 1”: the string match calls it hallucinated and the judge does not. Last column: largest change in any bank’s AUC on the audited rows when the judge’s labels replace the string match.

Setting	judged	disagree	match 0/judge 1	match 1/judge 0	max $ \Delta\text{AUC} $
TrQA Mis7B	200	14.0%	8	20	0.063
TrQA Qw7B	200	5.5%	7	4	0.027

G INTERVENING ON THE SINK

Proposition 1 states that TOHA should correlate more closely with the response sink score the stronger the sink. We can check this directly by intervening on the sink itself. For Qwen2.5-1.5B and TinyLlama on TruthfulQA, we add a constant $b \in \{-2, 0, 2, 4\}$ to the first token’s attention logit in every head and recompute all per-head quantities (Table 13). The intervention does not affect the identity (no violations in any run). It affects the structure the proposition says matters. Weakening the sink ($b = -2$) reduces the fraction of prompt-coned head graphs from 58% to 37% (Qwen) and from 74% to 63% (TinyLlama), the fraction where the sink is every response token’s top prompt token from 45% to 21% and from 63% to 21%, and the median rank correlation between d and the response sink score from 0.893 to 0.605 and from 0.967 to 0.406. Strengthening it ($b = 4$) increases the correlation to 0.962 and 0.996. In TinyLlama all three numbers increase monotonically with b . In Qwen they peak at $b = 2$ and drop a little at $b = 4$. TOHA’s own AUC hardly changes (0.65 to 0.67 and 0.65 to 0.70). The gap to the sink-score version is not monotone in b , showing that our intervention affects TOHA’s accuracy through the attention graph rather than being a dose-response.

Table 13: **Sink-bias intervention** (teacher-forced TruthfulQA; labels unchanged). A constant b is added to the attention logit of key 0 for every query after token 0, in every head and layer. $\delta_P=0$: share of prompt-coned head graphs. Sink top: share of head graphs in which token 0 is every response token’s most-attended prompt token. ρ : median per-head rank correlation between TOHA’s d and the response sink score. Remaining columns: AUCs as in Table 3.

Model	b	$\delta_P=0$	sink top	$\rho(d, 1 - \bar{a}_0)$	TOHA	TOHA $[\bar{a}_0]$	TOHA $_{\text{sup}}$	$\bar{a}_{0,\text{sup}}$
Qwen2.5-1.5B	-2	37%	21%	0.605	0.667	0.624	0.843	0.849
Qwen2.5-1.5B	0	58%	45%	0.893	0.646	0.628	0.858	0.857
Qwen2.5-1.5B	+2	72%	60%	0.970	0.667	0.637	0.857	0.857
Qwen2.5-1.5B	+4	69%	56%	0.962	0.647	0.640	0.839	0.838
TinyLlama-1.1B	-2	63%	21%	0.406	0.698	0.626	0.845	0.834
TinyLlama-1.1B	0	74%	63%	0.967	0.666	0.618	0.845	0.848
TinyLlama-1.1B	+2	81%	75%	0.992	0.652	0.610	0.848	0.848
TinyLlama-1.1B	+4	86%	83%	0.996	0.665	0.616	0.844	0.842

H MISTRAL-7B

Mistral-7B-Instruct-v0.2 is the most sink-dominated model in this study, and the reduction is correspondingly tightest there. In its [INST] format it puts 0.75 of its attention on token 0 — more than any smaller model, and 98.2% of its 52,288 layer graphs are exactly coned ($\delta_0 = 0$), with 0 violations of any bound in Theorem 1.

A previous check of this model’s accuracy level indicated that 99.99% of its 1D features are exactly zero and that the 0D and 0D+1D banks are equivalent. This also indicated that near-universal coning could explain it, but did not check δ_0 directly. On the graphs themselves, both halves now check out: 1D persistence is exactly zero on 99.99% of them, and $\delta_0 = 0$ checks out on 98.2%. The length-normalized rank correlation between $P_0/(N-1)$ and $1 - \bar{m}$ has median 1.000 over layers

and never drops below 0.994, so on this model 0D total persistence is, up to rank, an affine function of sink attention in every layer. Per head, 81% of its graphs are exactly prompt-coned. Its median $\rho(d, \bar{\pi})$ is 0.998, and Proposition 1 has 0 violations.

I RELATION TO ACCURACY-LEVEL AUDITS, AND COST

These are four facts about audits of attention-graph PH that stop at classification accuracy, now explained by the theory: (i) 0D persistence is an MST statistic (Fact 1); (ii) 1D adds little beyond 0D (Table 14, column “1D | 0D”) (Theorem 1); (iii) An earlier audit revealed that the 1D bank of Mistral-7B using its [INST] format is 99.99% zeros, as it would be for exact coning (directly verified on the graphs in Appendix H); (iv) HaluEval answers can be separated from text alone.

Cost. Median per-matrix wall-clock time on real attention from TinyLlama-1.1B-Chat-v1.0 (sink features / δ_0 / dense MST / ripser 0D / ripser 0D+1D) is 0.001/0.02/0.10/0.02/0.02 ms at $N=32$; 0.001/0.03/0.21/0.03/0.03 ms at $N=64$; 0.001/0.07/0.42/0.07/0.06 ms at $N=128$; 0.001/0.25/0.90/0.18/0.18 ms at $N=256$. On i.i.d. random matrices of the same size, ripser 0D+1D takes 0.171 ms ($N=32$), 3.46 ms ($N=64$), 115 ms ($N=128$), 4,631 ms ($N=256$). On real attention the same computation is four orders of magnitude cheaper at $N=256$, consistent with Theorem 1: sink-dominated graphs have few, short 1D bars to compute. Sink features, read off one column, are one to two orders of magnitude cheaper still.

J FULL RESULTS AND COMPLETE TABLES

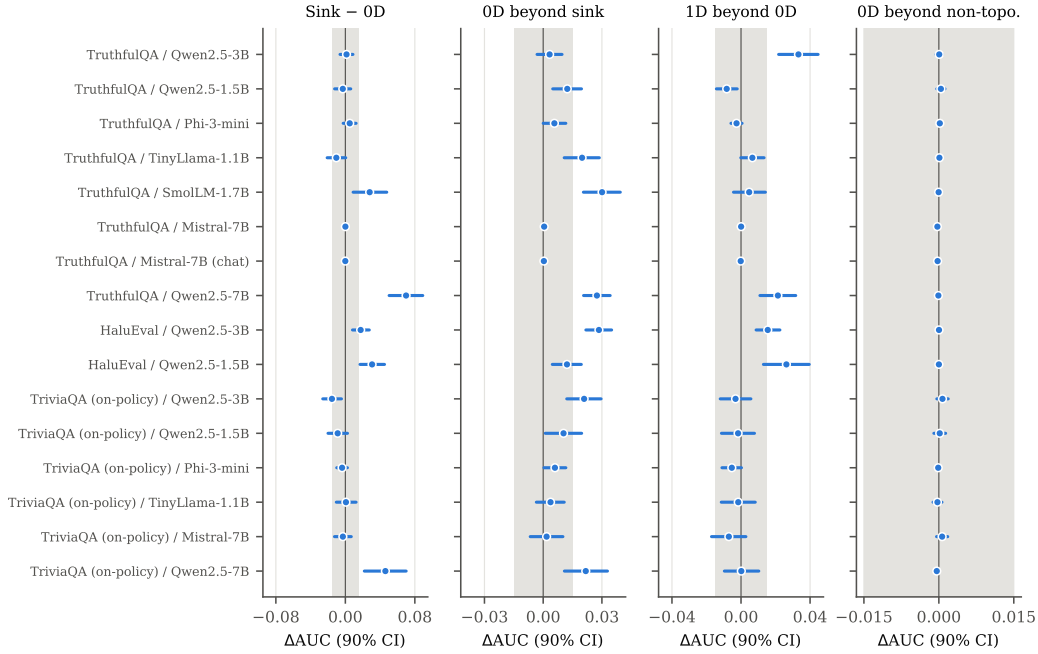


Figure 5: **Four comparisons across all settings** (ΔAUC with 90% cluster-bootstrap intervals; shaded band: ± 0.015). Sink features nearly reproduce 0D; 0D adds little beyond them; 1D adds little beyond 0D except in 4 settings; and 0D adds nothing beyond non-topological probes under early fusion (late fusion in Table 14).

Table 14: **Key paired comparisons** (Δ AUC). “X–Y” is $\text{AUC}(\text{X}) - \text{AUC}(\text{Y})$; “X|Y” is $\text{AUC}(\text{Y} + \text{X}) - \text{AUC}(\text{Y})$ under early fusion; “X $\oplus$ Y” is the same under late fusion (stacked out-of-fold logits). $P_{0,h}$: per-head MST total. Defl.: sink-deflated 0D+1D PH; NT: all non-topological features. $\equiv$: equivalent within ± 0.015 (90% interval inside the margin); *: 95% interval excludes 0. Intervals and TOST power for every cell are in Appendix J.

Setting	Sink–0D	0D Sink	$P_{0,h}$ Sink _h	Defl. Sink	1D 0D	0D NT	0D $\oplus$ NT	Defl. NT
TQA Qw3B	+0.001 $\equiv$	+0.003 $\equiv$	+0.009 $\equiv$	+0.029*	+0.033*	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$
TQA Qw1.5B	–0.003 $\equiv$	+0.012*	+0.011 $\equiv$	+0.034*	–0.008 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TQA Phi-3	+0.005 $\equiv$	+0.006 $\equiv$	0.000 $\equiv$	+0.033*	–0.003 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$
TQA TinyLL	–0.010	+0.020*	+0.003 $\equiv$	+0.045*	+0.007 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	–0.001 $\equiv$
TQA SmoLLM	+0.028*	+0.030*	+0.106*	+0.041*	+0.005 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	–0.001 $\equiv$
TQA Mis7B	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$	+0.055*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TQA Mis7B (chat)	0.000 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$	+0.067*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TQA Qw7B	+0.070*	+0.027*	+0.060*	+0.038*	+0.021*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
HE Qw3B	+0.018*	+0.028*	+0.004 $\equiv$	+0.058*	+0.015*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
HE Qw1.5B	+0.031*	+0.012*	+0.008 $\equiv$	+0.052*	+0.026*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TrQA Qw3B	–0.015*	+0.021*	+0.004 $\equiv$	+0.002	–0.003 $\equiv$	+0.001 $\equiv$	+0.002 $\equiv$	0.000 $\equiv$
TrQA Qw1.5B	–0.009	+0.010	+0.004 $\equiv$	+0.012	–0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	–0.001 $\equiv$
TrQA Phi-3	–0.004 $\equiv$	+0.006 $\equiv$	+0.002 $\equiv$	+0.029*	–0.005 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	+0.002 $\equiv$
TrQA TinyLL	+0.001 $\equiv$	+0.004 $\equiv$	–0.001 $\equiv$	+0.014	–0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TrQA Mis7B	–0.003 $\equiv$	+0.002 $\equiv$	+0.002 $\equiv$	+0.043*	–0.007	+0.001 $\equiv$	+0.003 $\equiv$	0.000 $\equiv$
TrQA Qw7B	+0.046*	+0.022*	+0.031*	+0.020*	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$	0.000 $\equiv$

Table 15: **Complete out-of-fold ROC-AUC results across all settings and feature banks.**

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA Qw3B	LEN	2	0.529	0.0003	0.567
TQA Qw3B	LEX	–	0.820	0.0014	0.829
TQA Qw3B	LOGPROB	6	0.630	0.0006	0.694
TQA Qw3B	LLMCHECK	36	0.693	0.0011	0.764
TQA Qw3B	ROWSTAT	72	0.750	0.0003	0.823
TQA Qw3B	SINK	72	0.737	0.0022	0.767
TQA Qw3B	0D	72	0.736	0.0013	0.778
TQA Qw3B	LEGACY MSTPROXY	36	0.713	0.0008	0.788
TQA Qw3B	1D	144	0.660	0.0001	0.674
TQA Qw3B	DEFL	216	0.715	0.0028	0.753
TQA Qw3B	ANS	216	0.748	0.0017	0.749
TQA Qw3B	PH SINK	576	0.851	0.0015	0.870
TQA Qw3B	PH 0DTOT	576	0.842	0.0011	0.869
TQA Qw3B	PH 0D	1152	0.853	0.0013	0.862
TQA Qw3B	PH ENT	576	0.857	0.0013	0.901
TQA Qw3B	LOOKBACK	576	0.897	0.0022	0.908
TQA Qw3B	HIDDEN	2048	0.913	0.0008	0.918
TQA Qw3B	NONTOPO	2738	0.926	0.0006	0.936
TQA Qw3B	0D+1D	216	0.769	0.0004	0.791
TQA Qw3B	0D+SINK	144	0.741	0.0015	0.776
TQA Qw3B	HIDDEN+0D	2120	0.913	0.0007	0.920
TQA Qw3B	LEX+0D	–	0.840	0.0015	0.848
TQA Qw3B	LEX+DEFL	–	0.830	0.0004	0.854
TQA Qw3B	LEX+SINK	–	0.843	0.0020	0.851
TQA Qw3B	LOGPROB+0D	78	0.776	0.0022	0.810
TQA Qw3B	NONTOPO+0D	2810	0.926	0.0004	0.936
TQA Qw3B	NONTOPO+DEFL	2954	0.928	0.0006	0.933
TQA Qw3B	PH SINK+PH 0DTOT	1152	0.860	0.0022	0.892
TQA Qw3B	SINK+1D	216	0.766	0.0007	0.787
TQA Qw3B	SINK+ANS	288	0.785	0.0031	0.797
TQA Qw3B	SINK+DEFL	288	0.766	0.0024	0.799
TQA Qw1.5B	LEN	2	0.529	0.0003	0.567
TQA Qw1.5B	LEX	–	0.820	0.0014	0.829
TQA Qw1.5B	LOGPROB	6	0.638	0.0007	0.682
TQA Qw1.5B	LLMCHECK	28	0.666	0.0005	0.743

Table 15: Complete out-of-fold ROC-AUC results across all settings and feature banks.

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA Qw1.5B	ROWSTAT	56	0.738	0.0014	0.815
TQA Qw1.5B	SINK	56	0.723	0.0019	0.771
TQA Qw1.5B	0D	56	0.726	0.0007	0.767
TQA Qw1.5B	LEGACY MSTPROXY	28	0.698	0.0010	0.786
TQA Qw1.5B	1D	112	0.586	0.0014	0.644
TQA Qw1.5B	DEFL	168	0.703	0.0042	0.755
TQA Qw1.5B	ANS	168	0.697	0.0040	0.700
TQA Qw1.5B	PH SINK	336	0.830	0.0008	0.851
TQA Qw1.5B	PH 0DTOT	336	0.819	0.0011	0.838
TQA Qw1.5B	PH 0D	672	0.838	0.0012	0.847
TQA Qw1.5B	PH ENT	336	0.833	0.0025	0.852
TQA Qw1.5B	LOOKBACK	336	0.851	0.0020	0.856
TQA Qw1.5B	HIDDEN	1536	0.916	0.0012	0.923
TQA Qw1.5B	NONTOPO	1962	0.917	0.0013	0.928
TQA Qw1.5B	0D+1D	168	0.718	0.0013	0.767
TQA Qw1.5B	0D+SINK	112	0.735	0.0012	0.766
TQA Qw1.5B	HIDDEN+0D	1592	0.916	0.0014	0.924
TQA Qw1.5B	LEX+0D	–	0.835	0.0017	0.841
TQA Qw1.5B	LEX+DEFL	–	0.825	0.0019	0.836
TQA Qw1.5B	LEX+SINK	–	0.833	0.0008	0.846
TQA Qw1.5B	LOGPROB+0D	62	0.740	0.0013	0.797
TQA Qw1.5B	NONTOPO+0D	2018	0.917	0.0014	0.927
TQA Qw1.5B	NONTOPO+DEFL	2130	0.917	0.0014	0.923
TQA Qw1.5B	PH SINK+PH 0DTOT	672	0.841	0.0013	0.867
TQA Qw1.5B	SINK+1D	168	0.715	0.0017	0.769
TQA Qw1.5B	SINK+ANS	224	0.761	0.0027	0.753
TQA Qw1.5B	SINK+DEFL	224	0.757	0.0028	0.802
TQA Phi-3	LEN	2	0.522	0.0004	0.562
TQA Phi-3	LEX	–	0.820	0.0014	0.829
TQA Phi-3	LOGPROB	6	0.578	0.0016	0.597
TQA Phi-3	LLMCHECK	32	0.653	0.0008	0.755
TQA Phi-3	ROWSTAT	64	0.661	0.0011	0.738
TQA Phi-3	SINK	64	0.651	0.0006	0.700
TQA Phi-3	0D	64	0.646	0.0011	0.709
TQA Phi-3	LEGACY MSTPROXY	32	0.613	0.0008	0.694
TQA Phi-3	1D	128	0.526	0.0030	0.551
TQA Phi-3	DEFL	192	0.639	0.0011	0.684
TQA Phi-3	ANS	192	0.629	0.0042	0.651
TQA Phi-3	PH SINK	1024	0.871	0.0008	0.901
TQA Phi-3	PH 0DTOT	1024	0.845	0.0014	0.882
TQA Phi-3	PH 0D	2048	0.868	0.0007	0.881
TQA Phi-3	PH ENT	1024	0.864	0.0010	0.895
TQA Phi-3	LOOKBACK	1024	0.892	0.0004	0.902
TQA Phi-3	HIDDEN	3072	0.933	0.0007	0.945
TQA Phi-3	NONTOPO	4198	0.936	0.0006	0.942
TQA Phi-3	0D+1D	192	0.643	0.0014	0.705
TQA Phi-3	0D+SINK	128	0.657	0.0007	0.704
TQA Phi-3	HIDDEN+0D	3136	0.933	0.0008	0.944
TQA Phi-3	LEX+0D	–	0.821	0.0003	0.842
TQA Phi-3	LEX+DEFL	–	0.802	0.0001	0.827
TQA Phi-3	LEX+SINK	–	0.820	0.0007	0.834
TQA Phi-3	LOGPROB+0D	70	0.655	0.0011	0.694
TQA Phi-3	NONTOPO+0D	4262	0.936	0.0006	0.942
TQA Phi-3	NONTOPO+DEFL	4390	0.937	0.0007	0.939
TQA Phi-3	PH SINK+PH 0DTOT	2048	0.871	0.0007	0.905
TQA Phi-3	SINK+1D	192	0.651	0.0011	0.696
TQA Phi-3	SINK+ANS	256	0.675	0.0015	0.698
TQA Phi-3	SINK+DEFL	256	0.684	0.0023	0.718
TQA TinyLl.	LEN	2	0.523	0.0005	0.556
TQA TinyLl.	LEX	–	0.820	0.0014	0.829

Table 15: Complete out-of-fold ROC-AUC results across all settings and feature banks.

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA TinyLl.	LOGPROB	6	0.637	0.0010	0.645
TQA TinyLl.	LLMCHECK	22	0.614	0.0005	0.705
TQA TinyLl.	ROWSTAT	44	0.661	0.0015	0.749
TQA TinyLl.	SINK	44	0.643	0.0006	0.718
TQA TinyLl.	0D	44	0.654	0.0011	0.729
TQA TinyLl.	LEGACY MSTPROXY	22	0.627	0.0009	0.741
TQA TinyLl.	1D	88	0.592	0.0006	0.673
TQA TinyLl.	DEFL	132	0.651	0.0019	0.705
TQA TinyLl.	ANS	132	0.644	0.0049	0.678
TQA TinyLl.	PH SINK	704	0.808	0.0015	0.841
TQA TinyLl.	PH 0DTOT	704	0.804	0.0021	0.830
TQA TinyLl.	PH 0D	1408	0.806	0.0015	0.820
TQA TinyLl.	PH ENT	704	0.817	0.0008	0.838
TQA TinyLl.	LOOKBACK	704	0.850	0.0028	0.846
TQA TinyLl.	HIDDEN	2048	0.883	0.0013	0.879
TQA TinyLl.	NONTOPO	2824	0.885	0.0011	0.887
TQA TinyLl.	0D+1D	132	0.660	0.0010	0.744
TQA TinyLl.	0D+SINK	88	0.663	0.0004	0.728
TQA TinyLl.	HIDDEN+0D	2092	0.883	0.0014	0.880
TQA TinyLl.	LEX+0D	–	0.830	0.0012	0.836
TQA TinyLl.	LEX+DEFL	–	0.826	0.0012	0.846
TQA TinyLl.	LEX+SINK	–	0.831	0.0011	0.832
TQA TinyLl.	LOGPROB+0D	50	0.683	0.0010	0.711
TQA TinyLl.	NONTOPO+0D	2868	0.885	0.0011	0.889
TQA TinyLl.	NONTOPO+DEFL	2956	0.884	0.0010	0.885
TQA TinyLl.	PH SINK+PH 0DTOT	1408	0.810	0.0031	0.841
TQA TinyLl.	SINK+1D	132	0.651	0.0016	0.729
TQA TinyLl.	SINK+ANS	176	0.685	0.0041	0.720
TQA TinyLl.	SINK+DEFL	176	0.688	0.0015	0.749
TQA SmolLM	LEN	2	0.524	0.0003	0.554
TQA SmolLM	LEX	–	0.820	0.0014	0.829
TQA SmolLM	LOGPROB	6	0.588	0.0010	0.592
TQA SmolLM	LLMCHECK	24	0.604	0.0004	0.684
TQA SmolLM	ROWSTAT	48	0.673	0.0014	0.741
TQA SmolLM	SINK	48	0.611	0.0019	0.628
TQA SmolLM	0D	48	0.583	0.0029	0.656
TQA SmolLM	LEGACY MSTPROXY	24	0.585	0.0011	0.674
TQA SmolLM	1D	96	0.543	0.0015	0.581
TQA SmolLM	DEFL	144	0.594	0.0023	0.650
TQA SmolLM	ANS	144	0.628	0.0041	0.630
TQA SmolLM	PH SINK	768	0.720	0.0008	0.775
TQA SmolLM	PH 0DTOT	768	0.819	0.0019	0.845
TQA SmolLM	PH 0D	1536	0.800	0.0012	0.818
TQA SmolLM	PH ENT	768	0.834	0.0007	0.841
TQA SmolLM	LOOKBACK	768	0.863	0.0017	0.864
TQA SmolLM	HIDDEN	2048	0.894	0.0022	0.892
TQA SmolLM	NONTOPO	2894	0.899	0.0015	0.903
TQA SmolLM	0D+1D	144	0.588	0.0019	0.643
TQA SmolLM	0D+SINK	96	0.641	0.0026	0.662
TQA SmolLM	HIDDEN+0D	2096	0.894	0.0020	0.891
TQA SmolLM	LEX+0D	–	0.827	0.0008	0.838
TQA SmolLM	LEX+DEFL	–	0.822	0.0001	0.845
TQA SmolLM	LEX+SINK	–	0.819	0.0015	0.826
TQA SmolLM	LOGPROB+0D	54	0.613	0.0009	0.630
TQA SmolLM	NONTOPO+0D	2942	0.899	0.0013	0.902
TQA SmolLM	NONTOPO+DEFL	3038	0.899	0.0012	0.900
TQA SmolLM	PH SINK+PH 0DTOT	1536	0.826	0.0023	0.854
TQA SmolLM	SINK+1D	144	0.629	0.0013	0.650
TQA SmolLM	SINK+ANS	192	0.646	0.0021	0.651
TQA SmolLM	SINK+DEFL	192	0.652	0.0022	0.681

Table 15: Complete out-of-fold ROC-AUC results across all settings and feature banks.

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA Mis7B	LEN	2	0.525	0.0002	0.571
TQA Mis7B	LEX	–	0.820	0.0008	0.828
TQA Mis7B	LOGPROB	6	0.641	0.0004	0.652
TQA Mis7B	LLMCHECK	32	0.652	0.0002	0.770
TQA Mis7B	ROWSTAT	64	0.694	0.0009	0.821
TQA Mis7B	SINK	64	0.685	0.0008	0.755
TQA Mis7B	0D	64	0.684	0.0008	0.755
TQA Mis7B	LEGACY MSTPROXY	32	0.646	0.0009	0.769
TQA Mis7B	1D	128	0.501	0.0000	0.499
TQA Mis7B	DEFL	192	0.710	0.0015	0.756
TQA Mis7B	ANS	192	0.677	0.0043	0.704
TQA Mis7B	PH SINK	1024	0.903	0.0009	0.938
TQA Mis7B	PH 0DTOT	1024	0.887	0.0020	0.928
TQA Mis7B	PH 0D	2048	0.893	0.0013	0.908
TQA Mis7B	PH ENT	1024	0.897	0.0010	0.929
TQA Mis7B	LOOKBACK	1024	0.933	0.0008	0.945
TQA Mis7B	HIDDEN	4096	0.956	0.0005	0.957
TQA Mis7B	NONTOPO	5222	0.959	0.0006	0.967
TQA Mis7B	0D+1D	192	0.685	0.0007	0.760
TQA Mis7B	0D+SINK	128	0.685	0.0007	0.754
TQA Mis7B	HIDDEN+0D	4160	0.956	0.0005	0.956
TQA Mis7B	LEX+0D	–	0.834	0.0013	0.859
TQA Mis7B	LEX+DEFL	–	0.823	0.0013	0.856
TQA Mis7B	LEX+SINK	–	0.833	0.0015	0.859
TQA Mis7B	LOGPROB+0D	70	0.722	0.0012	0.775
TQA Mis7B	NONTOPO+0D	5286	0.958	0.0006	0.967
TQA Mis7B	NONTOPO+DEFL	5414	0.959	0.0006	0.966
TQA Mis7B	PH SINK+PH 0DTOT	2048	0.903	0.0013	0.936
TQA Mis7B	SINK+1D	192	0.685	0.0008	0.758
TQA Mis7B	SINK+ANS	256	0.727	0.0023	0.763
TQA Mis7B	SINK+DEFL	256	0.740	0.0004	0.800
TQA Mis7B (chat)	LEN	2	0.525	0.0002	0.571
TQA Mis7B (chat)	LEX	–	0.820	0.0008	0.828
TQA Mis7B (chat)	LOGPROB	6	0.651	0.0014	0.668
TQA Mis7B (chat)	LLMCHECK	32	0.653	0.0001	0.769
TQA Mis7B (chat)	ROWSTAT	64	0.700	0.0008	0.823
TQA Mis7B (chat)	SINK	64	0.688	0.0010	0.754
TQA Mis7B (chat)	0D	64	0.688	0.0010	0.755
TQA Mis7B (chat)	LEGACY MSTPROXY	32	0.652	0.0009	0.783
TQA Mis7B (chat)	1D	128	0.501	0.0000	0.499
TQA Mis7B (chat)	DEFL	192	0.733	0.0013	0.781
TQA Mis7B (chat)	ANS	192	0.748	0.0012	0.767
TQA Mis7B (chat)	PH SINK	1024	0.907	0.0011	0.934
TQA Mis7B (chat)	PH 0DTOT	1024	0.890	0.0021	0.931
TQA Mis7B (chat)	PH 0D	2048	0.900	0.0017	0.919
TQA Mis7B (chat)	PH ENT	1024	0.901	0.0010	0.928
TQA Mis7B (chat)	LOOKBACK	1024	0.934	0.0008	0.942
TQA Mis7B (chat)	HIDDEN	4096	0.958	0.0004	0.957
TQA Mis7B (chat)	NONTOPO	5222	0.960	0.0006	0.965
TQA Mis7B (chat)	0D+1D	192	0.688	0.0009	0.754
TQA Mis7B (chat)	0D+SINK	128	0.689	0.0008	0.755
TQA Mis7B (chat)	HIDDEN+0D	4160	0.958	0.0004	0.957
TQA Mis7B (chat)	LEX+0D	–	0.834	0.0013	0.860
TQA Mis7B (chat)	LEX+DEFL	–	0.835	0.0020	0.864
TQA Mis7B (chat)	LEX+SINK	–	0.834	0.0015	0.859
TQA Mis7B (chat)	LOGPROB+0D	70	0.733	0.0021	0.798
TQA Mis7B (chat)	NONTOPO+0D	5286	0.960	0.0005	0.965
TQA Mis7B (chat)	NONTOPO+DEFL	5414	0.961	0.0006	0.966
TQA Mis7B (chat)	PH SINK+PH 0DTOT	2048	0.907	0.0012	0.938
TQA Mis7B (chat)	SINK+1D	192	0.688	0.0009	0.756

Table 15: Complete out-of-fold ROC-AUC results across all settings and feature banks.

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA Mis7B (chat)	SINK+ANS	256	0.767	0.0028	0.792
TQA Mis7B (chat)	SINK+DEFL	256	0.756	0.0018	0.803
TQA Qw7B	LEN	2	0.529	0.0003	0.567
TQA Qw7B	LEX	–	0.820	0.0008	0.828
TQA Qw7B	LOGPROB	6	0.674	0.0007	0.717
TQA Qw7B	LLMCHECK	28	0.667	0.0011	0.729
TQA Qw7B	ROWSTAT	56	0.736	0.0008	0.807
TQA Qw7B	SINK	56	0.749	0.0001	0.761
TQA Qw7B	0D	56	0.679	0.0010	0.774
TQA Qw7B	LEGACY MSTPROXY	28	0.678	0.0011	0.785
TQA Qw7B	1D	112	0.610	0.0005	0.634
TQA Qw7B	DEFL	168	0.697	0.0009	0.748
TQA Qw7B	ANS	168	0.741	0.0019	0.749
TQA Qw7B	PH SINK	784	0.821	0.0025	0.843
TQA Qw7B	PH 0DTOT	784	0.872	0.0016	0.909
TQA Qw7B	PH 0D	1568	0.850	0.0022	0.867
TQA Qw7B	PH ENT	784	0.887	0.0003	0.901
TQA Qw7B	LOOKBACK	784	0.922	0.0004	0.923
TQA Qw7B	HIDDEN	3584	0.933	0.0008	0.934
TQA Qw7B	NONTOPO	4458	0.950	0.0013	0.950
TQA Qw7B	0D+1D	168	0.701	0.0002	0.756
TQA Qw7B	0D+SINK	112	0.777	0.0004	0.794
TQA Qw7B	HIDDEN+0D	3640	0.933	0.0008	0.933
TQA Qw7B	LEX+0D	–	0.834	0.0008	0.858
TQA Qw7B	LEX+DEFL	–	0.828	0.0014	0.844
TQA Qw7B	LEX+SINK	–	0.848	0.0001	0.849
TQA Qw7B	LOGPROB+0D	62	0.727	0.0002	0.792
TQA Qw7B	NONTOPO+0D	4514	0.949	0.0013	0.949
TQA Qw7B	NONTOPO+DEFL	4626	0.949	0.0012	0.952
TQA Qw7B	PH SINK+PH 0DTOT	1568	0.881	0.0005	0.920
TQA Qw7B	SINK+1D	168	0.767	0.0023	0.780
TQA Qw7B	SINK+ANS	224	0.794	0.0010	0.802
TQA Qw7B	SINK+DEFL	224	0.788	0.0006	0.797
HE Qw3B	LEN	2	0.962	0.0001	0.970
HE Qw3B	LEX	–	0.981	0.0004	0.977
HE Qw3B	LOGPROB	6	0.993	0.0001	0.991
HE Qw3B	LLMCHECK	36	0.745	0.0004	0.956
HE Qw3B	ROWSTAT	72	0.829	0.0002	0.976
HE Qw3B	SINK	72	0.829	0.0008	0.930
HE Qw3B	0D	72	0.811	0.0006	0.927
HE Qw3B	LEGACY MSTPROXY	36	0.731	0.0002	0.960
HE Qw3B	1D	144	0.710	0.0013	0.891
HE Qw3B	DEFL	216	0.834	0.0009	0.959
HE Qw3B	ANS	216	0.978	0.0009	0.980
HE Qw3B	PH SINK	576	0.958	0.0007	0.999
HE Qw3B	PH 0DTOT	576	0.958	0.0005	0.995
HE Qw3B	PH 0D	1152	0.971	0.0006	0.988
HE Qw3B	PH ENT	576	0.964	0.0005	0.993
HE Qw3B	LOOKBACK	576	0.998	0.0001	0.998
HE Qw3B	HIDDEN	2048	1.000	0.0000	1.000
HE Qw3B	NONTOPO	2738	1.000	0.0002	1.000
HE Qw3B	0D+1D	216	0.826	0.0013	0.945
HE Qw3B	0D+SINK	144	0.857	0.0010	0.948
HE Qw3B	HIDDEN+0D	2120	1.000	0.0000	1.000
HE Qw3B	LEX+0D	–	0.979	0.0001	0.983
HE Qw3B	LEX+DEFL	–	0.971	0.0003	0.975
HE Qw3B	LEX+SINK	–	0.980	0.0003	0.981
HE Qw3B	LOGPROB+0D	78	0.993	0.0003	0.992
HE Qw3B	NONTOPO+0D	2810	1.000	0.0002	1.000
HE Qw3B	NONTOPO+DEFL	2954	1.000	0.0002	1.000

Table 15: Complete out-of-fold ROC-AUC results across all settings and feature banks.

Setting	Bank	dim	AUC	seed SD	pair acc.
HE Qw3B	PH SINK+PH 0DTOT	1152	0.962	0.0006	0.997
HE Qw3B	SINK+1D	216	0.843	0.0011	0.938
HE Qw3B	SINK+ANS	288	0.978	0.0003	0.982
HE Qw3B	SINK+DEFL	288	0.887	0.0020	0.957
HE Qw1.5B	LEN	2	0.967	0.0002	0.976
HE Qw1.5B	LEX	–	0.979	0.0004	0.980
HE Qw1.5B	LOGPROB	6	0.995	0.0002	0.988
HE Qw1.5B	LLMCHECK	28	0.791	0.0005	0.978
HE Qw1.5B	ROWSTAT	56	0.788	0.0017	0.984
HE Qw1.5B	SINK	56	0.820	0.0015	0.934
HE Qw1.5B	0D	56	0.789	0.0014	0.918
HE Qw1.5B	LEGACY MSTPROXY	28	0.719	0.0003	0.976
HE Qw1.5B	1D	112	0.747	0.0006	0.878
HE Qw1.5B	DEFL	168	0.825	0.0021	0.952
HE Qw1.5B	ANS	168	0.972	0.0006	0.978
HE Qw1.5B	PH SINK	336	0.922	0.0007	0.998
HE Qw1.5B	PH 0DTOT	336	0.921	0.0007	0.996
HE Qw1.5B	PH 0D	672	0.949	0.0027	0.974
HE Qw1.5B	PH ENT	336	0.933	0.0011	0.996
HE Qw1.5B	LOOKBACK	336	0.999	0.0002	0.998
HE Qw1.5B	HIDDEN	1536	1.000	0.0001	1.000
HE Qw1.5B	NONTOPO	1962	1.000	0.0002	1.000
HE Qw1.5B	0D+1D	168	0.815	0.0032	0.922
HE Qw1.5B	0D+SINK	112	0.832	0.0015	0.924
HE Qw1.5B	HIDDEN+0D	1592	1.000	0.0001	1.000
HE Qw1.5B	LEX+0D	–	0.971	0.0004	0.986
HE Qw1.5B	LEX+DEFL	–	0.956	0.0002	0.986
HE Qw1.5B	LEX+SINK	–	0.968	0.0005	0.984
HE Qw1.5B	LOGPROB+0D	62	0.994	0.0002	0.990
HE Qw1.5B	NONTOPO+0D	2018	1.000	0.0000	1.000
HE Qw1.5B	NONTOPO+DEFL	2130	1.000	0.0001	1.000
HE Qw1.5B	PH SINK+PH 0DTOT	672	0.931	0.0024	0.998
HE Qw1.5B	SINK+1D	168	0.826	0.0035	0.940
HE Qw1.5B	SINK+ANS	224	0.975	0.0015	0.990
HE Qw1.5B	SINK+DEFL	224	0.872	0.0028	0.962
TrQA Qw3B	LEN	2	0.569	0.0014	–
TrQA Qw3B	LEX	–	0.596	0.0039	–
TrQA Qw3B	LOGPROB	6	0.839	0.0002	–
TrQA Qw3B	LLMCHECK	36	0.688	0.0012	–
TrQA Qw3B	ROWSTAT	72	0.724	0.0046	–
TrQA Qw3B	SINK	72	0.717	0.0038	–
TrQA Qw3B	0D	72	0.732	0.0038	–
TrQA Qw3B	LEGACY MSTPROXY	36	0.691	0.0033	–
TrQA Qw3B	1D	144	0.601	0.0031	–
TrQA Qw3B	DEFL	216	0.684	0.0043	–
TrQA Qw3B	ANS	216	0.719	0.0020	–
TrQA Qw3B	PH SINK	576	0.820	0.0021	–
TrQA Qw3B	PH 0DTOT	576	0.807	0.0012	–
TrQA Qw3B	PH 0D	1152	0.807	0.0017	–
TrQA Qw3B	PH ENT	576	0.810	0.0007	–
TrQA Qw3B	LOOKBACK	576	0.861	0.0013	–
TrQA Qw3B	HIDDEN	2048	0.841	0.0042	–
TrQA Qw3B	NONTOPO	2738	0.870	0.0028	–
TrQA Qw3B	0D+1D	216	0.729	0.0022	–
TrQA Qw3B	0D+SINK	144	0.738	0.0050	–
TrQA Qw3B	HIDDEN+0D	2120	0.843	0.0040	–
TrQA Qw3B	LEX+0D	–	0.733	0.0011	–
TrQA Qw3B	LEX+DEFL	–	0.690	0.0031	–
TrQA Qw3B	LEX+SINK	–	0.724	0.0005	–
TrQA Qw3B	LOGPROB+0D	78	0.843	0.0022	–

Table 15: Complete out-of-fold ROC-AUC results across all settings and feature banks.

Setting	Bank	dim	AUC	seed SD	pair acc.
TrQA Qw3B	NONTOPO+0D	2810	0.870	0.0027	—
TrQA Qw3B	NONTOPO+DEFL	2954	0.870	0.0029	—
TrQA Qw3B	PH SINK+PH 0DTOT	1152	0.824	0.0017	—
TrQA Qw3B	SINK+1D	216	0.718	0.0035	—
TrQA Qw3B	SINK+ANS	288	0.752	0.0010	—
TrQA Qw3B	SINK+DEFL	288	0.719	0.0038	—
TrQA Qw1.5B	LEN	2	0.588	0.0012	—
TrQA Qw1.5B	LEX	—	0.609	0.0033	—
TrQA Qw1.5B	LOGPROB	6	0.823	0.0006	—
TrQA Qw1.5B	LLMCHECK	28	0.676	0.0029	—
TrQA Qw1.5B	ROWSTAT	56	0.720	0.0008	—
TrQA Qw1.5B	SINK	56	0.707	0.0000	—
TrQA Qw1.5B	0D	56	0.716	0.0004	—
TrQA Qw1.5B	LEGACY MSTPROXY	28	0.685	0.0014	—
TrQA Qw1.5B	1D	112	0.574	0.0024	—
TrQA Qw1.5B	DEFL	168	0.655	0.0016	—
TrQA Qw1.5B	ANS	168	0.675	0.0035	—
TrQA Qw1.5B	PH SINK	336	0.767	0.0034	—
TrQA Qw1.5B	PH 0DTOT	336	0.759	0.0051	—
TrQA Qw1.5B	PH 0D	672	0.768	0.0013	—
TrQA Qw1.5B	PH ENT	336	0.768	0.0025	—
TrQA Qw1.5B	LOOKBACK	336	0.824	0.0017	—
TrQA Qw1.5B	HIDDEN	1536	0.855	0.0031	—
TrQA Qw1.5B	NONTOPO	1962	0.873	0.0023	—
TrQA Qw1.5B	0D+1D	168	0.714	0.0015	—
TrQA Qw1.5B	0D+SINK	112	0.717	0.0006	—
TrQA Qw1.5B	HIDDEN+0D	1592	0.855	0.0032	—
TrQA Qw1.5B	LEX+0D	—	0.721	0.0031	—
TrQA Qw1.5B	LEX+DEFL	—	0.672	0.0047	—
TrQA Qw1.5B	LEX+SINK	—	0.715	0.0036	—
TrQA Qw1.5B	LOGPROB+0D	62	0.828	0.0007	—
TrQA Qw1.5B	NONTOPO+0D	2018	0.873	0.0025	—
TrQA Qw1.5B	NONTOPO+DEFL	2130	0.872	0.0024	—
TrQA Qw1.5B	PH SINK+PH 0DTOT	672	0.772	0.0022	—
TrQA Qw1.5B	SINK+1D	168	0.709	0.0028	—
TrQA Qw1.5B	SINK+ANS	224	0.726	0.0059	—
TrQA Qw1.5B	SINK+DEFL	224	0.719	0.0023	—
TrQA Phi-3	LEN	2	0.581	0.0022	—
TrQA Phi-3	LEX	—	0.635	0.0083	—
TrQA Phi-3	LOGPROB	6	0.877	0.0002	—
TrQA Phi-3	LLMCHECK	32	0.676	0.0018	—
TrQA Phi-3	ROWSTAT	64	0.728	0.0006	—
TrQA Phi-3	SINK	64	0.707	0.0018	—
TrQA Phi-3	0D	64	0.710	0.0022	—
TrQA Phi-3	LEGACY MSTPROXY	32	0.669	0.0002	—
TrQA Phi-3	1D	128	0.546	0.0031	—
TrQA Phi-3	DEFL	192	0.698	0.0021	—
TrQA Phi-3	ANS	192	0.710	0.0037	—
TrQA Phi-3	PH SINK	1024	0.826	0.0010	—
TrQA Phi-3	PH 0DTOT	1024	0.810	0.0016	—
TrQA Phi-3	PH 0D	2048	0.807	0.0024	—
TrQA Phi-3	PH ENT	1024	0.817	0.0021	—
TrQA Phi-3	LOOKBACK	1024	0.840	0.0016	—
TrQA Phi-3	HIDDEN	3072	0.867	0.0035	—
TrQA Phi-3	NONTOPO	4198	0.883	0.0030	—
TrQA Phi-3	0D+1D	192	0.705	0.0011	—
TrQA Phi-3	0D+SINK	128	0.713	0.0024	—
TrQA Phi-3	HIDDEN+0D	3136	0.867	0.0037	—
TrQA Phi-3	LEX+0D	—	0.721	0.0031	—
TrQA Phi-3	LEX+DEFL	—	0.712	0.0022	—

Table 15: Complete out-of-fold ROC-AUC results across all settings and feature banks.

Setting	Bank	dim	AUC	seed SD	pair acc.
TrQA Phi-3	LEX+SINK	–	0.718	0.0044	–
TrQA Phi-3	LOGPROB+0D	70	0.878	0.0013	–
TrQA Phi-3	NONTOPO+0D	4262	0.883	0.0030	–
TrQA Phi-3	NONTOPO+DEFL	4390	0.885	0.0030	–
TrQA Phi-3	PH SINK+PH 0DTOT	2048	0.828	0.0009	–
TrQA Phi-3	SINK+1D	192	0.699	0.0012	–
TrQA Phi-3	SINK+ANS	256	0.736	0.0046	–
TrQA Phi-3	SINK+DEFL	256	0.736	0.0016	–
TrQA TinyLl.	LEN	2	0.631	0.0005	–
TrQA TinyLl.	LEX	–	0.740	0.0003	–
TrQA TinyLl.	LOGPROB	6	0.724	0.0006	–
TrQA TinyLl.	LLMCHECK	22	0.688	0.0010	–
TrQA TinyLl.	ROWSTAT	44	0.752	0.0001	–
TrQA TinyLl.	SINK	44	0.720	0.0020	–
TrQA TinyLl.	0D	44	0.720	0.0020	–
TrQA TinyLl.	LEGACY MSTPROXY	22	0.729	0.0006	–
TrQA TinyLl.	1D	88	0.659	0.0007	–
TrQA TinyLl.	DEFL	132	0.726	0.0024	–
TrQA TinyLl.	ANS	132	0.701	0.0021	–
TrQA TinyLl.	PH SINK	704	0.814	0.0016	–
TrQA TinyLl.	PH 0DTOT	704	0.792	0.0018	–
TrQA TinyLl.	PH 0D	1408	0.775	0.0027	–
TrQA TinyLl.	PH ENT	704	0.807	0.0003	–
TrQA TinyLl.	LOOKBACK	704	0.823	0.0024	–
TrQA TinyLl.	HIDDEN	2048	0.827	0.0028	–
TrQA TinyLl.	NONTOPO	2824	0.842	0.0031	–
TrQA TinyLl.	0D+1D	132	0.718	0.0002	–
TrQA TinyLl.	0D+SINK	88	0.724	0.0015	–
TrQA TinyLl.	HIDDEN+0D	2092	0.828	0.0027	–
TrQA TinyLl.	LEX+0D	–	0.764	0.0010	–
TrQA TinyLl.	LEX+DEFL	–	0.757	0.0032	–
TrQA TinyLl.	LEX+SINK	–	0.762	0.0021	–
TrQA TinyLl.	LOGPROB+0D	50	0.762	0.0021	–
TrQA TinyLl.	NONTOPO+0D	2868	0.842	0.0029	–
TrQA TinyLl.	NONTOPO+DEFL	2956	0.842	0.0032	–
TrQA TinyLl.	PH SINK+PH 0DTOT	1408	0.813	0.0016	–
TrQA TinyLl.	SINK+1D	132	0.719	0.0013	–
TrQA TinyLl.	SINK+ANS	176	0.725	0.0030	–
TrQA TinyLl.	SINK+DEFL	176	0.734	0.0032	–
TrQA Mis7B	LEN	2	0.635	0.0017	–
TrQA Mis7B	LEX	–	0.677	0.0050	–
TrQA Mis7B	LOGPROB	6	0.728	0.0007	–
TrQA Mis7B	LLMCHECK	32	0.686	0.0019	–
TrQA Mis7B	ROWSTAT	64	0.731	0.0016	–
TrQA Mis7B	SINK	64	0.696	0.0017	–
TrQA Mis7B	0D	64	0.699	0.0003	–
TrQA Mis7B	LEGACY MSTPROXY	32	0.693	0.0017	–
TrQA Mis7B	1D	128	0.611	0.0028	–
TrQA Mis7B	DEFL	192	0.721	0.0024	–
TrQA Mis7B	ANS	192	0.662	0.0027	–
TrQA Mis7B	PH SINK	1024	0.812	0.0024	–
TrQA Mis7B	PH 0DTOT	1024	0.808	0.0037	–
TrQA Mis7B	PH 0D	2048	0.775	0.0013	–
TrQA Mis7B	PH ENT	1024	0.812	0.0027	–
TrQA Mis7B	LOOKBACK	1024	0.829	0.0041	–
TrQA Mis7B	HIDDEN	4096	0.833	0.0008	–
TrQA Mis7B	NONTOPO	5222	0.841	0.0011	–
TrQA Mis7B	0D+1D	192	0.692	0.0035	–
TrQA Mis7B	0D+SINK	128	0.698	0.0018	–
TrQA Mis7B	HIDDEN+0D	4160	0.834	0.0011	–

Table 15: Complete out-of-fold ROC-AUC results across all settings and feature banks.

Setting	Bank	dim	AUC	seed SD	pair acc.
TrQA Mis7B	LEX+0D	—	0.723	0.0030	—
TrQA Mis7B	LEX+DEFL	—	0.744	0.0057	—
TrQA Mis7B	LEX+SINK	—	0.721	0.0027	—
TrQA Mis7B	LOGPROB+0D	70	0.755	0.0016	—
TrQA Mis7B	NONTOPO+0D	5286	0.842	0.0015	—
TrQA Mis7B	NONTOPO+DEFL	5414	0.841	0.0010	—
TrQA Mis7B	PH SINK+PH 0DTOT	2048	0.814	0.0022	—
TrQA Mis7B	SINK+1D	192	0.686	0.0042	—
TrQA Mis7B	SINK+ANS	256	0.693	0.0048	—
TrQA Mis7B	SINK+DEFL	256	0.739	0.0019	—
TrQA Qw7B	LEN	2	0.595	0.0004	—
TrQA Qw7B	LEX	—	0.594	0.0029	—
TrQA Qw7B	LOGPROB	6	0.847	0.0003	—
TrQA Qw7B	LLMCHECK	28	0.725	0.0015	—
TrQA Qw7B	ROWSTAT	56	0.734	0.0016	—
TrQA Qw7B	SINK	56	0.714	0.0023	—
TrQA Qw7B	0D	56	0.668	0.0006	—
TrQA Qw7B	LEGACY MSTPROXY	28	0.672	0.0005	—
TrQA Qw7B	1D	112	0.582	0.0022	—
TrQA Qw7B	DEFL	168	0.672	0.0016	—
TrQA Qw7B	ANS	168	0.755	0.0026	—
TrQA Qw7B	PH SINK	784	0.801	0.0009	—
TrQA Qw7B	PH 0DTOT	784	0.822	0.0011	—
TrQA Qw7B	PH 0D	1568	0.798	0.0029	—
TrQA Qw7B	PH ENT	784	0.841	0.0021	—
TrQA Qw7B	LOOKBACK	784	0.879	0.0010	—
TrQA Qw7B	HIDDEN	3584	0.854	0.0007	—
TrQA Qw7B	NONTOPO	4458	0.881	0.0008	—
TrQA Qw7B	0D+1D	168	0.668	0.0021	—
TrQA Qw7B	0D+SINK	112	0.735	0.0008	—
TrQA Qw7B	HIDDEN+0D	3640	0.853	0.0008	—
TrQA Qw7B	LEX+0D	—	0.678	0.0043	—
TrQA Qw7B	LEX+DEFL	—	0.678	0.0022	—
TrQA Qw7B	LEX+SINK	—	0.728	0.0007	—
TrQA Qw7B	LOGPROB+0D	62	0.842	0.0012	—
TrQA Qw7B	NONTOPO+0D	4514	0.881	0.0007	—
TrQA Qw7B	NONTOPO+DEFL	4626	0.881	0.0007	—
TrQA Qw7B	PH SINK+PH 0DTOT	1568	0.832	0.0010	—
TrQA Qw7B	SINK+1D	168	0.718	0.0023	—
TrQA Qw7B	SINK+ANS	224	0.781	0.0035	—
TrQA Qw7B	SINK+DEFL	224	0.734	0.0021	—

Table 16: Complete paired difference tests and equivalence tests across all settings and comparisons.

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\equiv_{.015}$	power
TQA Qw3B	0D $\rightarrow$ SINK	0.736	0.737	+0.001	[-0.006, +0.009]	yes	0.90
TQA Qw3B	0D $\rightarrow$ LEGACY_MSTPROXY	0.736	0.713	-0.023	[-0.034, -0.012]	no	0.42
TQA Qw3B	SINK $\rightarrow$ 0D+SINK	0.737	0.741	+0.003	[-0.003, +0.010]	yes	0.97
TQA Qw3B	SINK $\rightarrow$ SINK+DEFL	0.737	0.766	+0.029	[+0.017, +0.041]	no	0.30
TQA Qw3B	0D $\rightarrow$ 0D+1D	0.736	0.769	+0.033	[+0.022, +0.045]	no	0.38
TQA Qw3B	SINK $\rightarrow$ SINK+1D	0.737	0.766	+0.028	[+0.017, +0.039]	no	0.46
TQA Qw3B	SINK $\rightarrow$ SINK+ANS	0.737	0.785	+0.047	[+0.033, +0.062]	no	0.03
TQA Qw3B	PH_0DTOT $\rightarrow$ PH_SINK	0.842	0.851	+0.009	[+0.002, +0.016]	no	0.93
TQA Qw3B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.851	0.860	+0.009	[+0.005, +0.012]	yes	1.00
TQA Qw3B	PH_0D $\rightarrow$ PH_ENT	0.853	0.857	+0.004	[-0.006, +0.014]	yes	0.57
TQA Qw3B	0D $\rightarrow$ LOGPROB	0.736	0.630	-0.106	[-0.129, -0.083]	no	0.00
TQA Qw3B	0D $\rightarrow$ HIDDEN	0.736	0.913	+0.177	[+0.163, +0.192]	no	0.03
TQA Qw3B	0D $\rightarrow$ LOOKBACK	0.736	0.897	+0.161	[+0.146, +0.175]	no	0.02
TQA Qw3B	0D $\rightarrow$ LLMCHECK	0.736	0.693	-0.043	[-0.059, -0.026]	no	0.00
TQA Qw3B	0D $\rightarrow$ LEX	0.736	0.820	+0.084	[+0.066, +0.102]	no	0.00
TQA Qw3B	0D $\rightarrow$ ROWSTAT	0.736	0.750	+0.014	[+0.002, +0.026]	no	0.30
TQA Qw3B	0D $\rightarrow$ LEN	0.736	0.529	-0.207	[-0.229, -0.185]	no	0.00
TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO+0D	0.926	0.926	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.926	0.928	+0.001	[-0.000, +0.003]	yes	1.00
TQA Qw3B	LEX $\rightarrow$ LEX+0D	0.820	0.840	+0.020	[+0.011, +0.030]	no	0.63

Table 16: Complete paired difference tests and equivalence tests across all settings and comparisons.

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\equiv_{.015}$	power
TQA Qw3B	LEX $\rightarrow$ LEX+SINK	0.820	0.843	+0.022	[+0.013, +0.033]	no	0.59
TQA Qw3B	LEX $\rightarrow$ LEX+DEFL	0.820	0.830	+0.010	[-0.003, +0.023]	no	0.19
TQA Qw3B	LOGPROB $\rightarrow$ LOGPROB+0D	0.630	0.776	+0.146	[+0.127, +0.165]	no	0.00
TQA Qw3B	HIDDEN $\rightarrow$ HIDDEN+0D	0.913	0.913	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.926	0.926	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.926	0.926	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.926	0.926	+0.000	[-0.000, +0.001]	yes	1.00
TQA Qw3B	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.913	0.914	+0.001	[-0.000, +0.002]	yes	1.00
TQA Qw3B	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.630	0.757	+0.128	[+0.109, +0.147]	no	0.00
TQA Qw3B	SINK $\rightarrow$ SINK(+)0D	0.737	0.739	+0.002	[-0.001, +0.006]	yes	1.00
TQA Qw1.5B	0D $\rightarrow$ SINK	0.726	0.723	-0.003	[-0.012, +0.006]	yes	0.68
TQA Qw1.5B	0D $\rightarrow$ LEGACY_MSTPROXY	0.726	0.698	-0.028	[-0.040, -0.016]	no	0.36
TQA Qw1.5B	SINK $\rightarrow$ 0D+SINK	0.723	0.735	+0.012	[+0.005, +0.020]	no	0.91
TQA Qw1.5B	SINK $\rightarrow$ SINK+DEFL	0.723	0.757	+0.034	[+0.022, +0.045]	no	0.35
TQA Qw1.5B	0D $\rightarrow$ 0D+1D	0.726	0.718	-0.008	[-0.014, -0.002]	yes	0.99
TQA Qw1.5B	SINK $\rightarrow$ SINK+1D	0.723	0.715	-0.008	[-0.014, -0.002]	yes	0.99
TQA Qw1.5B	SINK $\rightarrow$ SINK+ANS	0.723	0.761	+0.038	[+0.024, +0.051]	no	0.11
TQA Qw1.5B	PH_0DTOT $\rightarrow$ PH_SINK	0.819	0.830	+0.011	[+0.003, +0.018]	no	0.90
TQA Qw1.5B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.830	0.841	+0.011	[+0.007, +0.014]	yes	1.00
TQA Qw1.5B	PH_0D $\rightarrow$ PH_ENT	0.838	0.833	-0.004	[-0.016, +0.007]	no	0.44
TQA Qw1.5B	0D $\rightarrow$ LOGPROB	0.726	0.638	-0.088	[-0.108, -0.067]	no	0.00
TQA Qw1.5B	0D $\rightarrow$ HIDDEN	0.726	0.916	+0.190	[+0.175, +0.204]	no	0.05
TQA Qw1.5B	0D $\rightarrow$ LOOKBACK	0.726	0.851	+0.125	[+0.109, +0.140]	no	0.00
TQA Qw1.5B	0D $\rightarrow$ LLMCHECK	0.726	0.666	-0.060	[-0.076, -0.044]	no	0.00
TQA Qw1.5B	0D $\rightarrow$ LEX	0.726	0.820	+0.094	[+0.076, +0.112]	no	0.00
TQA Qw1.5B	0D $\rightarrow$ ROWSTAT	0.726	0.738	+0.012	[-0.000, +0.025]	no	0.27
TQA Qw1.5B	0D $\rightarrow$ LEN	0.726	0.529	-0.197	[-0.218, -0.176]	no	0.00
TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+0D	0.917	0.917	+0.000	[-0.000, +0.001]	yes	1.00
TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.917	0.917	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw1.5B	LEX $\rightarrow$ LEX+0D	0.820	0.835	+0.015	[+0.006, +0.024]	no	0.71
TQA Qw1.5B	LEX $\rightarrow$ LEX+SINK	0.820	0.833	+0.013	[+0.004, +0.022]	no	0.73
TQA Qw1.5B	LEX $\rightarrow$ LEX+DEFL	0.820	0.825	+0.005	[-0.006, +0.016]	no	0.45
TQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB+0D	0.638	0.740	+0.101	[+0.086, +0.116]	no	0.00
TQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN+0D	0.916	0.916	+0.001	[-0.000, +0.002]	yes	1.00
TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.916	0.916	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.916	0.916	-0.000	[-0.001, +0.001]	yes	1.00
TQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.916	0.916	-0.000	[-0.001, +0.000]	yes	1.00
TQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.915	0.915	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.638	0.737	+0.100	[+0.084, +0.115]	no	0.00
TQA Qw1.5B	SINK $\rightarrow$ SINK(+)0D	0.722	0.730	+0.007	[+0.002, +0.012]	yes	1.00
TQA Phi-3	0D $\rightarrow$ SINK	0.646	0.651	+0.005	[-0.003, +0.012]	yes	0.90
TQA Phi-3	0D $\rightarrow$ LEGACY_MSTPROXY	0.646	0.613	-0.033	[-0.046, -0.020]	no	0.20
TQA Phi-3	SINK $\rightarrow$ 0D+SINK	0.651	0.657	+0.006	[-0.000, +0.012]	yes	0.99
TQA Phi-3	SINK $\rightarrow$ SINK+DEFL	0.651	0.684	+0.033	[+0.020, +0.048]	no	0.10
TQA Phi-3	0D $\rightarrow$ 0D+1D	0.646	0.643	-0.003	[-0.006, +0.001]	yes	1.00
TQA Phi-3	SINK $\rightarrow$ SINK+1D	0.651	0.651	+0.000	[-0.004, +0.005]	yes	1.00
TQA Phi-3	SINK $\rightarrow$ SINK+ANS	0.651	0.675	+0.024	[+0.007, +0.039]	no	0.00
TQA Phi-3	PH_0DTOT $\rightarrow$ PH_SINK	0.845	0.871	+0.026	[+0.020, +0.031]	no	1.00
TQA Phi-3	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.871	0.871	-0.000	[-0.001, +0.001]	yes	1.00
TQA Phi-3	PH_0D $\rightarrow$ PH_ENT	0.868	0.864	-0.004	[-0.012, +0.004]	yes	0.87
TQA Phi-3	0D $\rightarrow$ LOGPROB	0.646	0.578	-0.068	[-0.092, -0.044]	no	0.00
TQA Phi-3	0D $\rightarrow$ HIDDEN	0.646	0.933	+0.287	[+0.272, +0.302]	no	0.00
TQA Phi-3	0D $\rightarrow$ LOOKBACK	0.646	0.892	+0.246	[+0.230, +0.262]	no	0.00
TQA Phi-3	0D $\rightarrow$ LLMCHECK	0.646	0.653	+0.007	[-0.010, +0.024]	no	0.00
TQA Phi-3	0D $\rightarrow$ LEX	0.646	0.820	+0.174	[+0.154, +0.193]	no	0.00
TQA Phi-3	0D $\rightarrow$ ROWSTAT	0.646	0.661	+0.015	[+0.002, +0.029]	no	0.12
TQA Phi-3	0D $\rightarrow$ LEN	0.646	0.522	-0.124	[-0.145, -0.102]	no	0.00
TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+0D	0.936	0.936	+0.000	[-0.000, +0.001]	yes	1.00
TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.936	0.937	+0.001	[-0.000, +0.002]	yes	1.00
TQA Phi-3	LEX $\rightarrow$ LEX+0D	0.820	0.821	+0.001	[-0.007, +0.009]	yes	0.85
TQA Phi-3	LEX $\rightarrow$ LEX+SINK	0.820	0.820	-0.000	[-0.008, +0.007]	yes	0.89
TQA Phi-3	LEX $\rightarrow$ LEX+DEFL	0.820	0.802	-0.018	[-0.028, -0.007]	no	0.52
TQA Phi-3	LOGPROB $\rightarrow$ LOGPROB+0D	0.578	0.655	+0.077	[+0.058, +0.096]	no	0.00
TQA Phi-3	HIDDEN $\rightarrow$ HIDDEN+0D	0.933	0.933	+0.000	[-0.001, +0.001]	yes	1.00
TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.936	0.936	-0.000	[-0.000, -0.000]	yes	1.00
TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.936	0.936	-0.000	[-0.000, -0.000]	yes	1.00
TQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.936	0.936	-0.000	[-0.000, +0.000]	yes	1.00
TQA Phi-3	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.932	0.932	-0.000	[-0.000, -0.000]	yes	1.00
TQA Phi-3	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.577	0.650	+0.073	[+0.055, +0.091]	no	0.00
TQA Phi-3	SINK $\rightarrow$ SINK(+)0D	0.650	0.651	+0.000	[-0.003, +0.003]	yes	1.00
TQA TinyLl	0D $\rightarrow$ SINK	0.654	0.643	-0.010	[-0.021, +0.000]	no	0.51
TQA TinyLl	0D $\rightarrow$ LEGACY_MSTPROXY	0.654	0.627	-0.027	[-0.037, -0.016]	no	0.46
TQA TinyLl	SINK $\rightarrow$ 0D+SINK	0.643	0.663	+0.020	[+0.011, +0.029]	no	0.73
TQA TinyLl	SINK $\rightarrow$ SINK+DEFL	0.643	0.688	+0.045	[+0.032, +0.057]	no	0.28
TQA TinyLl	0D $\rightarrow$ 0D+1D	0.654	0.660	+0.007	[-0.000, +0.013]	yes	0.95
TQA TinyLl	SINK $\rightarrow$ SINK+1D	0.643	0.651	+0.008	[+0.001, +0.015]	no	0.93
TQA TinyLl	SINK $\rightarrow$ SINK+ANS	0.643	0.685	+0.041	[+0.025, +0.058]	no	0.00
TQA TinyLl	PH_0DTOT $\rightarrow$ PH_SINK	0.804	0.808	+0.004	[-0.002, +0.011]	yes	0.97
TQA TinyLl	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.808	0.810	+0.003	[+0.000, +0.005]	yes	1.00
TQA TinyLl	PH_0D $\rightarrow$ PH_ENT	0.806	0.817	+0.011	[+0.000, +0.022]	no	0.50
TQA TinyLl	0D $\rightarrow$ LOGPROB	0.654	0.637	-0.017	[-0.039, +0.004]	no	0.00
TQA TinyLl	0D $\rightarrow$ HIDDEN	0.654	0.883	+0.229	[+0.214, +0.244]	no	0.00
TQA TinyLl	0D $\rightarrow$ LOOKBACK	0.654	0.850	+0.197	[+0.182, +0.212]	no	0.00
TQA TinyLl	0D $\rightarrow$ LLMCHECK	0.654	0.614	-0.039	[-0.055, -0.024]	no	0.00
TQA TinyLl	0D $\rightarrow$ LEX	0.654	0.820	+0.166	[+0.149, +0.184]	no	0.00
TQA TinyLl	0D $\rightarrow$ ROWSTAT	0.654	0.661	+0.007	[-0.004, +0.019]	no	0.37
TQA TinyLl	0D $\rightarrow$ LEN	0.654	0.523	-0.131	[-0.152, -0.110]	no	0.00
TQA TinyLl	NONTOPO $\rightarrow$ NONTOPO+0D	0.885	0.885	+0.000	[-0.000, +0.000]	yes	1.00
TQA TinyLl	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.885	0.884	-0.001	[-0.002, +0.000]	yes	1.00

Table 16: Complete paired difference tests and equivalence tests across all settings and comparisons.

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\equiv_{.015}$	power
TQA TinyLL	LEX $\rightarrow$ LEX+0D	0.820	0.830	+0.010	[+0.003, +0.016]	no	0.97
TQA TinyLL	LEX $\rightarrow$ LEX+SINK	0.820	0.831	+0.011	[+0.005, +0.017]	no	0.98
TQA TinyLL	LEX $\rightarrow$ LEX+DEFL	0.820	0.826	+0.005	[-0.003, +0.014]	yes	0.75
TQA TinyLL	LOGPROB $\rightarrow$ LOGPROB+0D	0.637	0.683	+0.046	[+0.033, +0.060]	no	0.13
TQA TinyLL	HIDDEN $\rightarrow$ HIDDEN+0D	0.883	0.883	-0.000	[-0.001, +0.000]	yes	1.00
TQA TinyLL	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.884	0.884	-0.000	[-0.000, -0.000]	yes	1.00
TQA TinyLL	NONTOPO $\rightarrow$ NONTOPO(+0D)+DEFL	0.884	0.884	-0.000	[-0.001, +0.000]	yes	1.00
TQA TinyLL	NONTOPO $\rightarrow$ NONTOPO(+0D)+SINK	0.884	0.884	-0.000	[-0.000, -0.000]	yes	1.00
TQA TinyLL	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.883	0.882	-0.000	[-0.000, +0.000]	yes	1.00
TQA TinyLL	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.636	0.678	+0.043	[+0.030, +0.055]	no	0.25
TQA TinyLL	SINK $\rightarrow$ SINK(+0D)	0.643	0.657	+0.014	[+0.007, +0.021]	no	0.96
TQA SmoLLM	0D $\rightarrow$ SINK	0.583	0.611	+0.028	[+0.009, +0.048]	no	0.00
TQA SmoLLM	0D $\rightarrow$ LEGACY_MSTPROXY	0.583	0.585	+0.002	[-0.003, +0.006]	yes	1.00
TQA SmoLLM	SINK $\rightarrow$ 0D+SINK	0.611	0.641	+0.030	[+0.021, +0.039]	no	0.66
TQA SmoLLM	SINK $\rightarrow$ SINK+DEFL	0.611	0.652	+0.041	[+0.030, +0.053]	no	0.34
TQA SmoLLM	0D $\rightarrow$ 0D+1D	0.583	0.588	+0.005	[-0.004, +0.014]	yes	0.68
TQA SmoLLM	SINK $\rightarrow$ SINK+1D	0.611	0.629	+0.018	[+0.008, +0.028]	no	0.62
TQA SmoLLM	SINK $\rightarrow$ SINK+ANS	0.611	0.646	+0.036	[+0.018, +0.053]	no	0.00
TQA SmoLLM	PH_0DTOT $\rightarrow$ PH_SINK	0.819	0.720	-0.099	[-0.113, -0.086]	no	0.15
TQA SmoLLM	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.720	0.826	+0.106	[+0.094, +0.118]	no	0.29
TQA SmoLLM	PH_0D $\rightarrow$ PH_ENT	0.800	0.834	+0.034	[+0.024, +0.044]	no	0.60
TQA SmoLLM	0D $\rightarrow$ LOGPROB	0.583	0.588	+0.005	[-0.017, +0.027]	no	0.00
TQA SmoLLM	0D $\rightarrow$ HIDDEN	0.583	0.894	+0.311	[+0.296, +0.326]	no	0.01
TQA SmoLLM	0D $\rightarrow$ LOOKBACK	0.583	0.863	+0.280	[+0.264, +0.295]	no	0.00
TQA SmoLLM	0D $\rightarrow$ LLMCHECK	0.583	0.604	+0.021	[+0.006, +0.037]	no	0.00
TQA SmoLLM	0D $\rightarrow$ LEX	0.583	0.820	+0.237	[+0.218, +0.256]	no	0.00
TQA SmoLLM	0D $\rightarrow$ ROWSTAT	0.583	0.673	+0.090	[+0.079, +0.102]	no	0.37
TQA SmoLLM	0D $\rightarrow$ LEN	0.583	0.524	-0.059	[-0.074, -0.043]	no	0.00
TQA SmoLLM	NONTOPO $\rightarrow$ NONTOPO+0D	0.899	0.899	-0.000	[-0.001, +0.001]	yes	1.00
TQA SmoLLM	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.899	0.899	-0.001	[-0.002, +0.000]	yes	1.00
TQA SmoLLM	LEX $\rightarrow$ LEX+0D	0.820	0.827	+0.007	[+0.001, +0.012]	yes	1.00
TQA SmoLLM	LEX $\rightarrow$ LEX+SINK	0.820	0.819	-0.001	[-0.007, +0.004]	yes	1.00
TQA SmoLLM	LEX $\rightarrow$ LEX+DEFL	0.820	0.822	+0.002	[-0.005, +0.009]	yes	0.91
TQA SmoLLM	LOGPROB $\rightarrow$ LOGPROB+0D	0.588	0.613	+0.025	[+0.015, +0.035]	no	0.59
TQA SmoLLM	HIDDEN $\rightarrow$ HIDDEN+0D	0.894	0.894	+0.000	[-0.001, +0.001]	yes	1.00
TQA SmoLLM	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.899	0.899	-0.000	[-0.000, -0.000]	yes	1.00
TQA SmoLLM	NONTOPO $\rightarrow$ NONTOPO(+0D)+DEFL	0.899	0.899	-0.000	[-0.000, -0.000]	yes	1.00
TQA SmoLLM	NONTOPO $\rightarrow$ NONTOPO(+0D)+SINK	0.899	0.899	-0.000	[-0.000, +0.000]	yes	1.00
TQA SmoLLM	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.893	0.893	-0.000	[-0.001, -0.000]	yes	1.00
TQA SmoLLM	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.588	0.607	+0.019	[+0.010, +0.029]	no	0.67
TQA SmoLLM	SINK $\rightarrow$ SINK(+0D)	0.610	0.633	+0.022	[+0.014, +0.031]	no	0.81
TQA Mis7B	0D $\rightarrow$ SINK	0.684	0.685	+0.000	[-0.001, +0.001]	yes	1.00
TQA Mis7B	0D $\rightarrow$ LEGACY_MSTPROXY	0.684	0.646	-0.038	[-0.051, -0.026]	no	0.24
TQA Mis7B	SINK $\rightarrow$ 0D+SINK	0.685	0.685	+0.000	[-0.001, +0.002]	yes	1.00
TQA Mis7B	SINK $\rightarrow$ SINK+DEFL	0.685	0.740	+0.055	[+0.040, +0.070]	no	0.00
TQA Mis7B	0D $\rightarrow$ 0D+1D	0.684	0.685	+0.000	[-0.000, +0.001]	yes	1.00
TQA Mis7B	SINK $\rightarrow$ SINK+1D	0.685	0.685	+0.000	[-0.000, +0.001]	yes	1.00
TQA Mis7B	SINK $\rightarrow$ SINK+ANS	0.685	0.727	+0.043	[+0.027, +0.058]	no	0.00
TQA Mis7B	PH_0DTOT $\rightarrow$ PH_SINK	0.887	0.903	+0.016	[+0.011, +0.020]	no	1.00
TQA Mis7B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.903	0.903	+0.001	[-0.000, +0.002]	yes	1.00
TQA Mis7B	PH_0D $\rightarrow$ PH_ENT	0.893	0.897	+0.004	[-0.003, +0.011]	yes	0.94
TQA Mis7B	0D $\rightarrow$ LOGPROB	0.684	0.641	-0.044	[-0.067, -0.020]	no	0.00
TQA Mis7B	0D $\rightarrow$ HIDDEN	0.684	0.956	+0.271	[+0.256, +0.286]	no	0.00
TQA Mis7B	0D $\rightarrow$ LOOKBACK	0.684	0.933	+0.248	[+0.233, +0.263]	no	0.00
TQA Mis7B	0D $\rightarrow$ LLMCHECK	0.684	0.652	-0.032	[-0.048, -0.016]	no	0.00
TQA Mis7B	0D $\rightarrow$ LEX	0.684	0.820	+0.136	[+0.116, +0.155]	no	0.00
TQA Mis7B	0D $\rightarrow$ ROWSTAT	0.684	0.694	+0.010	[-0.004, +0.024]	no	0.08
TQA Mis7B	0D $\rightarrow$ LEN	0.684	0.525	-0.159	[-0.179, -0.140]	no	0.00
TQA Mis7B	NONTOPO $\rightarrow$ NONTOPO+0D	0.959	0.958	-0.000	[-0.000, -0.000]	yes	1.00
TQA Mis7B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.959	0.959	+0.000	[-0.000, +0.001]	yes	1.00
TQA Mis7B	LEX $\rightarrow$ LEX+0D	0.820	0.834	+0.013	[+0.004, +0.023]	no	0.66
TQA Mis7B	LEX $\rightarrow$ LEX+SINK	0.820	0.833	+0.013	[+0.004, +0.023]	no	0.66
TQA Mis7B	LEX $\rightarrow$ LEX+DEFL	0.820	0.823	+0.003	[-0.009, +0.015]	no	0.29
TQA Mis7B	LOGPROB $\rightarrow$ LOGPROB+0D	0.641	0.722	+0.081	[+0.066, +0.097]	no	0.00
TQA Mis7B	HIDDEN $\rightarrow$ HIDDEN+0D	0.956	0.956	+0.000	[-0.000, +0.001]	yes	1.00
TQA Mis7B	NONTOPO $\rightarrow$ NONTOPO(+0D)	0.958	0.958	-0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B	NONTOPO $\rightarrow$ NONTOPO(+0D)+DEFL	0.958	0.958	-0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B	NONTOPO $\rightarrow$ NONTOPO(+0D)+SINK	0.958	0.958	-0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B	HIDDEN $\rightarrow$ HIDDEN(+0D)	0.955	0.955	-0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B	LOGPROB $\rightarrow$ LOGPROB(+0D)	0.640	0.719	+0.079	[+0.064, +0.094]	no	0.00
TQA Mis7B	SINK $\rightarrow$ SINK(+0D)	0.684	0.684	+0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B (chat)	0D $\rightarrow$ SINK	0.688	0.688	-0.000	[-0.001, +0.000]	yes	1.00
TQA Mis7B (chat)	0D $\rightarrow$ LEGACY_MSTPROXY	0.688	0.652	-0.037	[-0.049, -0.024]	no	0.23
TQA Mis7B (chat)	SINK $\rightarrow$ 0D+SINK	0.688	0.689	+0.000	[-0.001, +0.001]	yes	1.00
TQA Mis7B (chat)	SINK $\rightarrow$ SINK+DEFL	0.688	0.756	+0.067	[+0.052, +0.082]	no	0.00
TQA Mis7B (chat)	0D $\rightarrow$ 0D+1D	0.688	0.688	-0.000	[-0.001, +0.000]	yes	1.00
TQA Mis7B (chat)	SINK $\rightarrow$ SINK+1D	0.688	0.688	-0.000	[-0.001, +0.000]	yes	1.00
TQA Mis7B (chat)	SINK $\rightarrow$ SINK+ANS	0.688	0.767	+0.079	[+0.061, +0.096]	no	0.00
TQA Mis7B (chat)	PH_0DTOT $\rightarrow$ PH_SINK	0.890	0.907	+0.016	[+0.012, +0.021]	no	1.00
TQA Mis7B (chat)	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.907	0.907	+0.001	[-0.001, +0.002]	yes	1.00
TQA Mis7B (chat)	PH_0D $\rightarrow$ PH_ENT	0.900	0.901	+0.001	[-0.006, +0.008]	yes	0.95
TQA Mis7B (chat)	0D $\rightarrow$ LOGPROB	0.688	0.651	-0.037	[-0.062, -0.013]	no	0.00
TQA Mis7B (chat)	0D $\rightarrow$ HIDDEN	0.688	0.958	+0.270	[+0.255, +0.284]	no	0.03
TQA Mis7B (chat)	0D $\rightarrow$ LOOKBACK	0.688	0.934	+0.246	[+0.231, +0.260]	no	0.03
TQA Mis7B (chat)	0D $\rightarrow$ LLMCHECK	0.688	0.653	-0.036	[-0.052, -0.019]	no	0.00
TQA Mis7B (chat)	0D $\rightarrow$ LEX	0.688	0.820	+0.132	[+0.112, +0.151]	no	0.00
TQA Mis7B (chat)	0D $\rightarrow$ ROWSTAT	0.688	0.700	+0.011	[-0.003, +0.025]	no	0.09
TQA Mis7B (chat)	0D $\rightarrow$ LEN	0.688	0.525	-0.163	[-0.184, -0.143]	no	0.00
TQA Mis7B (chat)	NONTOPO $\rightarrow$ NONTOPO+0D	0.960	0.960	-0.000	[-0.000, -0.000]	yes	1.00

Table 16: Complete paired difference tests and equivalence tests across all settings and comparisons.

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\equiv_{.015}$	power
TQA Mis7B (chat)	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.960	0.961	+0.000	[-0.000, +0.001]	yes	1.00
TQA Mis7B (chat)	LEX $\rightarrow$ LEX+0D	0.820	0.834	+0.014	[+0.004, +0.023]	no	0.65
TQA Mis7B (chat)	LEX $\rightarrow$ LEX+SINK	0.820	0.834	+0.014	[+0.004, +0.023]	no	0.65
TQA Mis7B (chat)	LEX $\rightarrow$ LEX+DEFL	0.820	0.835	+0.015	[+0.002, +0.027]	no	0.25
TQA Mis7B (chat)	LOGPROB $\rightarrow$ LOGPROB+0D	0.651	0.733	+0.082	[+0.067, +0.097]	no	0.01
TQA Mis7B (chat)	HIDDEN $\rightarrow$ HIDDEN+0D	0.958	0.958	-0.000	[-0.001, +0.000]	yes	1.00
TQA Mis7B (chat)	NONTOPO $\rightarrow$ NONTOPO(+)+0D	0.960	0.960	-0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B (chat)	NONTOPO $\rightarrow$ NONTOPO(+)+DEFL	0.960	0.960	-0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B (chat)	NONTOPO $\rightarrow$ NONTOPO(+)+SINK	0.960	0.960	-0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B (chat)	HIDDEN $\rightarrow$ HIDDEN(+)+0D	0.958	0.958	-0.000	[-0.000, +0.000]	yes	1.00
TQA Mis7B (chat)	LOGPROB $\rightarrow$ LOGPROB(+)+0D	0.651	0.728	+0.077	[+0.062, +0.092]	no	0.00
TQA Mis7B (chat)	SINK $\rightarrow$ SINK(+)+0D	0.688	0.688	+0.000	[-0.000, +0.000]	yes	1.00
TQA Qw7B	0D $\rightarrow$ SINK	0.679	0.749	+0.070	[+0.050, +0.089]	no	0.00
TQA Qw7B	0D $\rightarrow$ LEGACY_MSTPROXY	0.679	0.678	-0.001	[-0.007, +0.004]	yes	0.99
TQA Qw7B	SINK $\rightarrow$ 0D+SINK	0.749	0.777	+0.027	[+0.021, +0.034]	no	0.95
TQA Qw7B	SINK $\rightarrow$ SINK+DEFL	0.749	0.788	+0.038	[+0.029, +0.048]	no	0.64
TQA Qw7B	0D $\rightarrow$ 0D+1D	0.679	0.701	+0.021	[+0.011, +0.032]	no	0.52
TQA Qw7B	SINK $\rightarrow$ SINK+1D	0.749	0.767	+0.017	[+0.010, +0.025]	no	0.89
TQA Qw7B	SINK $\rightarrow$ SINK+ANS	0.749	0.794	+0.045	[+0.030, +0.060]	no	0.02
TQA Qw7B	PH_0DTOT $\rightarrow$ PH_SINK	0.872	0.821	-0.051	[-0.062, -0.040]	no	0.45
TQA Qw7B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.821	0.881	+0.060	[+0.051, +0.069]	no	0.71
TQA Qw7B	PH_0D $\rightarrow$ PH_ENT	0.850	0.887	+0.037	[+0.027, +0.048]	no	0.52
TQA Qw7B	0D $\rightarrow$ LOGPROB	0.679	0.674	-0.006	[-0.028, +0.017]	no	0.00
TQA Qw7B	0D $\rightarrow$ HIDDEN	0.679	0.933	+0.254	[+0.240, +0.268]	no	0.09
TQA Qw7B	0D $\rightarrow$ LOOKBACK	0.679	0.922	+0.242	[+0.229, +0.256]	no	0.17
TQA Qw7B	0D $\rightarrow$ LLMCHECK	0.679	0.667	-0.012	[-0.027, +0.004]	no	0.00
TQA Qw7B	0D $\rightarrow$ LEX	0.679	0.820	+0.141	[+0.122, +0.159]	no	0.00
TQA Qw7B	0D $\rightarrow$ ROWSTAT	0.679	0.736	+0.057	[+0.048, +0.066]	no	0.74
TQA Qw7B	0D $\rightarrow$ LEN	0.679	0.529	-0.151	[-0.171, -0.130]	no	0.00
TQA Qw7B	NONTOPO $\rightarrow$ NONTOPO+0D	0.950	0.949	-0.000	[-0.000, +0.000]	yes	1.00
TQA Qw7B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.950	0.949	-0.000	[-0.001, +0.000]	yes	1.00
TQA Qw7B	LEX $\rightarrow$ LEX+0D	0.820	0.834	+0.014	[+0.007, +0.021]	no	0.94
TQA Qw7B	LEX $\rightarrow$ LEX+SINK	0.820	0.848	+0.028	[+0.016, +0.040]	no	0.29
TQA Qw7B	LEX $\rightarrow$ LEX+DEFL	0.820	0.828	+0.008	[-0.001, +0.016]	no	0.78
TQA Qw7B	LOGPROB $\rightarrow$ LOGPROB+0D	0.674	0.727	+0.053	[+0.042, +0.065]	no	0.40
TQA Qw7B	HIDDEN $\rightarrow$ HIDDEN+0D	0.933	0.933	-0.000	[-0.001, +0.000]	yes	1.00
TQA Qw7B	NONTOPO $\rightarrow$ NONTOPO(+)+0D	0.949	0.949	-0.000	[-0.000, +0.000]	yes	1.00
TQA Qw7B	NONTOPO $\rightarrow$ NONTOPO(+)+DEFL	0.949	0.949	+0.000	[-0.000, +0.000]	yes	1.00
TQA Qw7B	NONTOPO $\rightarrow$ NONTOPO(+)+SINK	0.949	0.949	+0.000	[-0.001, +0.001]	yes	1.00
TQA Qw7B	HIDDEN $\rightarrow$ HIDDEN(+)+0D	0.933	0.933	+0.000	[-0.000, +0.001]	yes	1.00
TQA Qw7B	LOGPROB $\rightarrow$ LOGPROB(+)+0D	0.673	0.725	+0.052	[+0.039, +0.064]	no	0.23
TQA Qw7B	SINK $\rightarrow$ SINK(+)+0D	0.749	0.775	+0.026	[+0.018, +0.034]	no	0.88
HE Qw3B	0D $\rightarrow$ SINK	0.811	0.829	+0.018	[+0.008, +0.028]	no	0.64
HE Qw3B	0D $\rightarrow$ LEGACY_MSTPROXY	0.811	0.731	-0.080	[-0.091, -0.069]	no	0.46
HE Qw3B	SINK $\rightarrow$ 0D+SINK	0.829	0.857	+0.028	[+0.022, +0.035]	no	0.97
HE Qw3B	SINK $\rightarrow$ SINK+DEFL	0.829	0.887	+0.058	[+0.049, +0.067]	no	0.73
HE Qw3B	0D $\rightarrow$ 0D+1D	0.811	0.826	+0.015	[+0.009, +0.023]	no	0.94
HE Qw3B	SINK $\rightarrow$ SINK+1D	0.829	0.843	+0.014	[+0.007, +0.021]	no	0.97
HE Qw3B	SINK $\rightarrow$ SINK+ANS	0.829	0.978	+0.150	[+0.139, +0.161]	no	0.44
HE Qw3B	PH_0DTOT $\rightarrow$ PH_SINK	0.958	0.958	+0.000	[-0.005, +0.006]	yes	1.00
HE Qw3B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.958	0.962	+0.004	[+0.003, +0.005]	yes	1.00
HE Qw3B	PH_0D $\rightarrow$ PH_ENT	0.971	0.964	-0.006	[-0.012, -0.000]	yes	0.99
HE Qw3B	0D $\rightarrow$ LOGPROB	0.811	0.993	+0.182	[+0.170, +0.194]	no	0.33
HE Qw3B	0D $\rightarrow$ HIDDEN	0.811	1.000	+0.189	[+0.177, +0.201]	no	0.32
HE Qw3B	0D $\rightarrow$ LOOKBACK	0.811	0.998	+0.187	[+0.175, +0.199]	no	0.32
HE Qw3B	0D $\rightarrow$ LLMCHECK	0.811	0.745	-0.066	[-0.078, -0.054]	no	0.27
HE Qw3B	0D $\rightarrow$ LEX	0.811	0.981	+0.170	[+0.159, +0.182]	no	0.37
HE Qw3B	0D $\rightarrow$ ROWSTAT	0.811	0.829	+0.018	[+0.007, +0.030]	no	0.35
HE Qw3B	0D $\rightarrow$ LEN	0.811	0.962	+0.151	[+0.140, +0.163]	no	0.36
HE Qw3B	NONTOPO $\rightarrow$ NONTOPO+0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	NONTOPO $\rightarrow$ NONTOPO+DEFL	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	LEX $\rightarrow$ LEX+0D	0.981	0.979	-0.002	[-0.005, +0.000]	yes	1.00
HE Qw3B	LEX $\rightarrow$ LEX+SINK	0.981	0.980	-0.001	[-0.003, +0.001]	yes	1.00
HE Qw3B	LEX $\rightarrow$ LEX+DEFL	0.981	0.971	-0.010	[-0.014, -0.007]	yes	1.00
HE Qw3B	LOGPROB $\rightarrow$ LOGPROB+0D	0.993	0.993	+0.000	[-0.001, +0.001]	yes	1.00
HE Qw3B	HIDDEN $\rightarrow$ HIDDEN+0D	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)+0D	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)+DEFL	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)+SINK	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	HIDDEN $\rightarrow$ HIDDEN(+)+0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw3B	LOGPROB $\rightarrow$ LOGPROB(+)+0D	0.993	0.993	+0.001	[+0.000, +0.001]	yes	1.00
HE Qw3B	SINK $\rightarrow$ SINK(+)+0D	0.828	0.839	+0.011	[+0.007, +0.015]	yes	1.00
HE Qw1.5B	0D $\rightarrow$ SINK	0.789	0.820	+0.031	[+0.017, +0.045]	no	0.08
HE Qw1.5B	0D $\rightarrow$ LEGACY_MSTPROXY	0.789	0.719	-0.070	[-0.086, -0.053]	no	0.00
HE Qw1.5B	SINK $\rightarrow$ 0D+SINK	0.820	0.832	+0.012	[+0.005, +0.019]	no	0.91
HE Qw1.5B	SINK $\rightarrow$ SINK+DEFL	0.820	0.872	+0.052	[+0.038, +0.065]	no	0.17
HE Qw1.5B	0D $\rightarrow$ 0D+1D	0.789	0.815	+0.026	[+0.013, +0.040]	no	0.17
HE Qw1.5B	SINK $\rightarrow$ SINK+1D	0.820	0.826	+0.006	[-0.005, +0.017]	no	0.45
HE Qw1.5B	SINK $\rightarrow$ SINK+ANS	0.820	0.975	+0.155	[+0.138, +0.172]	no	0.00
HE Qw1.5B	PH_0DTOT $\rightarrow$ PH_SINK	0.921	0.922	+0.002	[-0.009, +0.013]	yes	0.46
HE Qw1.5B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.922	0.931	+0.008	[+0.006, +0.011]	yes	1.00
HE Qw1.5B	PH_0D $\rightarrow$ PH_ENT	0.949	0.933	-0.016	[-0.028, -0.004]	no	0.33
HE Qw1.5B	0D $\rightarrow$ LOGPROB	0.789	0.995	+0.205	[+0.188, +0.222]	no	0.00
HE Qw1.5B	0D $\rightarrow$ HIDDEN	0.789	1.000	+0.210	[+0.193, +0.227]	no	0.00
HE Qw1.5B	0D $\rightarrow$ LOOKBACK	0.789	0.999	+0.210	[+0.193, +0.226]	no	0.00
HE Qw1.5B	0D $\rightarrow$ LLMCHECK	0.789	0.791	+0.002	[-0.017, +0.021]	no	0.00
HE Qw1.5B	0D $\rightarrow$ LEX	0.789	0.979	+0.190	[+0.172, +0.207]	no	0.00
HE Qw1.5B	0D $\rightarrow$ ROWSTAT	0.789	0.788	-0.001	[-0.018, +0.016]	no	0.00
HE Qw1.5B	0D $\rightarrow$ LEN	0.789	0.967	+0.178	[+0.160, +0.196]	no	0.00

Table 16: Complete paired difference tests and equivalence tests across all settings and comparisons.

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\equiv_{.015}$	power
HE Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+0D	1.000	1.000	+0.000	[+0.000, +0.000]	yes	1.00
HE Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+DEFL	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw1.5B	LEX $\rightarrow$ LEX+0D	0.979	0.971	-0.008	[-0.014, -0.002]	yes	0.99
HE Qw1.5B	LEX $\rightarrow$ LEX+SINK	0.979	0.968	-0.011	[-0.017, -0.005]	no	0.98
HE Qw1.5B	LEX $\rightarrow$ LEX+DEFL	0.979	0.956	-0.022	[-0.030, -0.016]	no	0.94
HE Qw1.5B	LOGPROB $\rightarrow$ LOGPROB+0D	0.995	0.994	-0.001	[-0.003, +0.001]	yes	1.00
HE Qw1.5B	HIDDEN $\rightarrow$ HIDDEN+0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)+0D	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)+DEFL	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
HE Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)+SINK	1.000	1.000	+0.000	[-0.000, +0.000]	yes	1.00
HE Qw1.5B	HIDDEN $\rightarrow$ HIDDEN(+)+0D	1.000	1.000	-0.000	[-0.000, +0.000]	yes	1.00
HE Qw1.5B	LOGPROB $\rightarrow$ LOGPROB(+)+0D	0.994	0.994	-0.000	[-0.001, +0.001]	yes	1.00
HE Qw1.5B	SINK $\rightarrow$ SINK(+)+0D	0.819	0.823	+0.004	[-0.000, +0.009]	yes	1.00
TrQA Qw3B	0D $\rightarrow$ SINK	0.732	0.717	-0.015	[-0.026, -0.004]	no	0.48
TrQA Qw3B	0D $\rightarrow$ LEGACY_MSTPROXY	0.732	0.691	-0.041	[-0.057, -0.025]	no	0.00
TrQA Qw3B	SINK $\rightarrow$ 0D+SINK	0.717	0.738	+0.021	[+0.012, +0.030]	no	0.76
TrQA Qw3B	SINK $\rightarrow$ SINK+DEFL	0.717	0.719	+0.002	[-0.012, +0.015]	no	0.13
TrQA Qw3B	0D $\rightarrow$ 0D+1D	0.732	0.729	-0.003	[-0.012, +0.006]	yes	0.74
TrQA Qw3B	SINK $\rightarrow$ SINK+1D	0.717	0.718	+0.001	[-0.009, +0.010]	yes	0.65
TrQA Qw3B	SINK $\rightarrow$ SINK+ANS	0.717	0.752	+0.035	[+0.019, +0.051]	no	0.00
TrQA Qw3B	PH_0DTOT $\rightarrow$ PH_SINK	0.807	0.820	+0.012	[+0.004, +0.021]	no	0.82
TrQA Qw3B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.820	0.824	+0.004	[+0.002, +0.006]	yes	1.00
TrQA Qw3B	PH_0D $\rightarrow$ PH_ENT	0.807	0.810	+0.003	[-0.011, +0.017]	no	0.08
TrQA Qw3B	0D $\rightarrow$ LOGPROB	0.732	0.839	+0.107	[+0.088, +0.125]	no	0.00
TrQA Qw3B	0D $\rightarrow$ HIDDEN	0.732	0.841	+0.109	[+0.090, +0.129]	no	0.00
TrQA Qw3B	0D $\rightarrow$ LOOKBACK	0.732	0.861	+0.129	[+0.111, +0.147]	no	0.00
TrQA Qw3B	0D $\rightarrow$ LLMCHECK	0.732	0.688	-0.044	[-0.065, -0.023]	no	0.00
TrQA Qw3B	0D $\rightarrow$ LEX	0.732	0.596	-0.137	[-0.163, -0.110]	no	0.00
TrQA Qw3B	0D $\rightarrow$ ROWSTAT	0.732	0.724	-0.009	[-0.025, +0.008]	no	0.00
TrQA Qw3B	0D $\rightarrow$ LEN	0.732	0.569	-0.163	[-0.189, -0.137]	no	0.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO+0D	0.870	0.870	+0.001	[-0.000, +0.002]	yes	1.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.870	0.870	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw3B	LEX $\rightarrow$ LEX+0D	0.596	0.733	+0.137	[+0.117, +0.158]	no	0.00
TrQA Qw3B	LEX $\rightarrow$ LEX+SINK	0.596	0.724	+0.128	[+0.108, +0.149]	no	0.00
TrQA Qw3B	LEX $\rightarrow$ LEX+DEFL	0.596	0.690	+0.095	[+0.073, +0.116]	no	0.00
TrQA Qw3B	LOGPROB $\rightarrow$ LOGPROB+0D	0.839	0.843	+0.004	[-0.002, +0.011]	yes	0.95
TrQA Qw3B	HIDDEN $\rightarrow$ HIDDEN+0D	0.841	0.843	+0.002	[+0.000, +0.004]	yes	1.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)+0D	0.869	0.871	+0.002	[-0.000, +0.004]	yes	1.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)+DEFL	0.869	0.870	+0.001	[-0.001, +0.002]	yes	1.00
TrQA Qw3B	NONTOPO $\rightarrow$ NONTOPO(+)+SINK	0.869	0.871	+0.001	[-0.001, +0.004]	yes	1.00
TrQA Qw3B	HIDDEN $\rightarrow$ HIDDEN(+)+0D	0.841	0.849	+0.008	[+0.003, +0.012]	yes	1.00
TrQA Qw3B	LOGPROB $\rightarrow$ LOGPROB(+)+0D	0.838	0.844	+0.006	[+0.002, +0.010]	yes	1.00
TrQA Qw3B	SINK $\rightarrow$ SINK(+)+0D	0.716	0.733	+0.017	[+0.010, +0.025]	no	0.89
TrQA Qw1.5B	0D $\rightarrow$ SINK	0.716	0.707	-0.009	[-0.020, +0.002]	no	0.43
TrQA Qw1.5B	0D $\rightarrow$ LEGACY_MSTPROXY	0.716	0.685	-0.031	[-0.046, -0.015]	no	0.00
TrQA Qw1.5B	SINK $\rightarrow$ 0D+SINK	0.707	0.717	+0.010	[+0.001, +0.020]	no	0.68
TrQA Qw1.5B	SINK $\rightarrow$ SINK+DEFL	0.707	0.719	+0.012	[-0.003, +0.027]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ 0D+1D	0.716	0.714	-0.002	[-0.011, +0.008]	yes	0.65
TrQA Qw1.5B	SINK $\rightarrow$ SINK+1D	0.707	0.709	+0.002	[-0.008, +0.013]	yes	0.49
TrQA Qw1.5B	SINK $\rightarrow$ SINK+ANS	0.707	0.726	+0.019	[+0.001, +0.036]	no	0.00
TrQA Qw1.5B	PH_0DTOT $\rightarrow$ PH_SINK	0.759	0.767	+0.008	[-0.003, +0.018]	no	0.53
TrQA Qw1.5B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.767	0.772	+0.004	[+0.001, +0.008]	yes	1.00
TrQA Qw1.5B	PH_0D $\rightarrow$ PH_ENT	0.768	0.768	-0.000	[-0.017, +0.017]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LOGPROB	0.716	0.823	+0.107	[+0.085, +0.129]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ HIDDEN	0.716	0.855	+0.140	[+0.119, +0.160]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LOOKBACK	0.716	0.824	+0.108	[+0.088, +0.129]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LLMCHECK	0.716	0.676	-0.039	[-0.062, -0.016]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LEX	0.716	0.609	-0.106	[-0.136, -0.078]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ ROWSTAT	0.716	0.720	+0.005	[-0.011, +0.020]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LEN	0.716	0.588	-0.128	[-0.154, -0.102]	no	0.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+0D	0.873	0.873	+0.000	[-0.001, +0.001]	yes	1.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.873	0.872	-0.001	[-0.002, +0.001]	yes	1.00
TrQA Qw1.5B	LEX $\rightarrow$ LEX+0D	0.609	0.721	+0.112	[+0.091, +0.134]	no	0.00
TrQA Qw1.5B	LEX $\rightarrow$ LEX+SINK	0.609	0.715	+0.106	[+0.085, +0.127]	no	0.00
TrQA Qw1.5B	LEX $\rightarrow$ LEX+DEFL	0.609	0.672	+0.063	[+0.040, +0.085]	no	0.00
TrQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB+0D	0.823	0.828	+0.006	[-0.001, +0.013]	yes	0.92
TrQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN+0D	0.855	0.855	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)+0D	0.872	0.873	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)+DEFL	0.872	0.873	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO(+)+SINK	0.872	0.873	+0.001	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN(+)+0D	0.855	0.857	+0.002	[-0.000, +0.004]	yes	1.00
TrQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB(+)+0D	0.822	0.828	+0.006	[+0.001, +0.010]	yes	1.00
TrQA Qw1.5B	SINK $\rightarrow$ SINK(+)+0D	0.706	0.717	+0.011	[+0.003, +0.018]	no	0.93
TrQA Phi-3	0D $\rightarrow$ SINK	0.710	0.707	-0.004	[-0.010, +0.002]	yes	0.98
TrQA Phi-3	0D $\rightarrow$ LEGACY_MSTPROXY	0.710	0.669	-0.041	[-0.057, -0.025]	no	0.00
TrQA Phi-3	SINK $\rightarrow$ 0D+SINK	0.707	0.713	+0.006	[+0.000, +0.012]	yes	0.99
TrQA Phi-3	SINK $\rightarrow$ SINK+DEFL	0.707	0.736	+0.029	[+0.013, +0.045]	no	0.00
TrQA Phi-3	0D $\rightarrow$ 0D+1D	0.710	0.705	-0.005	[-0.011, +0.000]	yes	0.99
TrQA Phi-3	SINK $\rightarrow$ SINK+1D	0.707	0.699	-0.008	[-0.013, -0.002]	yes	1.00
TrQA Phi-3	SINK $\rightarrow$ SINK+ANS	0.707	0.736	+0.029	[+0.012, +0.046]	no	0.00
TrQA Phi-3	PH_0DTOT $\rightarrow$ PH_SINK	0.810	0.826	+0.016	[+0.009, +0.022]	no	0.96
TrQA Phi-3	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.826	0.828	+0.002	[-0.001, +0.004]	yes	1.00
TrQA Phi-3	PH_0D $\rightarrow$ PH_ENT	0.807	0.817	+0.010	[-0.002, +0.022]	no	0.31
TrQA Phi-3	0D $\rightarrow$ LOGPROB	0.710	0.877	+0.167	[+0.148, +0.186]	no	0.00
TrQA Phi-3	0D $\rightarrow$ HIDDEN	0.710	0.867	+0.157	[+0.137, +0.176]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LOOKBACK	0.710	0.840	+0.129	[+0.109, +0.150]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LLMCHECK	0.710	0.676	-0.034	[-0.055, -0.014]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LEX	0.710	0.635	-0.076	[-0.102, -0.050]	no	0.00
TrQA Phi-3	0D $\rightarrow$ ROWSTAT	0.710	0.728	+0.018	[+0.002, +0.034]	no	0.00

Table 16: Complete paired difference tests and equivalence tests across all settings and comparisons.

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\equiv_{.015}$	power
TrQA Phi-3	0D $\rightarrow$ LEN	0.710	0.581	-0.130	[-0.156, -0.104]	no	0.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+0D	0.883	0.883	-0.000	[-0.001, +0.000]	yes	1.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.883	0.885	+0.002	[+0.000, +0.004]	yes	1.00
TrQA Phi-3	LEX $\rightarrow$ LEX+0D	0.635	0.721	+0.086	[+0.069, +0.104]	no	0.00
TrQA Phi-3	LEX $\rightarrow$ LEX+SINK	0.635	0.718	+0.083	[+0.066, +0.101]	no	0.00
TrQA Phi-3	LEX $\rightarrow$ LEX+DEFL	0.635	0.712	+0.077	[+0.057, +0.098]	no	0.00
TrQA Phi-3	LOGPROB $\rightarrow$ LOGPROB+0D	0.877	0.878	+0.000	[-0.002, +0.002]	yes	1.00
TrQA Phi-3	HIDDEN $\rightarrow$ HIDDEN+0D	0.867	0.867	-0.000	[-0.001, +0.000]	yes	1.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.883	0.884	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.883	0.884	+0.001	[-0.001, +0.003]	yes	1.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.883	0.883	+0.000	[-0.001, +0.001]	yes	1.00
TrQA Phi-3	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.867	0.869	+0.002	[-0.000, +0.004]	yes	1.00
TrQA Phi-3	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.877	0.878	+0.001	[-0.001, +0.003]	yes	1.00
TrQA Phi-3	SINK $\rightarrow$ SINK(+)0D	0.706	0.710	+0.004	[-0.001, +0.009]	yes	1.00
TrQA TinyLL	0D $\rightarrow$ SINK	0.720	0.720	+0.001	[-0.010, +0.012]	yes	0.40
TrQA TinyLL	0D $\rightarrow$ LEGACY_MSTPROXY	0.720	0.729	+0.010	[-0.000, +0.020]	no	0.55
TrQA TinyLL	SINK $\rightarrow$ 0D+SINK	0.720	0.724	+0.004	[-0.003, +0.011]	yes	0.93
TrQA TinyLL	SINK $\rightarrow$ SINK+DEFL	0.720	0.734	+0.014	[-0.000, +0.028]	no	0.10
TrQA TinyLL	0D $\rightarrow$ 0D+1D	0.720	0.718	-0.002	[-0.011, +0.008]	yes	0.60
TrQA TinyLL	SINK $\rightarrow$ SINK+1D	0.720	0.719	-0.001	[-0.010, +0.008]	yes	0.73
TrQA TinyLL	SINK $\rightarrow$ SINK+ANS	0.720	0.725	+0.005	[-0.011, +0.021]	no	0.00
TrQA TinyLL	PH_0DTOT $\rightarrow$ PH_SINK	0.792	0.814	+0.022	[+0.015, +0.029]	no	0.94
TrQA TinyLL	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.814	0.813	-0.001	[-0.002, +0.000]	yes	1.00
TrQA TinyLL	PH_0D $\rightarrow$ PH_ENT	0.775	0.807	+0.032	[+0.017, +0.047]	no	0.01
TrQA TinyLL	0D $\rightarrow$ LOGPROB	0.720	0.724	+0.004	[-0.019, +0.027]	no	0.00
TrQA TinyLL	0D $\rightarrow$ HIDDEN	0.720	0.827	+0.108	[+0.088, +0.128]	no	0.00
TrQA TinyLL	0D $\rightarrow$ LOOKBACK	0.720	0.823	+0.104	[+0.087, +0.121]	no	0.00
TrQA TinyLL	0D $\rightarrow$ LLMCHECK	0.720	0.688	-0.032	[-0.050, -0.013]	no	0.00
TrQA TinyLL	0D $\rightarrow$ LEX	0.720	0.740	+0.021	[-0.003, +0.044]	no	0.00
TrQA TinyLL	0D $\rightarrow$ ROWSTAT	0.720	0.752	+0.032	[+0.020, +0.045]	no	0.26
TrQA TinyLL	0D $\rightarrow$ LEN	0.720	0.631	-0.089	[-0.109, -0.068]	no	0.00
TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO+0D	0.842	0.842	-0.000	[-0.001, +0.001]	yes	1.00
TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.842	0.842	+0.000	[-0.003, +0.003]	yes	1.00
TrQA TinyLL	LEX $\rightarrow$ LEX+0D	0.740	0.764	+0.024	[+0.012, +0.036]	no	0.33
TrQA TinyLL	LEX $\rightarrow$ LEX+SINK	0.740	0.762	+0.021	[+0.010, +0.032]	no	0.45
TrQA TinyLL	LEX $\rightarrow$ LEX+DEFL	0.740	0.757	+0.016	[+0.002, +0.031]	no	0.04
TrQA TinyLL	LOGPROB $\rightarrow$ LOGPROB+0D	0.724	0.762	+0.038	[+0.025, +0.052]	no	0.17
TrQA TinyLL	HIDDEN $\rightarrow$ HIDDEN+0D	0.827	0.828	+0.001	[-0.001, +0.002]	yes	1.00
TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.842	0.842	+0.000	[-0.001, +0.002]	yes	1.00
TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.842	0.843	+0.001	[-0.001, +0.004]	yes	1.00
TrQA TinyLL	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.842	0.842	+0.001	[-0.001, +0.002]	yes	1.00
TrQA TinyLL	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.827	0.828	+0.001	[-0.001, +0.004]	yes	1.00
TrQA TinyLL	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.723	0.755	+0.031	[+0.020, +0.043]	no	0.40
TrQA TinyLL	SINK $\rightarrow$ SINK(+)0D	0.720	0.725	+0.005	[+0.001, +0.009]	yes	1.00
TrQA Mis7B	0D $\rightarrow$ SINK	0.699	0.696	-0.003	[-0.013, +0.007]	yes	0.62
TrQA Mis7B	0D $\rightarrow$ LEGACY_MSTPROXY	0.699	0.693	-0.006	[-0.019, +0.007]	no	0.21
TrQA Mis7B	SINK $\rightarrow$ 0D+SINK	0.696	0.698	+0.002	[-0.007, +0.010]	yes	0.80
TrQA Mis7B	SINK $\rightarrow$ SINK+DEFL	0.696	0.739	+0.043	[+0.026, +0.060]	no	0.00
TrQA Mis7B	0D $\rightarrow$ 0D+1D	0.699	0.692	-0.007	[-0.017, +0.003]	no	0.59
TrQA Mis7B	SINK $\rightarrow$ SINK+1D	0.696	0.686	-0.010	[-0.020, +0.001]	no	0.52
TrQA Mis7B	SINK $\rightarrow$ SINK+ANS	0.696	0.693	-0.004	[-0.020, +0.014]	no	0.00
TrQA Mis7B	PH_0DTOT $\rightarrow$ PH_SINK	0.808	0.812	+0.004	[-0.004, +0.011]	yes	0.88
TrQA Mis7B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.812	0.814	+0.002	[+0.000, +0.004]	yes	1.00
TrQA Mis7B	PH_0D $\rightarrow$ PH_ENT	0.775	0.812	+0.037	[+0.023, +0.051]	no	0.09
TrQA Mis7B	0D $\rightarrow$ LOGPROB	0.699	0.728	+0.029	[+0.005, +0.053]	no	0.00
TrQA Mis7B	0D $\rightarrow$ HIDDEN	0.699	0.833	+0.134	[+0.113, +0.155]	no	0.00
TrQA Mis7B	0D $\rightarrow$ LOOKBACK	0.699	0.829	+0.130	[+0.110, +0.150]	no	0.00
TrQA Mis7B	0D $\rightarrow$ LLMCHECK	0.699	0.686	-0.013	[-0.034, +0.008]	no	0.00
TrQA Mis7B	0D $\rightarrow$ LEX	0.699	0.677	-0.023	[-0.047, +0.002]	no	0.00
TrQA Mis7B	0D $\rightarrow$ ROWSTAT	0.699	0.731	+0.032	[+0.017, +0.047]	no	0.03
TrQA Mis7B	0D $\rightarrow$ LEN	0.699	0.635	-0.064	[-0.086, -0.043]	no	0.00
TrQA Mis7B	NONTOPO $\rightarrow$ NONTOPO+0D	0.841	0.842	+0.001	[-0.000, +0.002]	yes	1.00
TrQA Mis7B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.841	0.841	-0.000	[-0.001, +0.001]	yes	1.00
TrQA Mis7B	LEX $\rightarrow$ LEX+0D	0.677	0.723	+0.046	[+0.030, +0.062]	no	0.00
TrQA Mis7B	LEX $\rightarrow$ LEX+SINK	0.677	0.721	+0.045	[+0.029, +0.060]	no	0.00
TrQA Mis7B	LEX $\rightarrow$ LEX+DEFL	0.677	0.744	+0.068	[+0.049, +0.087]	no	0.00
TrQA Mis7B	LOGPROB $\rightarrow$ LOGPROB+0D	0.728	0.755	+0.027	[+0.012, +0.043]	no	0.00
TrQA Mis7B	HIDDEN $\rightarrow$ HIDDEN+0D	0.833	0.834	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Mis7B	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.841	0.843	+0.003	[+0.000, +0.005]	yes	1.00
TrQA Mis7B	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.841	0.846	+0.005	[+0.002, +0.008]	yes	1.00
TrQA Mis7B	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.841	0.843	+0.002	[-0.000, +0.005]	yes	1.00
TrQA Mis7B	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.833	0.836	+0.003	[+0.000, +0.006]	yes	1.00
TrQA Mis7B	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.727	0.755	+0.028	[+0.014, +0.042]	no	0.11
TrQA Mis7B	SINK $\rightarrow$ SINK(+)0D	0.696	0.700	+0.005	[-0.001, +0.011]	yes	0.98
TrQA Qw7B	0D $\rightarrow$ SINK	0.668	0.714	+0.046	[+0.022, +0.070]	no	0.00
TrQA Qw7B	0D $\rightarrow$ LEGACY_MSTPROXY	0.668	0.672	+0.004	[-0.002, +0.010]	yes	0.98
TrQA Qw7B	SINK $\rightarrow$ 0D+SINK	0.714	0.735	+0.022	[+0.011, +0.033]	no	0.46
TrQA Qw7B	SINK $\rightarrow$ SINK+DEFL	0.714	0.734	+0.020	[+0.008, +0.033]	no	0.27
TrQA Qw7B	0D $\rightarrow$ 0D+1D	0.668	0.668	+0.000	[-0.010, +0.010]	yes	0.59
TrQA Qw7B	SINK $\rightarrow$ SINK+1D	0.714	0.718	+0.004	[-0.004, +0.013]	yes	0.73
TrQA Qw7B	SINK $\rightarrow$ SINK+ANS	0.714	0.781	+0.067	[+0.051, +0.083]	no	0.00
TrQA Qw7B	PH_0DTOT $\rightarrow$ PH_SINK	0.822	0.801	-0.020	[-0.035, -0.006]	no	0.07
TrQA Qw7B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.801	0.832	+0.031	[+0.024, +0.039]	no	0.91
TrQA Qw7B	PH_0D $\rightarrow$ PH_ENT	0.798	0.841	+0.044	[+0.032, +0.056]	no	0.33
TrQA Qw7B	0D $\rightarrow$ LOGPROB	0.668	0.847	+0.180	[+0.158, +0.201]	no	0.00
TrQA Qw7B	0D $\rightarrow$ HIDDEN	0.668	0.854	+0.186	[+0.165, +0.207]	no	0.00
TrQA Qw7B	0D $\rightarrow$ LOOKBACK	0.668	0.879	+0.211	[+0.191, +0.231]	no	0.00
TrQA Qw7B	0D $\rightarrow$ LLMCHECK	0.668	0.725	+0.058	[+0.038, +0.078]	no	0.00
TrQA Qw7B	0D $\rightarrow$ LEX	0.668	0.594	-0.074	[-0.101, -0.046]	no	0.00

Table 16: Complete paired difference tests and equivalence tests across all settings and comparisons.

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\equiv_{.015}$	power
TrQA Qw7B	0D $\rightarrow$ ROWSTAT	0.668	0.734	+0.067	[+0.053, +0.081]	no	0.10
TrQA Qw7B	0D $\rightarrow$ LEN	0.668	0.595	-0.073	[-0.097, -0.049]	no	0.00
TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO+0D	0.881	0.881	-0.000	[-0.001, +0.000]	yes	1.00
TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.881	0.881	+0.000	[-0.001, +0.001]	yes	1.00
TrQA Qw7B	LEX $\rightarrow$ LEX+0D	0.594	0.678	+0.085	[+0.067, +0.102]	no	0.00
TrQA Qw7B	LEX $\rightarrow$ LEX+SINK	0.594	0.728	+0.134	[+0.113, +0.154]	no	0.00
TrQA Qw7B	LEX $\rightarrow$ LEX+DEFL	0.594	0.678	+0.085	[+0.066, +0.103]	no	0.00
TrQA Qw7B	LOGPROB $\rightarrow$ LOGPROB+0D	0.847	0.842	-0.005	[-0.010, +0.001]	yes	1.00
TrQA Qw7B	HIDDEN $\rightarrow$ HIDDEN+0D	0.854	0.853	-0.001	[-0.001, -0.000]	yes	1.00
TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO(+)0D	0.881	0.881	-0.000	[-0.001, +0.001]	yes	1.00
TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO(+)DEFL	0.881	0.881	-0.000	[-0.001, +0.001]	yes	1.00
TrQA Qw7B	NONTOPO $\rightarrow$ NONTOPO(+)SINK	0.881	0.882	+0.001	[-0.001, +0.003]	yes	1.00
TrQA Qw7B	HIDDEN $\rightarrow$ HIDDEN(+)0D	0.853	0.854	+0.001	[-0.001, +0.002]	yes	1.00
TrQA Qw7B	LOGPROB $\rightarrow$ LOGPROB(+)0D	0.847	0.846	-0.000	[-0.003, +0.002]	yes	1.00
TrQA Qw7B	SINK $\rightarrow$ SINK(+)0D	0.713	0.732	+0.019	[+0.009, +0.029]	no	0.60